Curriculum Vitae: updated October 11, 2026

I. Identifying Data

Name	M. Brandon Westover, MD, PhD
Current Position	Professor of Neurology and Neurological Sciences, Stanford University
Current Affiliation	Stanford University

II. Education History

Colleges and Universities Attended

7/1999	BS in Physics, Brigham Young University
7/2006	PhD, Physics, Washington University in St. Louis
7/2006	MD, Washington University School of Medicine

Residency and Fellowship Training

2006 – 2007	Internship, Barnes Jewish Hospital, Washington University School of Medicine, St. Louis, MO
2007 – 2010	Residency, Partners Neurology Program (Massachusetts General Hospital and Brigham and Women's Hospital), Boston, MA
2010 – 2012	Fellowship, Epilepsy and Clinical Neurophysiology, Massachusetts General Hospital, Boston, MA

Board Certification

2011	American Board of Psychiatry and Neurology (No. 56586)

III. Employment

Academic Appointments:

07/12-05/14	Instructor in Neurology, Harvard Medical School, Boston, MA
05/14-05/18	Assistant Professor in Neurology, Harvard Medical School, Boston, MA
05/18-10/22	Associate Professor of Neurology, Harvard Medical School, Boston, MA
10/22 – 5/26	Professor of Neurology, Harvard Medical School, Boston, MA
11/22 – 5/26	Emily Fisher Landau Professor of Neurology, Harvard Medical School, Boston, MA
5/26 – present	Professor of Neurology and Neurological Sciences, Stanford University, Stanford, CA

Other Appointments:

2022 to present	Director, Critical Care EEG Monitoring Service, BIDMC

IV. Honors and Awards

2011	Best Paper Award	Statistical Literacy (www.statlit.org)	Statistical Education
2012	Best Paper Award	Association for Computing Machinery: Special Interest Group on Knowledge Discovery and Data Mining (SIGKDD)	Basic Research
2014	Rappaport Fellowship	Massachusetts General Hospital	Nominated award
2018	#1 Most Disruptive AI Innovator	Partners Healthcare	Partners World Innovation Forum
2018	Innovation Discovery Grant	Partners Innovation Office	Artificial Intelligence Research / Development
2018	MGH-MIT Grand Challenge Winner	ECOR	Artificial Intelligence Research
2018	Best Neurology Clinical Research Abstract	MGH Neurology Department	Clinical Research
2018	Best Paper Award	CyberWorlds 2018, Singapore	Computer Science
2019	Disruptive Dozen AI Innovator	Massachusetts General Hospital Partners Healthcare	World Innovation Forum
2019	Innovation Discovery Award (IDG)	Partners Healthcare	World Innovation Forum
2019	Innovative Applications in Analytics Award (IAAA)	INFORMS Conference on Business Analytics & Operations Research	For achievements featuring novel combinations of descriptive, predictive and prescriptive analytics.
2019	Best Oral Abstract: Translational	American Delirium Society (ADS)	ADS annual meeting
2019	Derek Denny-Brown Young Neurological Scholar Award	American Neurological Association	ANA's Highest award. For outstanding scientific advances toward the prevention, diagnosis, treatment, and cure of neurological diseases

V. BIBLIOGRAPHY

Peer-reviewed original research (442 total, 5 in press, 10 submitted, 5 in review)

1. Eliasmith C, **Westover MB**, Anderson CH. "A general framework for neurobiological modeling: An application to the vestibular system." *Neurocomputing*. 2002. 46:1071-1076. PMCID: PMC6788744.
2. **Westover MB**, Eliasmith C, Anderson CH. "Linearly decodable functions from neural population codes." *Neurocomputing*. 2002. 45: 691-696. PMCID: PMC7062372.
3. Fischer B, **Westover MB**. "The neural multiple access channel." *Neurocomputing*. 2003. 52-54: 512-518. PMCID: PMC7062373.
4. **Westover MB**, Anderson CH. "Layer 4C of monkey V1 may linearize the output of the LGN." *Neurocomputing*. 2003. 52-54: 671-676. PMCID: PMC7062374.
5. **Westover MB**, O'Sullivan, JA. "Achievable rates for pattern recognition." *IEEE Transactions on Information Theory*, Jan 2008; 54(1): 299-320. PMCID: PMC7062371.
6. **Westover, MB**. "Asymptotic geometry of multiple hypothesis testing." *IEEE Transactions on Information Theory*, July 2008; 54(7): 3327-3329. PMCID: PMC6788803.
7. Woeltje KF, Butler AM, Goris AJ, Tutlam NT, Doherty JA, **Westover MB**, Ferris B, Bailey TC. "Automated Surveillance for Central Line Associated Bloodstream Infection in Intensive Care Units." *Infection Control and Hospital Epidemiology*, Sept.2008; 29:842-846. PMCID: PMC6788749.
8. Mueen A, Keogh E, Zhu Q, Cash SS, **Westover MB**. "Exact Discovery of Time Series Motifs." *Proceedings of the SIAM International Conference on Data Mining*, SDM 2009, April 30-May 2, 2008, Reno Nevada USA. SIAM 2008. PMCID: PMC6814436.
9. Chu-Shore J, **Westover MB**, Bianchi MT. "Power Law versus Exponential State Transition Dynamics: Application to Sleep-Wake Architecture." *PLoSONE* 5(12): e14204. Dec. 2, 2010. PMCID: PMC2996311.
10. Mueen A, Keogh E, Zhu Q, Cash SS, **Westover MB**, Bigdely-Shamlo N. "A disk-aware Algorithm for Time Series Motif Discovery." *DataMining and Knowledge Discovery*, 2011, 22(1-2): 73-105. PMCID: PMC7062370.

11. Zhang W-J, Keary C, **Westover MB**. "Premortem diagnosis of sporadic Creutzfeldt-Jakob disease aided by positron-emission tomography imaging." *American Journal of Neuroradiology*, Neuroradiology. 2011 Jan;32(1):E18. doi: 10.3174/ajnr.A2292. Epub 2010 Nov 11. PMCID: PMC6788750.
12. **Westover MB** , Bianchi MT, Eckman MH, Greenberg SM. "Statin Use Following Intracerebral Hemorrhage: A Decision Analysis." *Archives of Neurology*. 2011;68(5):573-579. PMCID: PMC3158138.
    - ***Featured article with accompanying editorial:*** *Arch Neurol. 2011;68(5):565-566. "Statins After Intracerebral Hemorrhage: To Treat or Not to Treat".*
    - ***Covered in over 150 news feeds****, including CBS News, ABC News, US News and World Report, and Medscape Medical News.*
13. **Westover MB**, Westover KD, Bianchi MT. "Significance Testing as Perverse Probabilistic Reasoning." *BMC Medicine*, 9(20). Published Online Feb. 28, 2011. PMCID: PMC3058025.
14. Westover KD, **Westover MB**, Winer EP, Richardson AL, Iglehart JD, Punglia RS. "Should a sentinel node biopsy be performed in patients with high risk breast cancer?" *International Journal of Breast Cancer*. 2011;2011:973245. doi: 10.4061/2011/973245. PMCID: PMC3262582.
15. Eiseman NA, **Westover MB**, Mietus JE, Thomas RJ, Bianchi MT. "Classification algorithms for predicting sleepiness and sleep apnea severity." *Journal of Sleep Research.* 2012 Feb;21(1):101-12. PMCID: PMC3698244.
16. Pascual-Leone A, Freitas C, Oberman L, Horvath JC, Halko M, Eldaief M, Bashir S, Vernet M, Shafi M, **Westover MB**, Vahabzadeh-Hagh AM, Rotenberg A. "Characterizing Brain Cortical Plasticity and Network Dynamics Across the Age-Span in Health and Disease with TMS-EEG and TMS-fMRI." *Brain Topography*, 2011 Oct;24(3-4):302-15. PMCID: PMC3374641.
17. Thakur K, **Westover MB**, "Cerebral Infarction Due to Smoker's Polycythemia." *BMJ Case Reports* 2011; doi:10.1136/bcr.08.2011.4714. PMCID: PMC3207785.
18. **Westover MB**, Cormier J, Bianchi MT, Shafi M, Kilbride R, Cole AJ, Cash SS. "Revising the rule of three for inferring seizure freedom." *Epilepsia*. 2012 Feb;53(2):368-76. PMCID: PMC3267849.
19. **Westover MB**, Bianchi MT. "Should risky treatments be reserved for secondary prevention? A decision theory approach to risk-benefit balance." *Journal of Clinical Epidemiology,* Aug 2012; 65(8):877-86. PMCID: PMC6794097.
20. Chu-Shore C, Kramer MA, Pathmanathan J, Bianchi MT, **Westover MB**, Wizon L, Cash SS, "Emergence of stable functional networks in long-term human EEG". *The Journal of Neuroscience*, Feb 2012; 32(8): 2703-2713. PMCID: PMC3361717.
21. Shafi MM, **Westover MB**, Fox MD, Pascual-Leone A, "Exploration and modulation of brain network interactions with noninvasive brain stimulation in combination with neuroimaging." *European Journal of Neuroscience*, 2012 Mar;35(6):805-25. PMCID: PMC3313459.
22. Rakthanmanon T, Campana B, Mueen A, Batista G, **Westover MB**, Zhu Q, Zakaria J, Keogh E, "Searching and Mining Trillions of Time Series Subsequences under Dynamic Time Warping". *Proceedings of ACM SIGKDD 2012*. PMCID: PMC6816304.
    - ***Best Paper Award at Association for Computing Machinery's annual SIGKDD (Special Interest Group on Knowledge Discovery and Data Mining) meeting, 2012.***
23. Eiseman NA, **Westover MB**, Ellenbogen JM, Bianchi MT, "The Impact of Body Posture and Sleep Stages on Sleep Apnea Severity in Adults". *Journal of Clinical Sleep Medicine*. 2012 Dec 15;8(6):655-66A. PMCID: PMC3501662.
24. Van Esbroeck A, **Westover MB**, "Data-Driven Modeling of Sleep States from EEG". *Proceedings of the International Conference of the IEEE Engineering in Medicine and Biology Society* (EMBC'12). 2012;2012:5090-3. PMCID: PMC6788787.
25. Shafi, MM*, **Westover MB*,** Cole AJ, Kilbride RD, Hoch DB, Cash SS. "Absence of Early Epileptiform Abnormalities Predicts Lack of Seizures on Continuous *EEG". Neurology. Oct 2012; 79:1796-801.* PMCID: PMC3475619. ***joint first authorship.***
26. *Srinivasan P,* ***Westover*** **MB**, Bianchi MT. "Propagation of Uncertainty in Bayesian Diagnostic Test Interpretation". *Southern Medical Journal.* 2012 Sep;105(9):452-9. PMCID: PMC6785978.

27. **Westover MB**, Eiseman NA, Cash SS, Bianchi MT. "An Information Theoretic Approach to Uncertainty in Diagnostic Test Interpretation." *The Open Medical Informatics Journal.* 2012 Dec; 6: 36–50. PMCID: PMC3537080.
28. **Westover MB,** Cohen AB. "Reversible Vasoconstriction Syndrome with Bilateral Basal Ganglia Hemorrhages". *Journal of Neuroimaging*. 2013 Jan; 23(1): 122-125. PMCID: PMC6788746.
29. **Westover MB**, Bianchi MT, Shafi MM, Hoch DB, Cole AJ Chiappa K, Cash SS, "Inferring Seizure Frequency from Brief EEG Recordings". *Journal of Clinical Neurophysiology.* Apr 2013;30:174-7. PMCID: PMC3616271.
30. Ferguson M, Bianchi MT, Sutter R, Rosenthal ES, Cash SS, Kaplan PW, **Westover MB**, "Calculating the Risk Benefit Equation for Aggressive Treatment of Nonconvulsive Status Epilepticus." *Neurocritical Care.* 2013 Apr;18(2):216-27. PMCID: PMC3767472.
31. Rakthanmanon T, Campana B, Mueen A, Batista G, **Westover MB**, Zhu Q, Zakaria J, Keogh E. "Addressing Big Data Time Series: Mining Trillions of Time Series Subsequences under Dynamic Time Warping." *ACM Transactions on Knowledge Discovery from Data (TKDD)* 2013 Feb; 7.3, 1. PMCID: PMC6790126.
32. **Westover MB**, Ching S, Shafi MM, Cash SS, Brown EN "Real-time segmentation and tracking of brain metabolic state in ICU EEG recordings of burst suppression." Conf Proc IEEE Eng. Med Biol Soc. 2013;2013:7108-11. PMCID: PMC3939432.
33. Ching S*, Liberman MY*, Chemali JJ*, **Westover MB***, Kenny J, Purdon PL, Solt K, Brown EN. "Real-Time Closed Loop Control in a Rodent Model of Medically-Induced Coma." *Anesthesiology*. 2013 Oct;119(4):848-60. ***joint first authorship.** PMCID: PMC3857134.
34. **Westover MB**, Bianchi MT, Yang C, Schneider JA, Greenberg SM. "Eestimating Total Cerebral Microinfarct Burden From Autopsy Samples". *Neurology*. 2013 Apr 9;80(15):1365-9. PMCID: PMC3662273.
    - ***Featured article with accompanying editorial**: Neurology. 2013 Apr 9;80(15):1358-9, "Cerebral Microinfarcts: Enumerating the Innumerable".*
    - ***Recognized as work of special significance by Faculty of 1000**: Baird A and Adamski M: F1000Prime Recommendation of [Westover MB et al., Neurology 2013, 80(15):1365-9]. In F1000Prime, 30 Apr 2013; DOI: 10.3410/f.718000104.793474863. F1000Prime.com/718000104#eval793474863.*
35. **Westover MB***, Shafi MM*, Ching S*, Chemali JJ, Purdon PL, Cash SS, Brown EN. "Real-Time Segmentation of Burst Suppression Patterns in Critical Care EEG Monitoring." *Journal of Neuroscience Methods* 2013 Aug 8. (219):131-141. PMCID: PMC3939433.
36. Ng MC, **Westover MB**, Cole AJ. "Treating Seizures in Creutzfeldt-Jakob Disease." *Epilepsy and Behavior Case Reports* 2014 Jan; 2:75-79. PMCID: PMC4308028.
37. Williamson CA, Wahlster S, Shafi MM, **Westover MB.** "Sensitivity of Compressed Spectral Arrays for Detecting Seizures in Acutely Ill Adults". *Neurocritical Care*. 2014 Feb; 20(1):32-9. PMCID PMC6794096.
38. Eiseman, A, Bianchi MT, **Westover MB.** "The Information Theoretic Perspective on Medical Diagnostic Inference." *Hospital Practice.* 2014 Apr;42(2):125-38. PMCID PMC6993929.
39. Bai Z, Huang GB, Wang D, Wang H, **Westover MB.** "Sparse Extreme Learning Machine for Classification." *IEEE Transactions on Cybernetics*, "Sparse Extreme Learning Machine for Classification," *Cybernetics, IEEE Transactions on* , vol.PP, no.99, pp.1,1 doi: 10.1109/TCYB.2014.2298235. PMCID: PMC4883115.
40. Shafi MM*, **Westover MB***, Oberman L, Cash SS, Pascual-Leone A. "Modulation of EEG functional connectivity networks in subjects undergoing repetitive transcranial magnetic stimulation." *Brain Topography.* 2014 Jan;27(1):172-91. doi: 10.1007/s10548-013-0277-y. Epub 2013 Mar 8. PMCID: PMC4026010. ***joint first authorship.**
41. Kheder A, Bianchi MT, **Westover MB.** "Burst Suppression in Sleep in a Routine Outpatient EEG." *Epilepsy and Behavior Case Reports* 2014 Jan; 2:71-74. PMCID: PMC4308090.
42. Bianchi MT, **Westover MB.** "Insomnia and morning motor vehicle accidents: A decision analysis of the risk of hypnotics versus the risk of untreated insomnia." *Journal of Clinical Psychopharmacology*. 2014

Jun;34(3):400-402. PMCID: PMC6794095.

43. Moura LMVR, Shafi MM, Ng M, Pati S, Cash SS, Cole AJ, Hoch DB, Rosenthal ES, **Westover MB**. "Spectrogram Screening of Adult Electroencephalograms is Sensitive and Efficient." *Neurology*. 2014 Jul 1;83(1):56-64. PMCID: PMC4114174.
    - ***Reviewed in Epilepsy Currents, official journal of the American Epilepsy Society:*** *Epilepsy Currents. 2015 Jan-Feb;15(1):24-5. doi: 10.5698/1535-7597-15.1.24. PMCID: PMC4320952.*
44. Berkowitz AL, **Westover MB**, Bianchi MT, Chou SHY. "Aspirin for Acute Stroke of Unknown Etiology in Resource-Limited Settings: A Decision Analysis." *Neurology.* 2014; 83:1–7. PMCID: PMC4155044.
    - ***Featured article with accompanying editorial****: Neurology 2014; 83:1–2. "Expanding antiplatelet use for patients with stroke: The case for stroke of unknown type*."
45. Gaspard N, Hirsch LJ, LaRoche SM, Hahn CD, **Westover MB.** "Inter-rater agreement for Critical Care EEG Terminology." *Epilepsia*. 2014 Sep;55(9):1366-73. PMCID: PMC4879939.
46. Akeju O, Pavone KJ, **Westover MB**, Vazquez R, Harrell PG, Hartnack KE, Rhee J, Sampson AL, Habeeb K, Lei G, Pierce ET, Walsh JL, Brown EN, Purdon PL. "Comparison of Propofol- and Dexmedetomidine-induced Electroencephalogram Dynamics Using Spectral and Coherence Analysis." *Anesthesiology 2014 Nov;121(5):978-89.* PMCID: PMC4304638.
47. Bianchi MT, Lipoma T, Darling C, Alameddine Y, **Westover MB**. "Automated Sleep Apnea Quantification Based on Respiratory Movement." International Journal of Medical Sciences 2014; 11(8):796-802. PMCID: PMC4057486.
48. Jin Jing, Dauwels J, Cash SS, **Westover MB**, "SpikeGUI: Software for Rapid Interictal Discharge Annotation Via Template Matching and Online Machine Learning." *Proceedings of the International Conference of the IEEE Engineering in Medicine and Biology Society* (EMBC), 2014: 4435-4438. PMCID: PMC4416962.
49. Akeju O, **Westover MB**, Pavone KJ, Sampson A, Hartnack K, Brown EN, Purdon PL. "Effects of sevoflurane and propofol on frontal electroencephalogram power and coherence." *Anesthesiology*. 2014 Nov;121(5):990-8. PMCID: PMC4206606.
50. Akama-Garren EH, Bianchi MT, Leveroni C, Cole AJ, Cash SS, **Westover MB**. "Weighing the value of memory loss in the surgical evaluation of left temporal lobe epilepsy: A decision analysis." *Epilepsia.* 2014 Nov;55(11):1844-53. PMCID: PMC4877127.
51. Bianchi MT, Hershman S, Bahadoran M, Ferguson M, **Westover MB**. "The Challenge of Undiagnosed Sleep Apnea in Low-Risk Populations: A Decision Analysis." *Military Medicine* 2014 Aug;179(8S):47-54. PMCID: PMC6788752.
52. Weng Y, **Westover MB**, Speier C, Sharp G, Bianchi MT, Westover KD. "Applications of a Capacitor-Based Respiratory Position Sensing Device: Implications for Radiation Therapy." Austin Journal of Medical Oncology 2014;1(2). PMCID: PMC6956860.
53. **Westover MB***, Shafi MM*, Bianchi MT, Moura LMVR, O'Rourke D, Rosenthal ES, Chu CJ, Donovan S, Hoch DB, Kilbride RD, Cole AJ, Cash SS. "The Probability of Seizures During EEG Monitoring in Critically Ill Adults." *Clinical Neurophysiology*. 2014 Jul 11. pii: S1388-2457(14)00362-9. doi: 10.1016/j.clinph.2014.05.037. [Epub ahead of print]. PMCID: PMC4289643. ***joint first authorship.***
    - ***Featured article with accompanying editorial****: "Early epileptiform discharges and the yield of prolonged EEG monitoring."* Dangayach N, Claassen J., *Clinical Neurophysiology* (2014), doi: http://dx.doi.org/10.1016/j.clinph.2014.07.002.
54. Berkowitz AL*, **Westover MB***, Bianchi MT, Chou SH. "Aspirin for secondary stroke prevention after stroke of unknown etiology in resource-limited settings: A decision analysis." *Neurology* 2014 Sep 9;83(11):1004-11. PMCID: PMC4162302. ***joint first authorship***
55. Alvarez V, **Westover MB**, Drislane FW, Dworetzky BA, Curley D, Lee JW, Rossetti AO. "Evaluation of a clinical tool for early etiology identification in status epilepticus." *Epilepsia*. 2014 Dec;55(12):2059-2068. PMCID: PMC4870016.
56. O'Connor KL, **Westover MB**, Shafi MM, Rosenthal ES. "High Risk for Seizures Following Subarachnoid Hemorrhage Regardless of Referral Bias." *Neurocritical Care.* 2014 Dec;21(3):476-82. PMCID: PMC4878846.

57. Kenny JD, **Westover MB**, Ching S, Brown EN, Solt K. "Propofol and sevoflurane induce distinct burst suppression patterns in rats." *Frontiers in Neuroscience* Front Syst Neurosci. 2014 Dec 18;8:237. doi: 10.3389/fnsys.2014.00237. eCollection 2014. PMCID: PMC4270179.
58. Ng MC, Gaspard N, Cole AJ, Hoch DB, Cash SS, Bianchi MT, O'Rourke DA, Rosenthal ES, Chu CJ, **Westover MB**. "The Standardization Debate: A Conflation Trap in Critical Care Electroencephalography." *Seizure: European Journal of Epilepsy* 2015 Jan;24C:52-58. PMCID: PMC4465375.
59. de Gusmão CM, Berkowitz AL, Hung AY, **Westover MB**. "Cerebrospinal fluid shunt-induced chorea: case report and review of the literature on shunt-related movement disorders." Practical Neurology. 2015 Feb;15(1):42-4. PMCID: PMC4870013.
60. Shafi MM, Vernet M, Klooster D, Chu CJ, Boric K, Barnard ME, Romatoski K, **Westover MB**, Christodoulou JA, Gabrieli JDE, Whitfield-Gabrieli S, Pascual-Leone A, Chang BS. "Physiological consequences of abnormal connectivity in a developmental epilepsy." *Annals of Neurology* 2015 Mar;77(3):487-503. PMCID: PMC4394240.
61. Dranias MR, **Westover MB**, Cash SS, VanDongen AMJ, "Stimulus information stored in lasting active and hidden network states is destroyed by network bursts." *Frontiers in Integrative Neuroscience* 2015 Feb 23;9:14. doi: 10.3389/fnint.2015.00014. eCollection 2015. PMCID: PMC4337383.
62. **Westover MB**, Kim SE, Ching S, Purdon PL, Brown EN, "Robust Control of Burst Suppression for Medical Coma." *Journal of Neural Engineering.* 2015 Aug;12(4):046004. doi: 10.1088/1741-2560/12/4/046004. Epub 2015 May 28. PMCID: PMC4517835.
63. Alvarez V, Drislane FW, **Westover MB**, Dworetzky BA, Lee JW. "Characteristics and role in outcome prediction of continuous EEG after status epilepticus: A prospective observational cohort." *Epilepsia.* 2015. Jun;56(6):933-41. PMCID: PMC4878827.
64. Alvarez V, Drislane FW, **Westover MB**, Dworetzky BA, Lee JW. "Practice variability and efficacy of Clonazepam, Lorazepam and Midazolam in Status Epilepticus: A Multicenter Comparison." *Epilepsia 2015 Aug;56(8):1275-85.* PMCID: PMC4877129.
65. Auriel E, **Westover MB**, Bianchi MT, Reijmer Y, Martinez-Ramirez S, Ni J, Van Etten E, Frosch MP, Fotiadis P, Schwab K, Vashkevich A, Boulouis G, Younger AP, Johnson KA, Sperling RA, Hedden T, Gurol ME, Viswanathan A, Greenberg SM. "Estimating Total Cerebral Microinfarct Burden From Diffusion-Weighted Imaging." *Stroke*. 2015 Aug;46(8):2129-35. PMCID: PMC4519384.
66. Akeju O, Pavone KJ, Thum JA, Firth PG, **Westover MB**, Puglia M, Shank ES, Brown EN, Purdon PL. Age-dependency of sevoflurane-induced electroencephalogram dynamics in children. Br J Anaesth. 2015 Jul;115 Suppl 1(Suppl 1):i66-i76. doi: 10.1093/bja/aev114. PMID: 26174303; PMCID: PMC4501917.
67. Biswal B, Nip Z, Moura JV, Bianchi MT, Rosenthal ES, **Westover MB**. "Automated Information Extraction from Free-Text EEG Reports." *Conf Proc IEEE Eng Med Biol Soc.* 2015 Aug;2015:6804-7. doi: 10.1109/EMBC.2015.7319956. PMCID: PMC4872711.
68. Zhou DW*, **Westover MB***, McClain LM, Nagaraj SB, Bajwa EK, Quraishi SA, Akeju O, Cobb JP, Purdon PL. "Clustering analysis to identify distinct spectral components of encephalogram burst suppression in critically ill patients." ***joint first authorship.*** *Conf Proc IEEE Eng Med Biol Soc.* 2015 Aug;2015:7258-61. doi: 10.1109/EMBC.2015.7320067. PMCID: PMC4870011.
69. An JZ, Jonnalagadda D, Junior VM, Purdon PL, Brown EN, **Westover MB**. "Spatial Variation in Automated Burst Suppression Detection in Pharmacologically Induced Coma." *Conf Proc IEEE Eng Med Biol Soc.* 2015 Aug;2015:7430-3. PMCID: PMC4876722.
70. Ghassemi M, Amorim E, Pati A, Mark R, Brown EN, Purdon PL, **Westover MB**. "An Enhanced Cerebral Recovery Index for Coma Prognostication Following Cardiac Arrest." *Conf Proc IEEE Eng Med Biol Soc.* 2015 Aug;2015:534-7. doi: 10.1109/EMBC.2015.7318417. PMCID: PMC4870018.
71. Struck A, **Westover MB.** "Variability in Clinical Assessment of Neuroimaging in Temporal Lobe Epilepsy." *Seizure: European Journal of Epilepsy.* 2015 Aug;30:132-5. doi: 10.1016/ j.seizure.2015.06.011. Epub 2015 Jul 2. PMCID: PMC4887849.
72. **Westover MB**, Ching S, Kumaraswamy VM, Akeju O, Pierce E, Cash SS, Kilbride R, Brown EN, Purdon PL. "The Human Burst Suppression Electroencephalogram of Deep Hypothermia." *Clinical*

Neurophysiology. 2015 Oct;126(10):1901-1914. PMCID: PMC4504839.

73. Gaspard N, Foreman BP, Alvarez V, Cabrera Kang CM, Probasco JC, Jongeling AC, Meyers E, Espinera A, Haas KF, Schmitt SE, Gerard EE, Gofton T, Kaplan PW, Lee JW, Legros B, Szaflarski JP, **Westover MB**, LaRoche SM, Hirsch LJ, "New-onset refractory status epilepticus: etiology, clinical features and outcome." *Neurology. 2015 Nov 3;85(18):1604-13.* PMCID: PMC4642147.
74. Struck A, Cash SS, Cole AJ, **Westover MB**. "The Number of Seizures needed in the EMU." *Epilepsia.* 2015 Nov;56(11):1753-9. PMCID: PMC4877132.
75. Hermans MC, **Westover MB**, van Putten MJAM, Hirsch LJ, Gaspard N. "Quantification of EEG reactivity in comatose patients." *Clinical Neurophysiology.* 2016 Jan;127(1):571-580. PMCID: PMC4885124.
76. Ustun B, **Westover MB**, Rudin C, Bianchi MT. "Clinical Prediction Models for Sleep Apnea: The Importance of Medical History over Symptoms." *Journal of Clinical Sleep Medicine*. 2016 Feb;12(2):161-8. doi: 10.5664/jcsm.5476. PMCID: PMC4751423.
77. Moro M, **Westover MB,** Kelly J, Bianchi MT. "Decision modeling in sleep apnea: the critical roles of pre-test probability, cost of untreated OSA, and time horizon." *Journal of Clinical Sleep Medicine.* 2016 Mar;12(3):409-418. PMCID: PMC4773629.
78. Bagheri E, Jing J, Dauwels J, Cash SS, **Westover MB.** "Fast and Efficient Rejection of Background Waveforms in Interictal EEG." *Proceedings of The IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2016),* 2016 Mar, p744-748. PMCID: PMC5835012.
79. Johansen AR, Jing J, Maszczyk T, Dauwels J, Cash SS, **Westover MB**. "Epileptiform Spike Detection Via Convolutional Neural Networks." *Proceedings of The IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2016),* 2016 Mar, p744-748. PMCID: PMC5842703.
80. Thomas J, Jing J, Dauwels J, Cash SS, **Westover MB**. "Clustering of Interictal Spikes by Dynamic Time Warping and Affinity Propagation." *Proceedings of The IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2016),* 2016 Mar, p744-748. PMCID: PMC5842698.
81. Fürbass F, Herta J, Koren J, **Westover MB**, Gruber A, Baumgartner C, Kluge T. "Monitoring burst suppression in critically ill patients: Multi-centric evaluation of a novel method Clinical Neurophysiology." *Clinical Neurophysiology.* 2016 Apr;127(4):2038-46. PMCID: PMC4879619.
82. Jing J, Dauwels J, Rakthanmanon T, Keogh E, Cash SS, **Westover MB**. "Rapid Annotation of Interictal Epileptiform Discharges via Template Matching under Dynamic Time Warping." J Neurosci Methods. 2016 Dec 1;274:179-190. PMCID: PMC5519352.
83. Lee JW, LaRoche S, Choi H, Rodriguez Ruiz AA, Fertig E, Politsky J, Herman S, Loddenkemper T, Sansevere A, Korb PJ, Abend N, Goldstein JL, Sinha SR, Dombrowski KE, Ritzl EK, **Westover MB**, Gavvala J, Schmitt S, Szaflarski JP, Ding K, Haas KF, Buchsbaum R, Hirsch LJ, Wusthoff CJ, Hopp JL, Hahn CD. "Development and validation of a critical care EEG monitoring database for standardized clinical reporting and multicenter collaborative research." *Journal of Clinical Neurophysiology*. 2016 Apr;33(2):133-40. PMCID: PMC4878836.
84. O'Rourke D, Chen PM, Nicolas G, Foreman B, McClain L, Karakis I, Mahulikar A, **Westover MB.** "Response Rates to Anticonvulsant Trials in Patients with Triphasic-Wave EEG Patterns of Uncertain Significance." *Neurocritical Care.* 2016 Apr;24(2):233-239. PMCID: PMC4870012
85. Struck AF, **Westover MB,** Deck GM, Cole AJ, Rosenthal ES. "Metabolic Correlates of the Ictal-Interictal Continuum: FDG-PET during Continuous EEG." *Neurocritical Care* 2016 Jun;24(3):324-31. PMCID: PMC5478419.
86. Arif Haider H, Esteller R, Hahn CD, **Westover MB**, Halford JJ, Lee JW, Shafi MM, Gaspard N, Herman ST, Gerard EE, Hirsch LJ, Ehrenberg JA, LaRoche SM. "Sensitivity of Quantitative EEG for Seizure Identification in the Intensive Care Unit." *Neurology.* 2016 Aug 30;87(9):935-44. PMCID: PMC5035158.
87. Nagaraj SB, Ramaswamy SM, Biswal S, Boyle EJ, Zhou DW, McClain LM, Rosenthal ES, Purdon PL, **Westover MB**. "Heart rate variability as a biomarker for sedation depth estimation in ICU patients." *Conf Proc IEEE Eng Med Biol Soc*. 2016 Aug;2016:6397-6400. PMCID: PMC5478422.
88. Zafar SF, **Westover MB**, Gaspard N, Gilmore EJ, Foreman BP, O'Connor KL, Rosenthal ES. "Interrater Agreement for Consensus Definitions of Delayed Ischemic Events After Aneurysmal Subarachnoid Hemorrhage." J Clin Neurophysiol. 2016 Jun;33(3):235-40. PMCID: PMC4894325.

89. Wickering E, Gaspard N, Zafar S, Moura VJ, Biswal S, Bechek S, O'Connor K, Rosenthal ES, **Westover MB.** "Automation of Classical QEEG Trending Methods for Early Detection of Delayed Cerebral Ischemia: More Work to Do." J Clin Neurophysiol. 2016 Jun;33(3):227-34. PMCID: PMC4894333.
90. Muniz CF, Shenoy AV, O'Connor KL, Bechek SC, Boyle EJ, Guanci MM, Tehan TM, Zafar SF, Cole AJ, Patel AB, **Westover MB**, Rosenthal ES. J Clin Neurophysiol. "Clinical Development and Implementation of an Institutional Guideline for Prospective EEG Monitoring and Reporting of Delayed Cerebral Ischemia." 2016 Jun;33(3):217-26. PMCID: PMC4907266.
91. Hotan GC, Struck AF, Bianchi MT, Eskandar EN, Cole AJ, **Westover MB**. "Decision analysis of intracranial monitoring in non-lesional epilepsy." Seizure. 2016 Aug;40:59-70. doi: 10.1016/j.seizure.2016.06.010. Epub 2016 Jun 18. PMCID: PMC4967015.
92. Nagaraj SB, Mcclain LM, Zhou DW, Biswal S, Rosenthal ES, Purdon PL, **Westover MB**. "Automatic Classification of Sedation Levels in ICU Patients using Heart Rate Variability." Critical Care Medicine. 2016 Sep;44(9):e782-9. PMCID: PMC4987179.
93. Alvarez V, Lee JW, **Westover MB**, Drislane FW, Novy J, Faouzi M, Marchi NA, Dworetzky BA, Rossetti AO. "Therapeutic coma for status epilepticus: differing practices in a prospective multicenter study." Neurology. 2016 Oct 18;87(16):1650-1659. Epub 2016 Sep 24. PMCID: PMC5085074.
94. Nelson S, Edlow BL, Wu O, Rosenthal ES, **Westover MB**, Rordorf G. "Default Mode Network Perfusion in Aneurysmal Subarachnoid Hemorrhage." *Neurocritical care.* 2016 October;25(2):237-42. PMCID: PMC4958026.
95. Hartings JA, Shuttleworth CW, Kirov SA, Ayata C, Hinzman JM, Foreman B, Andrew RD, Boutelle MG, Brennan KC, Carlson AP, Dahlem MA, Drenckhahn C, Dohmen C, Fabricius M, Farkas E, Feuerstein D, Graf R, Helbok R, Lauritzen M, Major S, Oliveira-Ferreira AI, Richter F, Rosenthal ES, Sakowitz OW, Sánchez-Porras R, Santos E, Schöll M, Strong AJ, Urbach A, **Westover MB**, Winkler MK, Witte OW, Woitzik J, Dreier JP. The continuum of spreading depolarizations in acute cortical lesion development: Examining Leão's legacy. J Cereb Blood Flow Metab. 2017 May;37(5):1571-1594. doi: 10.1177/0271678X16654495. Epub 2016 Jan 01. PMID: 27328690; PMCID: PMC5435288.
96. Amorim E, Rittenberger JC, Zheng JJ, **Westover MB**, Baldwin ME, Callaway CW, Popescu A. "Continuous EEG monitoring enhances multimodal outcome prediction in hypoxic-ischemic brain injury." Resuscitation. 2016 Dec;109:121-126. PMCID: PMC5124407.
97. Kim JA, Rosenthal ES, Biswal S, Zafar S, Shenoy AV, O'Connor KL, Bechek SC, Valdery Moura J, Shafi MM, Patel AB, Cash SS, **Westover MB**. "Epileptiform abnormalities predict delayed cerebral ischemia in subarachnoid hemorrhage." Clin Neurophysiol. 2017 Jun;128(6):1091-1099. doi: 10.1016/j.clinph.2017.01.016. Epub 2017 Jan 29. PMCID: PMC5476529.
98. Sinha N, Dauwels J, Kaiser M, Cash SS, **Westover MB**, Wang Y, Taylor PN. "Predicting neurosurgical outcomes in focal epilepsy patients using computational modelling." Brain. 2017 Feb;140(Pt 2):319-332. PMCID: PMC5278304.
99. Moura LMVR, Carneiro T, Scott C, Moura VJ, Cohen J, Guanci M, Hoch DB, Cole AJ, Hsu H, **Westover MB**. "cEEG-electrode-related pressure ulcers in acutely hospitalized patients." Neurology, Clinical Practice, 2017 Feb;7(1):15-25. PMCID: PMC5310208.
100. Bargiela D, Bianchi MT, **Westover MB**, Chibnik LB, Healy BC, De Jager PL, Xia Z. "Selection of first-line therapy in multiple sclerosis using risk-benefit decision analysis." Neurology. 2017 Feb 14;88(7):677-684. PMCID: PMC5317380.
101. Kempfle J, **Westover MB**, Bianchi MT. "A cost-effectiveness analysis of nasal surgery to increase CPAP compliance in sleep apnea patients with nasal obstruction." *The Laryngoscope.* 2017 Apr;127(4):977-983. PMCID: PMC5483184.
102. Dreier J, Fabricius M, Cenk A, Sakowitz O, Shuttleworth C, Dohmen C, Graf R, Vajkoczy P, Helbok R, Suzuki M; Schiefecker A,; Major S, Winkler M, Kang EJ, Milakara D, Oliveira-Ferreira A, Reiffurth C, Revankar G, Sugimoto K, Dengler N, Hecht N, Foreman B, Feyen B, Kondziella D, Friberg C. Piilgaard H, Rosenthal ES, **Westover MB**, Maslarova A, Santos E, Hertle D, Sanchez-Porras R, Jewell S, Balança B, Platz J, Hinzman J, Luckl J, Schoknecht K, Schöll M, Drenckhahn C, Feuerstein D, Eriksen N, Horst V, Bretz J, Jahnke P, Scheel M, Bohner G, Rostrup E, Pakkenberg B, Heinemann U, Claassen J, Carlson A,

Kowoll C, Lublinsky S, Chassidim Y, Shelef I, Friedman A, Brinker G, Reiner M, Kirov S, Andrew R, Farkas E, Gueresir, E, Vatter H, Chung L, Brennan K Lieutaud T, Marinesco S, Maas A, Sahuquillo J, Dahlem M, Richter F, Herreras O, Boutelle M, Okonkwo D, Bullock R, Witte OW, Martus P, Van den Maagdenberg A, Ferrari M, Dijkhuizen R, Shutter L, Andaluz N, Schulte A, MacVicar B, Watanabe T, Woitzik J, Lauritzen M, Strong A, Hartings J. "Recording, analysis, and interpretation of spreading depolarizations in neurointensive care: review and recommendations of the COSBID research group." Journal of Cerebral Blood Flow and Metabolism. 2017 May;37(5):1595-1625. PMCID: PMC5435289.

103. Amorim E, Williamson C, Moura LMVRD, Shafi MM, Gaspard N, Rosenthal ES, Guanci M, Venkata K, **Westover MB**. "Performance of spectrogram-based seizure identification of adult EEGs by critical care nurses and neurophysiologists." J Clin Neurophysiol. 2017 Jul;34(4):359-364. PMCID: PMC5482787.

- ***Cosimo Ajmone-Marsan Award*** **for best paper of 2017 in the Journal of Clinical Neurophysiology.**

104. Moura LMVRD, **Westover MB**, Kwasnik D, Cole AJ, Hsu J. "Causal Inference as an Emerging Statistical Approach in Neurology: An Example for Epilepsy in the Elderly." Clinical Epidemiology. 2016 Dec 30;9:9-18. PMCID: PMC5221551.

105. Bianchi MT, Russo K, Gabbidon H, Smith T, Goparaju B, **Westover MB**. Big data in sleep medicine: prospects and pitfalls in phenotyping. Nat Sci Sleep. 2017;9:11-29. doi: 10.2147/NSS.S130141. Epub 2017 Feb 16. PMID: 28243157; PMCID: PMC5317347.

106. Kim JA, Rosenthal ES, Biswal S, Zafar S, Shenoy AV, O'Connor KL, Bechek SC, Valdery Moura J, Shafi MM, Patel AB, Cash SS, **Westover MB**. "Epileptiform Abnormalities Predict Delayed Cerebral Ischemia in Subarachnoid Hemorrhage." *Clinical Neurophysiology* 2017 Jun;128(6):1091-1099. PMCID: PMC5476529

107. An JZ, Solt K, Sims N, Brown EN, Purdon PL, **Westover MB**, "Towards Clinical Translation of a Physiological Closed-loop Control Device for Medical Coma" *Proceedings of the International Conference of the IEEE Engineering in Medicine and Biology Society* (EMBC'17), 2017 Jul;2017:4313-4316. PMCID: PMC5835010.

108. Nagaraj SB, Biswal S, Boyle EJ, Zhou DW, McClain LM, Bajwa EK, Quraishi SA, Oluwaseun A, Barbieri R, Purdon PL, **Westover MB**. "Patient-Specific Classification of ICU Sedation Levels from Heart Rate Variability." Critical Care Medicine, 2017 Jul;45(7):e683-e690. PMCID: PMC5474145.

109. Thomas J, Cash SS, Dauwels J, Jin J, **Westover MB**, "Automated Epileptiform Spike Detection via Affinity Propagation-Based Template Matching" Conf *Proc IEEE Eng Med Biol Soc*. 2017 Jul;2017:4313-4316. doi: 10.1109/EMBC.2017.8037810. PMCID: PMC5835014.

110. Struck AF, Osman G, Rampal N, Biswal S, Legros B, Hirsch LJ, **Westover MB**, Gaspard N, "Time-Dependent Risk of Seizures in Critically Ill Patients on Continuous EEG." Annals of Neurology. 2017 Aug;82(2):177-185. doi: 10.1002/ana.24985. Epub 2017 Jul 19. PMCID: PMC5842678.

111. Bagheri E, Dauwels J, Dean BC, Waters CG, **Westover MB**, Halford JJ, "Interictal epileptiform discharge characteristics underlying expert interrater agreement." Clin Neurophysiol. 2017 July 18; 128(10):1994-2005. doi: 10.1016/j.clinph.2017.06.252. PMCID: PMC5842710.

112. Pati BB, McClain L, Moura LMVRD, Fan Y, **Westover MB**, "Accuracy of Limited-Montage Electroencephalography in Monitoring Postanoxic Comatose Patients." Pati S, McClain L, Moura L, Fan Y, Westover MB. Clinical EEG and Neuroscience. 2017 Nov;48(6):422-427. PMCID: PMC5835011.

113. Romero K, Goparaju B, Russo K, **Westover MB,** Bianchi MT. "Alternative remedies for insomnia: a proposed method for personalized therapeutic trials". *Nature and Science of Sleep.* 2017;9:97-108. PMCID: PMC5364017.

114. Sun H, Jian J, Goparaju B, Huang GB, Sourina O, Bianchi MT, **Westover MB**, "Large-Scale Automated Sleep Staging." *Sleep*. 2017 Oct 1;40(10). doi: 10.1093/sleep/zsx139. PMCID: PMC6251659.

115. Struck AF, Ustun B, Ruiz AR, Lee JW, LaRoche S, Hirsch LJ, Gilmore EJ, Vlachy J, Haider HA, Rudin C, **Westover MB.** "Association of an Electroencephalography-Based Risk Score with Seizure Probability in Hospitalized Patients." *JAMA Neurology*. 2017 Dec 1;74(12):1419-1424. PMCID: PMC5822188.

116. An JZ, Purdon PL, Solt K, Sims NM, Brown EN, **Westover MB**. "Design, implementation, and evaluation of a physiological closed-loop control device for medically-induced coma." *Conf Proc IEEE*

Eng Med Biol Soc. 2017 Jul;2017:4313-4316. doi: 0.1109/EMBC.2017.8037810. PMCID: PMC5835010.
117. Rebergen DJ, Nagaraj SB, Rosenthal ES, Bianchi MT, van Putten MJAM, **Westover MB**. "ADARRI: A Novel Method to Detect Spurious R-peaks in the Electrocardiogram for Heart Rate Variability Analysis in the Intensive Care Unit." *Journal of Clinical Monitoring and Computing*, 2018 Feb;32(1):53-61. PMCID: PMC5559344.
118. Bagheri E, Jing J, Dauwels J, Cash SS, **Westover MB.** "Classifier Cascade to Aid in Detection of Epileptiform Transients in Interictal EEG." *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP).* April 15, 2018: 970-974. PMCID: PMC6775762.
119. Shelton KT, Qu J, Brown EN, Burns S, Cudemus G, Deng H, Demircioglu G, Dibiaso A, Hobbs LE, Houle TT, Ibala R, Loggia M, Pavone KJ, Shaefi S, **Westover MB**, Akeju O. "Minimizing ICU Neurological Dysfunction with Dexmedetomidine-induced Sleep (MINDDS): protocol for a randomized, double blind, parallel-arm, placebo-controlled *trial." British Medical Journal Open (BMJ Open)* 2018 Apr 20;8(4):e020316. PMCID: PMC5914725.
120. Kim JA, Boyle E, Wu AC, Cole AJ, Staley KJ, Zafar S, Cash SS, **Westover MB**. "Epileptiform activity in traumatic brain injury predicts post-traumatic epilepsy." *Annals of Neurology* 2018 Apr;83(4):858-862. PMCID: PMC5912971.
121. Zafar SF, Postma EN, Biswal S, Fleuren L, Boyle EJ, Bechek S, O'Connor K, Shenoy A, Jonnalagadda D, Kim J, Shafi MM, Patel AB, Rosenthal ES, **Westover MB**. "Electronic Health Data Predict Outcomes after Aneurysmal Subarachnoid Hemorrhage." Neurocritical Care. 2018 Apr;28(2):184-193 . PMCID: PMC5886829.
122. Rosenthal ES, Biswal S, Zafar SF, O'Connor KL, Bechek S, Shenoy AV, Boyle EJ, Shafi MM, Gilmore EJ, Foreman BP, Gaspard N, Leslie-Mazwi TM, Rosand J, Hoch DB, Ayata C, Cash SS, Cole AJ, Patel AB, **Westover MB**. "Continuous Electroencephalography Predicts Delayed Cerebral Ischemia after Subarachnoid Hemorrhage: A Prospective Study of Diagnostic Accuracy" *Annals of Neurology*. 2018 May;83(5):958-969. PMCID: PMC6021198.
123. Westover AM, Shapiro D, **Westover MB**, Bianchi MT. "The Rule of 100: a litmus test for informationless diagnostic tests." *Postgraduate Medical Journal.* 2018 Jun;94(1112):364-366 PMCID: PMC6771257.
124. Thomas J, Comoretto L, Jing J, Dauwels J, Cash SS, **Westover MB**. "EEG Classification via Convolutional Neural Network-Based Interictal Epileptiform Event Detection." Conf Proc IEEE Eng Med Biol Soc. 2018 Jul;2018:3148-3151. PMCID: PMC6775768.
125. Husain AM, Lee JW, Kolls BJ, Hirsch LJ, Halford JJ, Gupta PK, Minazad Y, Jones J, LaRoche SM, Herman ST, Swisher CB, Sinha SR, Palade A, Dombrowski KE, Gallentine WB, Hahn CD, Gerard EE, Bhapkar M, Lokhnygina Y, **Westover MB.** "Randomized trial of lacosamide vs fosphenytoin for nonconvulsive seizures." *Annals of Neurology*. 2018 Jun;83(6):1174-1185. PMCID: PMC3055953.
126. Thomas J, Comoretto L, Jing J, Dauwels J, Cash SS, **Westover MB**. "EEG Classification Via Convolutional Neural Network-Based Interictal Epileptiform Event Detection." *Proceedings of the International Conference of the IEEE Engineering in Medicine and Biology Society* (EMBC'18). 2018 Jul;2018:3148-3151. PMCID: PMC6775768.
127. Sun H, Nagaraj SB, Akeju O, Purdon PL, **Westover MB***. "Brain Monitoring of Sedation in the Intensive Care Unit Using a Recurrent Neural Network." *Proceedings of the International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC'18)*. 2018 Jul;2018:1-4. PMCID: PMC6775766.
128. Herlopian A, Dietrich J, Abramson JS, Cole AJ, **Westover MB.** "EEG findings in CAR T-cell therapy-related encephalopathy." Neurology. 2018 Jul 31;91(5):227-229. PMCID: PMC6093761
129. Song J, Sun H, Jing J, Carlos L, Chao L, Cash SS, Zhang R, **Westover MB**. "A Mean Field Model of Acute Hepatic Encephalopathy." *Conf Proc IEEE Eng Med Biol Soc*. 2018 Jul;2018:2366-2369. doi: 10.1109/EMBC.2018.8512786. PMCID: PMC7088433.
130. Jing J, drAngremont E, Zafar S, Rosenthal ES, Tabaeizadeh M, Ebrahim S, Dauwels J, **Westover MB**. "Rapid Annotation of Seizures and Interictal-ictal Continuum EEG Patterns." *Conf Proc IEEE Eng Med Biol Soc.* 2018 Jul;2018:3394-3397. PMCID: PMC6776236.

131. Herlopian A, Dietrich J, Abramson JS, Cole AJ, **Westover MB.** "EEG findings in CAR T-cell therapy-related encephalopathy." *Neurology*. 2018 Jul 31;91(5):227-229. doi: 10.1212/WNL.0000000000005910. PMCID: PMC6093761.
132. Shenoy ES, Rosenthal ES, Shao YP, Biswal S, Ghanta M, Ryan EE, Suslak D, Swanson N, Junior VM, Hooper DC, **Westover MB**. "Real-Time, Automated Detection of Ventilator Associated Events: Avoiding Missed Detections, Misclassifications, and False Detections due to Human Error." *Infection Control & Hospital Epidemiology.* 2018 May 17:1-8. doi: 10.1017/ice.2018.97. PMCID: PMC6776240.
133. Halford JJ, **Westover MB**, LaRoche SM, Macken MP, Kutluay E, Edwards JC, Bonilha L, Kalamangalam GP, Ding K, Hopp JL, Arain A, Dawson RA, Martz GU, Wolf BJ, Waters CG, Dean BC. "Interictal Epileptiform Discharge Detection in EEG in Different Practice Settings." *Journal of Clinical Neurophysiology*. 2018 Sep;35(5):375-380. PMCID: PMC6126936.
134. Ghassemi MM, Moody BE, Lehman LH, Song C, Li Q, Sun H, Mark RG, **Westover MB**, Clifford GD. You Snooze, You Win: the PhysioNet/Computing in Cardiology Challenge 2018. Comput Cardiol (2010). 2018 Sep;45. doi: 10.22489/cinc.2018.049. Epub 2019 Jun 24. PMID: 34796237; PMCID: PMC8596964.
135. Zafar SF, Postma EN, Biswal S, Boyle EJ, Bechek S, O'Connor K, Shenoy A, Kim J, Shafi MS, Patel AB, Rosenthal ES, **Westover MB**. "Effect of Epileptiform Abnormality Burden on Neurologic Outcome and Anticonvulsant Drug Management After Subarachnoid Hemorrhage." *Clinical Neurophysiology*. 2018 Sep 1;129(11):2219-2227. PMCID: PMC6478499.
136. Bao EL, Chao LY, Ni P, Moura, LMVR, Cole AJ, Cash SS, Hoch DB, Bianchi MT, **Westover MB**. "Antiepileptic drug treatment after an unprovoked first seizure: A decision analysis." *Neurology*. 2018 Oct 9;91(15):e1429-e1439. PMCID: PMC6177278.
137. Amorim E, Gilmore EJ, Abend NS, Hahn CD, Gaspard N, Herman ST, Hirsch LJ, Lee JW, Cash SS, **Westover MB**. "EEG Reactivity Evaluation Practices for Adult and Pediatric Hypoxic-Ischemic Coma Prognostication in North America." Journal of Clinical Neurophysiology. 2018 Nov;35(6):510-514. doi: 10.1097/WNP.0000000000000517. PMCID: PMC6424574.
138. An JZ, Jonnalagadda D, Junior VM, Purdon PL, Brown EN, **Westover MB**. "Variability in Pharmacologically-induced Coma for Treatment of Refractory Status Epilepticus." *PLoS One*. 2018 Oct 31;13(10):e0205789. doi: 10.1371/journal.pone.0205789. PMCID: PMC6209214.
139. Osman G, Rahangdale R, Britton JW, Gilmore EJ, Haider HA, Hantus S, Herlopian A, Hocker SE, Woo Lee J, Legros B, Mendoza M, Punia V, Rampal N, Szaflarski JP, Wallace AD, **Westover MB,** Hirsch LJ, Gaspard N. "Bilateral independent periodic discharges are associated with electrographic seizures and poor outcome: A case-control study." *Clinical Neurophysiology*. 2018 Nov;129(11):2284-2289. PMCID: PMC6785981.
140. An S, Malhotra K, Dilley C, Han-Burgess E, Valdez JN, Robertson J, Clark C, **Westover MB**, Sun J. "Early Prediction of Drug Resistant Epilepsy using Claims Data". Epilepsy Behav. 2018 Nov 6;89:118-125. PMCID: PMC6461470.
141. Herlopian A, Struck AF, Rosenthal E, **Westover MB**. Neuroimaging Correlates of Periodic Discharges. J Clin Neurophysiol. 2018 Jul;35(4):279-294. doi: 10.1097/WNP.0000000000000466. PMID: 29979286; PMCID: PMC9371611.
142. Amorim E, Ghassemi MM, Lee JW, Greer DM, Kaplan PW, Cole AJ, Cash SS, Bianchi MT, **Westover MB**. "Estimating the false positive rate of absent somatosensory evoked potentials in cardiac arrest prognostication." *Critical Care Medicine* 2018 Dec; 46(12):e1213-e1221. PMCID: PMC6424571.
143. Biswal S, Kulas J, Sun H, Goparaju B, **Westover MB***, Bianchi MT*, Sun J*. " Expert-level Sleep Scoring with Deep Neural Networks". *Journal of the American Medical Informatics Association (JAMIA)*. 2018 Dec 1;25(12):1643-1650. PMCID: PMC6289549. ***joint senior authorship.***
144. Fadayomi AB, Ibala R, Bilotta F, **Westover MB**, Akeju O. "A Systematic Review and Meta-Analysis Examining the Impact of Sleep Disturbance on Postoperative Delirium." *Critical Care Medicine*. 2018 Dec;46(12):e1204-e1212. PMCID: PMC6274586.
145. Nagaraj SB, McClain LM, Boyle EJ, Zhou DW, Ramaswamy SM, Biswal S, Akeju O, Purdon PL, **Westover MB**. "Electroencephalogram based Detection of Deep Sedation in ICU Patients Using Atomic Decomposition." IEEE Transactions on Biomedical Engineering. 2018 Dec;65(12):2684-2691. PMCID:

PMC6424570.

146. van Leeuwen KG, Sun H, Tabaeizadeh M, Struck AF, van Putten MJAM, **Westover MB**, "Detecting Abnormal Electroencephalograms Using Deep Convolutional Networks". *Clinical Neurophysiology*. 2019 Jan;130(1):77-84. doi: 10.1016/j.clinph.2018.10.012. PMCID: PMC63097079.
147. Subramaniam T, Jain A, Hall LT, Cole AJ, **Westover MB**, Rosenthal ES, Struck AF. "Lateralized Periodic Discharges Frequency Correlates with Glucose Metabolism." *Neurology*. 2019 Feb 12;92(7):e670-e674. PMCID: PMC6382363.
148. Sun H, Paixao L, Oliva JT, Goparaju B, Carvalho DZ, van Leeuwen KG, Akeju O, Thomas RJ, Cash SS, Bianchi MT, **Westover MB**. "Brain Age from the Electroencephalogram of Sleep". Neurobiol Aging. 2019 Feb;74:112-120. PMCID: PMC6478501.
149. Zhuo Ding J, Mallick R, Carpentier J, McBain K, Gaspard N, **Westover MB**, Fantaneanu TA. "Resident training and interrater agreements using the ACNS critical care EEG terminology." *Seizure*. 2019 Feb 20;66:76-80. doi: 10.1016/j.seizure.2019.02.013. PMCID: PMC6778405.
150. Song, JL, Qiang L, Pan M, Zhang B, **Westover MB*,** Zhang R*. "Seizure tracking of epileptic EEG using a model-driven approach". *** Co-senior authors**. *Journal of Neural Engineering.* J Neural Eng. 2019 May 23. doi: 10.1088/1741-2552/ab2409. PMCID: PMC6874715.
151. Struck AF, Rodriguez-Ruiz AA, Osman G, Gilmore EJ, Lee JW, Gaspard N, Hirsch LJ, **Westover MB.** "Comparison of Machine-Learning Models for Seizure Prediction in Hospitalized Patients." *Annals of Clinical and Translational Neurology. Annals of Clinical and Translational Neurology.* 2019; May 23. 6(7): 1239–1247. PMCID: PMC6649418.
152. Ghassemi MM, Amorim E, Hanai TA, Lee JW, Herman ST, Sivaraju A, Gaspard N, Hirsch LJ, Scirica BM, Biswal S, Junior VM, Cash SS, Brown EN, Mark RG, **Westover MB.** "Quantitative EEG Trends Predict Recovery in Hypoxic-Ischemic Encephalopathy." *Critical Care Medicine.* 2019 Jun 21. doi: 10.1097/CCM.0000000000003840. PMCID: PMC6746597.
153. Amorim E, van der Stoel M, Nagaraj SB, Ghassemi MM, Jing J, O'Reilly UM, Scirica BM, Lee JW, Cash SS, **Westover MB**, "Quantitative EEG reactivity and machine learning for prognostication in hypoxic-ischemic brain injury." *Clinical Neurophysiology* 2019 Jul 25; (130) 1908–1916. PMCID: PMC6751020.
154. Kimchi EY, Neelagiri A, Whitt W, Sagi AR, Ryan SL, Gadbois G, Groothuysen D, **Westover MB, "**Clinical EEG Slowing Correlates with Delirium Severity and Predicts Poor Clinical Outcomes." *Neurology.* 2019 Sep 24;93(13):e1260-e1271. PMCID: PMC7011865.
155. Sun H, Kimchi E, Akeju O, Nagaraj SB, McClain LM, Zhou DW, Boyle E, Zheng WL, Ge W, **Westover MB**. "Automated Tracking of Level of Consciousness and Delirium in Critical Illness using Deep Learning". NPJ Digital Medicine. 2019 Sep 9;2:89. doi: 10.1038/s41746-019-0167-0. PMCID: PMC6733797.
156. Song JL, Rui Z, **Westover MB**. "A Novel Neural Computational Model of Generalized Periodic Discharges in Acute Hepatic Encephalopathy. *Journal of Computational Neuroscience.* 2019 Sep 11. doi: 10.1007/s10827-019-00727-3. PMCID: PMC6881550.
157. Bagheri E, Jin J, Dauwels J, Cash S, **Westover MB**. "A Fast Machine Learning Approach to Facilitate the Detection of Interictal Epileptiform Discharges in the Scalp Electroencephalogram." *Journal of Neuroscience Methods.* 2019 Oct 1;326:108362. doi: 10.1016/j.jneumeth.2019.108362. PMCID: PMC6993930.
158. Zheng WL, Sun H, Akeju O, **Westover MB**. "Adaptive Sedation Monitoring from EEG in ICU Patients with Online Learning." *IEEE Transactions on Biomedical Engineering. 2019 Sep 23. doi: 10.1109/TBME.2019.2943062. [Epub ahead of print]* PMCID: PMC7085963.
159. Jing J, Herlopian A, Karakis I, Ng M, Halford JJ, Lam A, Maus D, Chan F, Dolatshahi M, Muniz C, Chu C, Sacca V, Pathmanathan J, Ge W, Sun H, Dauwels J, Cole AJ, Hoch DB, Cash SS, **Westover MB**. "Inter-rater Reliability of Experts in Identifying Epileptiform Discharges". *JAMA Neurology.* 2019 Oct 21. doi: 10.1001/jamaneurol.2019.3531. [Epub ahead of print]. PMCID: PMC6806666.
160. Jing J*, Sun H, Kim JA*, Herlopian A, Karakis I, Ng M, Halford JJ, Maus D, Chan F, Dolatshahi M, Muniz C, Chu C, Sacca V, Pathmanathan J, Ge W, Dauwels J, Lam A, Cole AJ, Cash SS, **Westover MB**.

"Development of Expert-Level Automated Detection of Epileptiform Discharges During Electroencephalogram Interpretation." *JAMA Neurol. 2019* Oct 21. doi: 10.1001/jamaneurol.2019.3485. [Epub ahead of print]. PMCID: PMC6806668.

161. Jaoude MA, Jing J, Sun H, Pellerin KR, **Westover MB**, Cash SS, Lam AD. "Detection of mesial temporal lobe epileptiform discharges on intracranial electrodes using deep learning". *Clinical Neurophysiology.* 2019 Nov 11;131(1):133-141. PMCID: PMC6879011.
162. Worden LT, Chinappen DM, Stoyell SM, Gold J, Paixao L, Krishnamoorthy K, Kramer MA, **Westover MB***, Chu CJ*. "The Probability of Seizures During EEG Monitoring in High-Risk Neonates." *Epilepsia.* ***** **Co-senior authors.** *Epilepsia*. Dec;60(12):2508-2518. PMCID: PMC7083278.
163. Reyna MA, Josef CS, Jeter R, Shashikumar SP, **Westover MB**, Nemati S, Clifford GD, Sharma A. "Early Prediction of Sepsis from Clinical Data: The PhysioNet/Computing in Cardiology Challenge 2019". Crit Care Med. 2019 Dec 17. doi: 10.1097/CCM.0000000000004145. PMCID: PMC6964870.
164. Moise AM, Karakis I, Herlopian A, Dhakar M, Cotsonis G, Hirsch L, LaRoche S, **Westover MB**, Rodriguez A, "Continuous EEG Findings in Autoimmune Encephalitis." *Journal of Clinical Neurophysiology*. 2019 Dec 2. doi: 10.1097/WNP.0000000000000654. PMCID: PMC7263965.
165. AF Struck, M Tabaeizadeh, S Schmitt, AR Ruiz, C Swisher, T Subramaniam, C Hernandez, H Haider, M Dhakar, LJ Hirsch, ES Rosenthal, SF Zafar, N. Gaspard, **MB Westover**. "Assessment of the Validity of the 2HELPS2B Score for Inpatient Seizure Risk Prediction." *JAMA Neurology.* 2020 Apr 1;77(4):500-507. doi: 10.1001/jamaneurol.2019.4656. PMCID: PMC6990873.
166. Kashkooli K, Murphy JM, Sun H, **Westover MB**, Akeju O, Polk SL, Chamadia S, Hahm E, Ethridge B, Gitlin J, Ibala R, Mekonnen J, Pedemonte J. "Drug-Specific Models Improve the Performance of an EEG-based Automated Brain-State Prediction System." Conf Proc IEEE Eng Med Biol Soc. 2019 Jul;2019:5808-5811. doi: 10.1109/EMBC.2019.8856935. PMCID: PMC7077760.
167. Polk SL, Kashkooli K, Nagaraj SB, Chamadia S, Murphy JM, Sun H, **Westover MB**, Barbieri R, Akeju O. "Automatic Detection of General Anesthetic-States using ECG-Derived Autonomic Nervous System Features." Conf Proc IEEE Eng Med Biol Soc. 2019 Jul; 2019: 5808–5811. PMCID: PMC7077759.
168. Paixao L, Sikka P, Sun H, Jain A, Hogan J, Thomas RJ, **Westover MB**. "Excess Brain Age Reflected in the Electroencephalogram of Sleep Predicts Reduced Life Expectancy." *Neurobiology of Aging.* 2019 Dec 23. pii: S0197-4580(19)30443-9. doi: 10.1016/j.neurobiolaging.2019.12.015. PMCID: PMC7085452.
169. **Westover MB**, Guruangan K, Markert MS, Blond BN, Lai S, Benard S, Bickel S, Hirsch LJ, Parvizi J. "Diagnostic Value of Electroencephalography with Ten Electrodes in Critically Ill Patients." *Neurocritical Care.* 2020 Feb 7. doi: 10.1007/s12028-019-00911-4. [Epub ahead of print]. PMCID: PMC7416437.
170. Li Q, Song JL, Li SH, **Westover MB**, Zhang R. "Effects of cholinergic neuromodulation on thalamocortical rhythms during NREM sleep: a model study." *Frontiers in Computational Neuroscience.* 2020 Jan 23;13:100. doi: 10.3389/fncom.2019.00100. eCollection 2019. PMCID: PMC6990259.
171. Goldstein CA, Berry RB, Kent DT, Kristo DA, Seixas AA, Redline S, **Westover MB**, Abbasi-Feinberg F, Aurora RN, Carden KA, Kirsch DB, Malhotra RK, Martin JL, Olson EJ, Ramar K, Rosen CL, Rowley JA, Shelgikar AV. "Artificial Intelligence in Sleep Medicine: An American Academy of Sleep Medicine Position Statement." *Journal of Clinical Sleep Medicine*. 2020 Feb 5. doi: 10.5664/jcsm.8288. PMCID: PMC7161449.
172. Vink JJT, Klooster DCW, Ozdemir RA, **Westover MB**, Pascual-Leone A, Shafi MM. "EEG Functional Connectivity is a Weak Predictor of Causal Brain Interactions." *Brain Topography.* 2020 Feb 24. doi: 10.1007/s10548-020-00757-6. PMCID: PMC7191224.
173. Goldstein CA, Berry RB, Kent DT, Kristo DA, Seixas AA, Redline S, **Westover MB**. "Artificial intelligence in sleep medicine: Background and implications for clinicians." Journal of Clinical Sleep Medicine. 2020 Feb 17. doi: 10.5664/jcsm.8388. PMCID: PMC7161463.
174. Hogan J, Sun H, Aboul-Nour H, Jing J, Tabaeizadeh M, Shoukat M, Javed F, Kassa S, Edhi MM, Bordbar E, Gallagher J, Valdery-Moura J, Ghanta M, Shao YP, Akeju O, Cole AJ, Rosenthal ES, Zafar S, **Westover MB**, "Burst Suppression: Causes and Effects on Mortality in Critical Illness." *Neurocritical Care.* 2020 Feb 24. doi: 10.1007/s12028-020-00932-4. PMCID: PMC7483190.

175. Elmer J, Coppler PJ, Solanki P, **Westover MB**, Struck AF, Baldwin MA, Kurz MC, Callaway CW. "Sensitivity of Continuous Electroencephalography to Detect Ictal Activity After Cardiac Arrest." *JAMA Network Open*. 2020 Apr 1;3(4):e203751. doi: 10.1001/jamanetworkopen.2020.3751. PMCID: PMC7189220.
176. Lissak IA, Zafar SF, **Westover MB**, Schleicher RL, Kim JA, Leslie-Mazwi T, Stapleton C, Patel AB, Kimberly T, Rosenthal ES. "Soluble ST2 is an Inflammatory Biomarker Associated with New Epileptiform Abnormalities following Nontraumatic Subarachnoid Hemorrhage." *Stroke.* 2020 Apr;51(4):1128-1134. PMCID: PMC7123848.
177. Rubin DB, Angelini B, Shoukat M, Chu CJ, Zafar SF, **Westover MB**, Cash SS, Rosenthal ES. "Quantitative EEG features predicting successful weaning from IV anesthetics in refractory status epilepticus." *Brain.* 2020 Apr 1;143(4):1143-1157. PMCID: PMC7174057.
178. Pedemonte JC, Plummer GS, Chamadia S, Locascio JJ, Hahm E, Ethridge B, Gitlin J, Ibala R, Mekonnen J, Colon KM, **Westover MB**, D'Alessandro DA, Tolis G, Houle T, Shelton KT, Qu J, Akeju O. "Electroencephalogram Burst-suppression during Cardiopulmonary Bypass in Elderly Patients Mediates Postoperative Delirium." *Anesthesiology*. 2020 Apr 24. doi: 10.1097/ALN.0000000000003328. PMCID: PMC7365754.
179. Kashkooli K; Polk S, Hahm E, Murphy J, Ethridge B, Gitlin J, Ibala R, Mekonnen J, Pedemonte J; Sun H, **Westover MB**; Barbieri R, Akeju Oluwaseun, Chamadia S. "Improved tracking of sevoflurane anesthetic states with drug-specific machine learning models." *Journal of Neural Engineering.* 2020 Aug 4;17(4):046020. PMCID: PMC7540939.
180. Tabaeizadeh M, Nour HA, Shoukat M, Sun H, Jin J, Javed F, Kassa S, Edhi M, Bordbar E, Gallagher J, Junior VM, Ghanta M, Shao YP, Cole AJ, Rosenthal ES, **Westover MB**, Zafar SZ. "Burden of epileptiform activity predicts discharge neurologic outcomes in severe acute ischemic stroke." *Neurocritical Care.* 2020 Jun; 32(3):697-706. PMCID: PMC7416505.
181. Eichenlaub JB, Biswal S, Peled N, Rivilis N, Golby AJ, Lee JW, **Westover MB**, Halgren E, Cash SS. Reactivation of Motor-Related Gamma Activity in Human NREM Sleep. Front Neurosci. 2020;14:449. doi: 10.3389/fnins.2020.00449. Epub 2020 May 12. PMID: 32477056; PMCID: PMC7235414.
182. Beuchat I, Sivaraju A, Amorim E, Gilmore EJ, Dunet V, Rossetti AO, **Westover MB**, Hsu L, Scirica B, Silva D, Tang K, Lee Jw. "MRI-EEG correlation for outcome prediction in post-anoxic myoclonus, a multicenter study." *Neurology.* 2020 June 1;10.1212/WNL.0000000000009610. doi: 10.1212/WNL.0000000000009610. PMCID: PMC7455317.
183. Amorim E, Mo SS, Palacios S, Ghassemi MM, Weng WH, Cash SS, Bianchi MT, **Westover MB**. "Cost-effectiveness analysis of multimodal prognostication in cardiac arrest with EEG monitoring." *Neurology*. 2020 Jul 13; 10.1212/WNL.0000000000009916. doi:10.1212/WNL.0000000000009916. [Epub ahead of print]. PMCID: PMC7455344.
184. Sun H, Ganglberger W, Panneerselvam E, Leone MJ, Quadri SA, Goparaju B, Tesh RA, Akeju O, Thomas RJ, **Westover MB**. "Sleep Staging from Electrocardiography and Respiration with Deep Learning." *SLEEP.* Sleep. 2020 Jul 13;43(7):zsz306. doi: 10.1093/sleep/zsz306. PMCID: PMC7355395.
185. Amorim E, McGraw C, **Westover MB.** "A Theoretical Paradigm for Evaluating Risk-Benefit of Status Epilepticus Treatment." Amorim E, McGraw CM, Westover MB. J Clin Neurophysiol. 2020 Sep;37(5):385-392. doi: 10.1097/WNP.0000000000000753. PMCID: PMC7516305.
186. Zafar SF, Amorim E, Williamsom CA, Jin J, Gilmore EJ, Haider HA, Swisher C, Struck A, Rosenthal ES, Ng M, Schmitt S, Lee JW, **Westover MB**. "A standardized nomenclature for spectrogram EEG patterns: Inter-rater agreement and correspondence with common ICU EEG patterns." *Clinical Neurophysiology.* 2020;131(9):2298-2306. PMCID: PMC7461156.
187. Foy BH, Carlson JCT, Reinertsen E, Valls RP, Lopez RP, Palanques-Tost E, Mow C, **Westover MB**, Aguirre AA, Higgins JM. "Association of Red Blood Cell Distribution Width with Mortality Risk in Adults Hospitalized with COVID-19 Infection." *JAMA Network Open.* 2020 Sep 1;3(9):e2022058. doi: 10.1001/jamanetworkopen.2020.22058. PMCID: PMC7512057.
188. Prasanth T, Thomas J, Yuvaraj R, Jing J, Cash SS, Chaudhari R, Leng TY, Rathakrishnan R, Rohit S, Saini V, **Westover MB**, 2020, July. "Deep Learning for Interictal Epileptiform Spike Detection from scalp

EEG frequency sub bands." In 2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) (pp. 3703-3706). IEEE. PMCID: PMC7545315.

189. Ye E, Sun H, Leone MJ, Paixao L, Thomas RJ, Lam AD, **Westover MB**. "Sleep EEG-based Brain Age Index is a Biomarker for Dementia". *JAMA Network Open.* 2020 Sep 1;3(9):e2017357. doi: 10.1001/jamanetworkopen.2020.17357. PMID: 32986106; PMCID: PMC7522697.
190. Sun H*, Jain A*, Leone MJ*, Alabsi HS, Brenner LN, Ye E, Ge W, Shao YP, Boutros CL, Wang R, Tesh RA, Magdamo C, Collens SI, Ganglberger W, Bassett IV, Meigs JB, Kalpathy-Cramer J, Li MD, Chu JT, Dougan ML, Stratton LW, Rosand J, Fischl B, Das S, Mukerji SS**, Robbins GK**, **Westover MB**. "CoVA: An Acuity Score for Outpatient Screening that Predicts COVID-19 Prognosis." *Journal of Infectious Diseases.* 2020 Oct 24:jiaa663. doi: 10.1093/infdis/jiaa663. Epub ahead of print. PMCID: PMC7665643. . ***Co-first authors. ** Co-senior authors.**
191. Thomas J, Jin J, Thangavel P, Bagheri E, Yuvaraj R, Dauwels J, Rathakrishnan R, Halford JJ, Cash SS, **Westover MB**. "Automated Detection of Interictal Epileptiform Discharges from Scalp Electroencephalograms by Convolutional Neural Networks." *International Journal of Neural Systems. Int J Neural Syst.* 2020; Aug 19; 2050030. doi:10.1142/S0129065720500306. PMCID: PMC7606586.
192. Jaoude MA, Sun H, Pellerin KR, Pavlova M, Sarkis RA, Cash SS, **Westover MB***, Lam AD* "Expert-level automated sleep staging of long-term scalp EEG recordings using deep learning." ***Co-senior authors.** *SLEEP.* June 2020;zsaa112. doi:10.1093/sleep/zsaa112. PMCID: PMC7686563.
193. Vespa PM, Olson DM, John S, Hobbs KS, Gururangan K, Nie K, Desai M, Markert M, Parvizi J, Bleck TP, Hirsch LJ, **Westover MB**. "Evaluating the Clinical Impact of Rapid Response Electroencephalography: The DECIDE Multicenter Prospective Observational Clinical Study." *Critical Care Medicine.* 2020 Jun 29. doi: 10.1097/CCM.0000000000004428. [Epub ahead of print]. PMCID: PMC7735649.
194. Ayub N, Cohen J, Jing J, Jain A, Tesh R, Mukerji SS, Zafar SF, **Westover MB**, Kimchi EY. Clinical Electroencephalography Findings and Considerations in Hospitalized Patients with Coronavirus SARS-CoV-2. medRxiv [Preprint]. 2020 Jul 15:2020.07.13.20152207. doi: 10.1101/2020.07.13.20152207. PMCID: PMC7373142.
195. Thorn EL, Ostrowski LM, Chinappen DM, Jing J, **Westover MB**, Stufflebeam SM, Kramer MA, Chu CJ. "Persistent abnormalities in Rolandic thalamocortical white matter circuits in Childhood Epilepsy with Centrotemporal Spikes." *Epilepsia.* 2020 Sep 18. doi: 10.1111/epi.16681. Online ahead of print. PMCID: PMC7722074.
196. Goldenholz DM, Goldenholz SR, Romero J, Moss R, Sun H, **Westover MB**. Development and Validation of Forecasting Next Reported Seizure Using e-Diaries. Ann Neurol. 2020 Sep;88(3):588-595. doi: 10.1002/ana.25812. Epub 2020 Jul 9. PMCID: PMC7720795.
197. Lam AD, Sarkis RA, Pellerin KR, Jing J, Dworetzky BA, Hoch DB, Jacobs CS, Lee JW, Maus D, Weisholtz DS, Zepeda R, **Westover MB**, Cole AJ, Cash SS . "Association of epileptiform abnormalities and seizures in Alzheimer disease." *Neurology.* [Epub ahead of print, 2020 Aug 6];10.1212/WNL.0000000000010612. doi:10.1212/WNL.0000000000010612. PMCID: PMC7713786.
198. Hogan J, Sun H, Thomas RT, **Westover MB.** "Night-to-Night Variability of Sleep Electroencephalography-Based Brain Age Measurements." *Clinical Neurophysiology.* 2020 Oct 29;132(1):1-12. doi: 10.1016/j.clinph.2020.09.029. Epub ahead of print. PMCID: PMC7855943.
199. Jing J, d'Angremont E, Ebrahim S, **Westover MB.** "Rapid Annotation of Seizures and Interictal-Ictal-Injury Continuum EEG Patterns." *Journal of Neuroscience Methods.* 2020 Oct 22:108966. doi: 10.1016/j.jneumeth.2020.108966. Epub ahead of print. PMCID: PMC7744406
200. Shashikumar SP, Wardi G, Paul P, Carlile M, Brenner LN, Hibbert KA, North CM, Mukerji SS, Robbins GK, Shao YP, **Westover MB**, Nemati S, Malhotra A. "Development and Prospective Validation of a Deep Learning Algorithm for Predicting Need for Mechanical Ventilation." *Chest*. 2020 Dec 17;. doi: 10.1016/j.chest.2020.12.009. [Epub ahead of print] PubMed PMID: 33345948; PubMed Central PMCID: PMC8027289.

201. Zafar SF, Khozein RJ, LaRoche S, **Westover MB**, Gilmore EJ. "Impact of the COVID-19 pandemic on continuous EEG utilization." *Journal of Clinical Neurophysiology*. 2020 Dec 22;Publish Ahead of Print. doi: 10.1097/WNP.0000000000000802. Epub ahead of print. PMCID: PMC8217411
202. Lissak IA, Locascio JJ, Zafar SF, Patel AB, Leslie-Mazwi T, Stapleton CJ, Kim JA, Anderson K, Schleicher RL, Rosand J, **Westover MB**, Kimberly T, Rosenthal ES. "Electroencephalography, Hospital Complications, and Longitudinal Outcomes after Subarachnoid Hemorrhage." *Neurocritical Care.* 2021 Jan 22:1–12. doi: 10.1007/s12028-020-01177-x. Epub ahead of print. PMCID: PMC7822587.
203. Conklin J, Frosch MP, Mukerji SS, Rapalino O, Maher MD, Schaefer PW, Lev MH, Gonzalez RG, Das S, Champion SN, Magdamo C, Sen P, Harrold GK, Alabsi H, Normandin E, Shaw B, Lemieux JE, Sabeti PC, Branda JA, Brown EN, **Westover MB**, Huang SY, Edlow BL. "Susceptibility-weighted imaging reveals cerebral microvascular injury in severe COVID-19." J Neurol Sci. 2021 Feb 15;421:117308. doi: 10.1016/j.jns.2021.117308. Epub 2021 Jan 15. PMCID: PMC7832284.
204. Fernandes MB, Sun H, Jain A, Alabsi HS, Brenner LN, Ye E, Ge W, Collens SI, Leone M, Das S, Robbins GK*, Mukerji SS*, **Westover MB*** "Classification of the Disposition of Patients Hospitalized with COVID-19: Reading Discharge Summaries using Natural Language Processing." *Co-senior authors. *JMIR Med Inform.* 2021 Feb 10;9(2):e25457. doi: 10.2196/25457. PMID: 33449908; PMCID: PMC7879729.
205. Janocko NJ, Jing J, Fan Z, Teagarden DL, Villarreal HK, Morton ML, Groover O, Loring DW, Drane DL, **Westover MB**, Karakis I. DDESVSFS: A simple, rapid and comprehensive screening tool for the Differential Diagnosis of Epileptic Seizures VS Functional Seizures. Epilepsy Res. 2021 Jan 21;171:106563. doi: 10.1016/j.eplepsyres.2021.106563. Online ahead of print. PMCID: PMC8092190.
206. Ozdemir RA, Tadayon E, Boucher P, Sun H, Momi D, Ganglberger W, **Westover MB**, Pascual-Leone A, Santarnecchi E, Shafi MM. "Cortical Responses to Noninvasive Perturbations Enable Individual Brain Fingerprinting." Brain Stimul. 2021 Feb 12:S1935-861X(21)00033-4. doi: 10.1016/j.brs.2021.02.005. Online ahead of print. PMC8108003.
207. Lin L, Al-Faraj A, Ayub N, Bravo P, Das S, Ferlini L, Karakis I, Lee JW, Mukerji SS, Newey CR, Pathmanathan J, Abdennadher M, Cassasa C, Gaspard N, Goldenholz DM, Gilmore EJ, Jin J, Kim JA, Kimchi EY, Ladha HS, Tobochnik S, Zafar S, Hirsch LJ, **Westover MB***, Shafi MM*. "EEG Abnormalities are Common in COVID-19 and are Associated with Outcomes." *Annals of Neurology*. Vol 89 (5), Pages 872-883. May 2021. PMCID: PMC8104061. * **Co-senior authors.**
208. Hirsch LJ, Fong MWK, Leitinger M, LaRoche SM, Beniczky S, Abend NS, Lee JW, Wusthoff CJ, Hahn CD, **Westover MB**, Gerard EE, Herman ST, Haider HA, Osman G, Rodriguez-Ruiz A, Maciel CB, Gilmore EJ, Fernandez A, Rosenthal ES, Claassen J, Husain AM, Yoo JY, So EL, Kaplan PW, Nuwer MR, van Putten M, Sutter R, Drislane FW, Trinka E, Gaspard N. "American Clinical Neurophysiology Society's Standardized Critical Care EEG Terminology: 2021 Version." *Journal of Clinical Neurophysiology.* 2021 Jan 1;38(1):1-29. PMCID: PMC8135051.
209. Kramer M, Stoyell S, Chinappen D, Ostrowski L, Spencer E, Morgan A, Emerton B, Jing J, **Westover MB**, Eden U, Stickgold R, Manoach D, Chu C. "Focal sleep spindle deficits reveal focal thalamocortical dysfunction and predict cognitive deficits in childhood epilepsy with centrotemporal spikes." *Journal of Neuroscience.* 2021 Jan 14:JN-RM-2009-20. doi: 10.1523/JNEUROSCI.2009-20.2020. Epub ahead of print. PMCID: PMC8115887.
210. Gaspard N, **Westover MB**, Hirsch LJ. "Assessment of a Study of Continuous vs Repeat-Spot Electroencephalography in Patients With Critical Illness." JAMA Neurol. 2021 Feb 1. doi: 10.1001/jamaneurol.2020.5348. PMCID: PMC8136812.
211. Ge W, Jing J, An S, Herlopian A, Ng M, Struck AF, Appavu B, Johnson EL, Osman G, Haider HA, Karakis I, Kim JA, Halford JJ, Dhakar MB, Sarkis RA, Swisher CB, Schmitt S, Lee JW, Tabaeizadeh M, Rodriguez A, Gaspard N, Gilmore E, Herman ST, Kaplan PW, Pathmanathan J, Hong S, Rosenthal ES, Zafar S, Sun J, **Westover MB**. "Deep active learning for Interictal Ictal Injury Continuum EEG patterns." *J Neurosci Methods*. 2021 Mar 1;351:108966. doi: 10.1016/j.jneumeth.2020.108966. Epub 2020 Oct 22. PMCID: PMC8135050.

212. Li Q, **Westover MB**, Zhang R, Chu CJ. Computational Evidence for a Competitive Thalamocortical Model of Spikes and Spindle Activity in Rolandic Epilepsy. Front Comput Neurosci. 2021 Jun 18;15:680549. doi: 10.3389/fncom.2021.680549. PMCID: PMC8249809. ***Co-senior authors.**
213. Mukerji S, Das S, Alabsi H, Brenner LN, Jain A, Magdamo C, Collens SI, Ye E, Keller K, Boutros CL, Leone M, Newhouse A, Foy B, Li MD, Lang M, Anahtar MN, Shao YP, Ge W, Sun H, Triant VA, Kalpathy-Cramer J, Higgins J, Rosand J, Robbins GK, **Westover MB.** "Prolonged intubation in patients with prior cerebrovascular disease and COVID-19". *Frontiers in Neurology, Neurocritical and Neurohospitalist Care.* 2021 Apr 9;12:642912. doi: 10.3389/fneur.2021.642912. PMCID: PMC8062773.
214. Jones FJS, Sanches PR, Smith JR, Zafar SF, Hernandez-Diaz S, Newhouse JP, Blacker D, Hsu J, Schwamm LH, **Westover MB*,** Moura LMVR*, "Seizure prophylaxis after spontaneous intracerebral hemorrhage: a decision analysis." *JAMA Neurology.* 2021 Jul 26:e212249. doi: 10.1001/jamaneurol.2021.2249. Epub ahead of print. PMCID: PMC8314179.
215. Pedemonte JC, Sun H, Franco-Garcia E, Zhou C, Heng M, Quraishi SA, **Westover MB**, Akeju O. "Postoperative delirium mediates 180-day mortality in orthopaedic trauma patients." *British Journal of Anesthesia.* 2021 Jul;127(1):102-109. doi: 10.1016/j.bja.2021.03.033. Epub 2021 May 29. PMCID: PMC8258970.
216. Yang JC, Harid NM, Nascimento FA, Kokkinos V, Shaughnessy A, Lam AD, **Westover MB**, Leslie-Mazwi TM, Hochberg LR, Rosenthal ES, Cole AJ, Richardson RM, Cash SS. "Responsive neurostimulation for focal motor status epilepticus." Ann Clin Transl Neurol. 2021 May 6. doi: 10.1002/acn3.51318. Online ahead of print. PMCID: PMC8164849.
217. Thomas J, Thangavel P, Peh WY, Jing J, Yuvaraj R, Cash SS, Chaudhari R, Karia S, Rathakrishnan R, Saini V, Shah N, Srivastava R, Tan YL, **Westover MB**, Dauwels J. "Automated Adult Epilepsy Diagnostic Tool Based on Interictal Scalp Electroencephalogram Characteristics: A Six-Center Study." Int J Neural Syst. 2021 May;31(5):2050074. doi: 10.1142/S0129065720500744. Epub 2021 Jan 12. PMCID: PMC9343226.
218. Leone MJ*, Sun H*, Boutros CL, Liu L, Ye E, Sullivan L, Thomas RJ, Robbins GK, Mukerji SS, **Westover MB**. "HIV Increases Sleep-based Brain Age Despite Antiretroviral Therapy." *SLEEP*. 2021 Mar 30:zsab058. doi: 10.1093/sleep/zsab058. Epub ahead of print. PMCID: PMC8361332.
219. Jones FJS, Sanches PR, Smith JR, Zafar SF, Hernandez-Diaz S, Newhouse JP, Blacker D, Hsu J, Schwamm LH, **Westover MB*,** Moura LMVR*. "Anticonvulsant primary and secondary prophylaxis for acute ischemic stroke patients: a decision analysis." **** Co-senior authors.*** *Stroke.* 2021 Jun 15:STROKEAHA120033299. doi: 10.1161/STROKEAHA.120.033299. Epub ahead of print. PMCID: PMC8384723.
220. Kim JA, Zheng WL, Elmer J, Jing J, Zafar SF, Ghanta M, Moura V Junior, Gilmore EJ, Hirsch LJ, Patel A, Rosenthal E, **Westover MB**. "High epileptiform discharge burden predicts delayed cerebral ischemia after subarachnoid hemorrhage." Clin Neurophysiol. 2021 Mar 10:S1388-2457(21)00063-8. doi: 10.1016/j.clinph.2021.01.022. Epub ahead of print. PMCID: PMC8429508.
221. Ligtenstein S; Song J, Jin J, Sun HB, Paixao L, Zafar S, **Westover MB**, "Do Triphasic Waves and Nonconvulsive Status Epilepticus Arise From Similar Mechanisms? A Computational Model." *Journal of Clinical Neurophysiology.* 2021 Jun 18. doi: 10.1097/WNP.0000000000000719. Epub ahead of print. PMCID: PMC8429048.
222. Nascimento FA, Chen P, Cohen J, **Westover MB**. "Generalized periodic discharges with triphasic morphology – to treat or not to treat?" *Epileptic Disorders.* 2021 Apr 1;23(2):435-436. doi: 10.1684/epd.2021.1272. PMCID: PMC8592021.
223. Oppersma E, Ganglberger W, Sun H, Thomas RJ*, **Westover MB***. "Algorithm for automatic detection of self-similarity and prediction of residual central respiratory events during continuous positive airway pressure." Sleep. 2021 Apr 9;44(4):zsaa215. doi: 10.1093/sleep/zsaa215. PMCID: PMC8631077. ***Co-senior authors.***
224. Zafar SF, Rosenthal ES, Jing J, Ge W, Tabaeizadeh M, Aboul Nour H, Shoukat M, Sun H, Javed F, Kassa S, Edhi M, Bordbar E, Gallagher J, Moura V Jr, Ghanta M, Shao YP, An S, Sun J, Cole AJ, **Westover MB**. "Automated Annotation of Epileptiform Burden and Its Association with Outcomes."

Annals of Neurology. 2021 Aug;90(2):300-311. doi: 10.1002/ana.26161. Epub 2021 Jul 20. PMID: 34231244; PMCID: PMC8516549.

225. Paixao L, Sun H, Hogan J, Hartnack K, Westmeijer M, Neelagiri A, Zhou DW, McClain LM, Kimchi EY, Purdon PL, Akeju O, **Westover MB**. "ICU Delirium Burden Predicts Functional Neurologic Outcomes." *PLoS One. 2021 Dec 2;16(12):e0259840. doi: 10.1371/journal.pone.0259840. eCollection 2021.* PMCID: PMC8638853.

226. van Sleuwen M, Sun H, Eckhardt C, Neelagiri A, Tesh RA, Westmeijer M, Paixao L, Rajan S, Velpula Krishnamurthy P, Sikka P, Leone MJ, Panneerselvam E, Quadri SA, Akeju O, Kimchi EY, **Westover MB**. "Physiological Assessment of Delirium Severity: The Electroencephalographic Confusion Assessment Method Severity Score (E-CAM-S)." *Crit Care Med*. 2021 Sep 24. doi: 10.1097/CCM.0000000000005224. Epub ahead of print. PMCID: PMC8678335.

227. Zheng WL, Amorim E, Jing J, Ge W, Hong S, Wu O, Ghassemi M, Lee JW, Sivaraju A, Pang T, Herman ST, Gaspard N, Ruijter BJ, Sun J, Tjepkema-Cloostermans MC, Hofmeijer J, van Putten MJAM, **Westover MB**. "Predicting Neurological Outcome in Comatose Patients after Cardiac Arrest with Multiscale Deep Neural Networks." *Resuscitation.* 2021 Oct 23:S0300-9572(21)00441-X. doi: 10.1016/j.resuscitation.2021.10.034. Epub ahead of print. PMCID: PMC8692444.

228. Tesh RA*, Sun H*, Jing J*, Westmeijer M, Neelagiri A, Rajan S, Krishnamurthy PV, Sikka P, Quadri SA, Leone MJ, Paixao L, Panneerselvam E, Eckhardt C, Struck AF, Kaplan PW, Akeju O, Jones D, Kimchi EY**, **Westover MB****. "VE-CAM-S: Visual EEG-based Grading of Delirium Severity and Associations with Clinical Outcomes". Critical Care Explorations. *co-first authors, ****co-senior authors**. 2022 Jan 18;4(1):e0611. doi: 10.1097/CCE.0000000000000611. PMCID: PMC8769081.

229. Amerineni R, Sun H, Lee H, Hsu J, Patorno E, **Westover MB**, Zafar SF. "Using electronic health data to explore effectiveness of ICU EEG and anti-seizure treatment." *Annals of Clinical and Translational Neurology.* 2021 Dec;8(12):2270-2279. doi: 10.1002/acn3.51478. Epub 2021 Nov 21. PMID: 34802196; PMCID: PMC8670316.

230. Song JL, Kim J, Struck A, Zhang R, **Westover MB**. "A Model of Metabolic Supply-Demand Mismatch Leading to Secondary Brain Injury." *Journal of Neurophysiology.* 2021 Jul 7. doi: 10.1152/jn.00674.2020. Online ahead of print. PMCID: PMC8815783.

231. Chen HY, Elmer J, Zafar SF, Ghanta M, Junior VM, Rosenthal ES, Gilmore EJ, Hirsch LJ, Zaveri HP, Sheth KN, Petersen NH, **Westover MB**, Kim JA. "Combining Transcranial Doppler and EEG Data to Predict Delayed Cerebral Ischemia After Subarachnoid Hemorrhage." *Neurology*. 2021 Nov 29:10.1212/WNL.0000000000013126. doi: 10.1212/WNL.0000000000013126. Epub ahead of print. PMCID: PMC8826465.

232. Ganglberger W, Bucklin AA, Tesh R, BSc, Cardoso MDS; Sun H, Leone M, Paixao L, Panneerselvam E, Ye E, Thompson T, Johnson-Akeju O, Kuller D, Thomas RJ*, **Westover MB***. Co-senior authors. "Sleep Apnea and Respiratory Anomaly Detection from a Wearable Band and Oxygen Saturation." *Sleep and Breathing.* 2021 Aug 18. doi: 10.1007/s11325-021-02465-2. Online ahead of print. PMCID: PMC8854446

233. Salami P, Borzello M, Kramer MA, **Westover MB**, Cash SS. "Quantifying seizure termination patterns reveals limited pathways to seizure end". *Neurobiology of Disease*. Neurobiol Dis. 2022 Jan 29;165:105645. doi: 10.1016/j.nbd.2022.105645. Epub 2022 Jan 29. PMCID: PMC8860887.

234. Nayan M, Salari K, Bozzo A, Ganglberger W, Lu G, , Carvalho F, Gusev A, **Westover MB**, Feldman AS. "A machine learning approach to predict progression on active surveillance for prostate cancer." *Urologic Oncology: Seminars and Original Investigations.* 2021 Aug 28:S1078-1439(21)00366-5. doi: 10.1016/j.urolonc.2021.08.007. Epub ahead of print. PMCID: PMC8882704.

235. Junior VM, **Westover MB**, Li F, Kimchi EY, Kennedy M, Benson NM, Moura LMVR, Hsu J. "Hospital Complications Among Older Adults: Better Processes Could Reduce the Risk of Delirium." *Health Services Management Research.* 2021 Jul 11:9514848211028707. doi: 10.1177/09514848211028707. Epub ahead of print. PMCID: PMC8748518.

236. Ong CS, Reinertsen E, Sun H, Moonsamy P, Mohan N, Funamoto M, Kaneko T, Shekar PS, Schena S, Lawton JS, D'Alessandro DA, **Westover MB**, Aguirre AD, Sundt TM. "Prediction of operative mortality

for patients undergoing cardiac surgical procedures without established risk scores." *J Thorac Cardiovasc Surg*. 2021 Sep 14:S0022-5223(21)01313-1. doi: 10.1016/j.jtcvs.2021.09.010. Epub ahead of print. PMCID: PMC8918430.

237. Ng M, Jing J, **Westover MB**. "A Primer of EEG Spectrograms." *Journal of Clinical Neurophysiology. J Clin Neurophysiol.* 2022 Mar 1;39(3):177-183. doi: 10.1097/WNP.0000000000000736. PMCID: PMC8901534.
238. Ng M, **Westover MB**. "Decoding the Spectrogram Rainbow." *Journal of Clinical Neurophysiology* 2022 Mar 1;39(3):176. doi: 10.1097/WNP.0000000000000741. PMCID: PMC8901547.
239. Kim J, Zafar S, Petersen N, Rosenthal ES, **Westover MB.** "The Utility of Quantitative EEG in Detecting Delayed Cerebral Ischemia After Aneurysmal Subarachnoid Hemorrhage." *Journal of Clinical Neurophysiology.* 2022 Mar 1;39(3):207-215. doi: 10.1097/WNP.0000000000000754. PMID: 34510093; PMCID: PMC8901442.
240. Nascimento FA, Jing J, Beniczky S, Benbadis SR, Gavvala JR, Yacubian EMT, Wiebe S, Rampp S, van Putten MJAM, Tripathi M, Cook MJ, Kaplan PW, Tatum WO, Trinka E, Cole AJ, **Westover MB.** "One EEG, one read - A manifesto towards reducing interrater variability among experts." Clin Neurophysiol. 2022 Jan;133:68-70. doi: 10.1016/j.clinph.2021.10.007. Epub 2021 Nov 5. PMID: 34814017; PMCID: PMC8926459.
241. Adra N, Sun H, Ganglberger W, Ye EM, Dümmer LW, Tesh RA, Westmeijer M, Cardoso MDS, Kitchener E, Ouyang A, Salinas J, Rosand J, Cash SS, Thomas RJ, Westover MB. "Optimal Spindle Detection Parameters for Predicting Cognitive Performance." *Sleep*. 2022 Jan 4:zsac001. doi: 10.1093/sleep/zsac001". Epub ahead of print. PMCID: PMC8996023.
242. Waldrop G, Safavynia SA, Barra ME, Agarwal S, Berlin DA, Boehme AK, Brodie D, Choi JM, Doyle K, Fins JJ, Ganglberger W, Hoffman K, Mittel AM, Roh D, Mukerji SS, Der Nigoghossian C, Park S, Schenck EJ, Salazar-Schicchi J, Shen Q, Sholle E, Velazquez AG, Walline MC, **Westover MB**, Brown EN, Victor J, Edlow BL, Schiff ND, Claassen J. Annals of Neurology . "Prolonged unconsciousness is common in COVID-19 and associated with hypoxemia." Ann Neurol. 2022 Mar 7. doi: 10.1002/ana.26342. Online ahead of print. PMCID: PMC9082460.
243. Abou Jaoude M, Jacobs CS, Sarkis RA, Jing J, Pellerin KR, Cole AJ, Cash SS, **Westover MB**, Lam AD. "Noninvasive Detection of Hippocampal Epileptiform Activity on Scalp Electroencephalogram." JAMA Neurol. 2022 May 2. doi: 10.1001/jamaneurol.2022.0888. Online ahead of print. PMCID: PMC9062772.
244. Zafar SF, Rosenthal ES, Postma EN, Sanches P, Ayub MA, Subapriya R, Kim JA, Rubin DB, Lee H, Patel AB, Hsu J, Patorno E, **Westover MB**. "Antiseizure medication treatment and outcomes in subarachnoid hemorrhage patients undergoing continuous EEG monitoring." *Neurocritical Care.* 2021 Nov 29. doi: 10.1007/s12028-021-01387-x. Online ahead of print. PMCID: PMC9117405.
245. Kural MA, Jing J, Fürbass F, Perko H, Qerama E, Johnsen B, Fuchs S, **Westover MB**, Beniczky S. "Accurate identification of EEG recordings with interictal epileptiform discharges using a hybrid approach: Artificial intelligence supervised by human experts." *Epilepsia*. 2022 Feb 20. doi: 10.1111/epi.17206. Epub ahead of print. PMCID: PMC9148170.
246. "Automated Scoring of Respiratory Events in Sleep with a Single Effort Belt and Deep Neural Networks." Nassi TE, Ganglberger W, Sun H, Bucklin AA, Biswal S, van Putten MJAM, Thomas RJ, **Westover MB**. *IEEE Transactions on Biomedical Engineering*. 2021 Dec 20;PP. doi: 10.1109/TBME.2021.3136753. Epub ahead of print. PMCID: PMC9119908.
247. Sheikh Z, Selioutski O, Taraschenko O, Gilmore EJ, **Westover MB***, Cohen AB*. ***co-Senior authors.** "Systematic Evaluation of Research Priorities in Critical Care Electroencephalography." *Journal of Clinical Neurophysiology.* 2022 Jan 20. doi: 10.1097/WNP.0000000000000916. Epub ahead of print. PMCID: PMC9296700.
248. Harid NM*, Jing J*, Hogan J*, Nascimento FA, Ouyang A, Zheng WL, Ge WD, Zafar SF, Kim JA, Lam AD, Herlopian A, Maus D, Karakis I, Ng M, Hong S, Zhu Y, Kaplan P, Cash SS, Shafi MM, Martz G, Halford JJ, **Westover MB**. "Measuring Expertise in Identifying Interictal Epileptiform Discharges".

Epileptic Disorders. 2022 Jan 14. doi: 10.1684/epd.2021.1409. Epub ahead of print. PMCID: PMC9340812.
249. Kulpanowski AM, Copen WA, Hancock B, Rosenthal ES, Schoenfeld D, Dodelson JA, Edlow BL, Taylor Kimberly W, Amorim E, **Westover MB**, Ming Ning M, Schaefer PW, Malhotra R, Giacino JT, Greer DM, Wu O. "Severe Cerebral Edema in Substance-Related Cardiac Arrest Patients." *Resuscitation*. 2022 Feb 8:S0300-9572(22)00035-1. doi: 10.1016/j.resuscitation.2022.01.033. Epub ahead of print. PMCID: PMC9282938.
250. Ge W, Alabsi H, Jain A, Ye E, Sun H, Fernandes M, Magdamo C, Tesh RA, Collens SI, Newhouse A, Mvr Moura L, Zafar S, Hsu J, Akeju O, Robbins GK, Mukerji SS, Das S, **Westover MB**. "Identifying Patients With Delirium Based on Unstructured Clinical Notes: Observational Study." *Journal of Medical Internet Research (JMIR)*. 2022 Jun 24;6(6):e33834. doi: 10.2196/33834. PMCID: PMC9270709.
251. Nascimento FA, McLaren JR, **Westover MB**, Zafar SF, Stufflebeam SM. "Teaching NeuroImage: Sturge-Weber Syndrome in an Adult." *Neurology*. 2022 May 10;98(19):814-815. doi: 10.1212/WNL.0000000000200512. Epub 2022 Apr 11. PMCID: PMC9141624.
252. Amorim E, Firme MS, Zheng WL, Shelton KT, Akeju O, Cudemus G, Yuval R, **Westover MB**. High incidence of epileptiform activity in adults undergoing extracorporeal membrane oxygenation. Clin Neurophysiol. 2022 Aug;140:4-11. doi: 10.1016/j.clinph.2022.04.018. Epub 2022 May 6. PMCID: PMC9340813.
253. Brink-Kjaer A, Leary EB, Sun H, **Westover MB**, Stone KL, Peppard PE, Lane NE, Cawthon PM, Redline S, Jennum P, Sorensen HBD, Mignot E. "Age estimation from sleep studies using deep learning predicts life expectancy." NPJ Digit Med. 2022 Jul 22;5(1):103. doi: 10.1038/s41746-022-00630-9. PMCID: PMC9307657.
254. Thangavel P, Thomas J, Peh WY, Jing J, Yuvaraj R, Cash SS, Chaudhari R, Karia S, Rathakrishnan R, Saini V, Shah N, Srivastava R, Tan YL, **Westover MB**, Dauwels J. "Time-Frequency Decomposition of Scalp Electroencephalograms Improves Deep Learning-Based Epilepsy Diagnosis." Int J Neural Syst. 2021 Aug;31(8):2150032. doi: 10.1142/S0129065721500325. Epub 2021 Jul 16. PMCID: PMC9340811.
255. Song JL, Li Q, Zhang B, **Westover MB**, Zhang R. A New Neural Mass Model Driven Method and Its Application in Early Epileptic Seizure Detection. IEEE Trans Biomed Eng. 2020 Aug;67(8):2194-2205. doi: 10.1109/TBME.2019.2957392. Epub 2019 Dec 3. PMCID: PMC9371613.
256. Singh NM, Harrod JB, Subramanian S, Robinson M, Chang K, Cetin-Karayumak S, Dalca AV, Eickhoff S, Fox M, Franke L, Golland P, Haehn D, Iglesias JE, O'Donnell LJ, Ou Y, Rathi Y, Siddiqi SH, Sun H, **Westover MB**, Whitfield-Gabrieli S, Gollub RL. How Machine Learning is Powering Neuroimaging to Improve Brain Health. Neuroinformatics. 2022 Mar 28:10.1007/s12021-022-09572-9. doi: 10.1007/s12021-022-09572-9. Epub ahead of print. PMID: 35347570; PMCID: PMC9515245.
257. Hanidziar D, **Westover MB**. Monitoring of sedation in mechanically ventilated patients using remote technology. Curr Opin Crit Care. 2022 Jun 01;28(3):360-366. doi: 10.1097/MCC.0000000000000940. PMID: 35653256; PMCID: PMC9434805.
258. Torres-Lopez VM, Rovenolt GE, Olcese AJ, Garcia GE, Chacko SM, Robinson A, Gaiser E, Acosta J, Herman AL, Kuohn LR, Leary M, Soto AL, Zhang Q, Fatima S, Falcone GJ, Payabvash MS, Sharma R, Struck AF, Sheth KN, **Westover MB**, Kim JA. Development and Validation of a Model to Identify Critical Brain Injuries Using Natural Language Processing of Text Computed Tomography Reports. JAMA Netw Open. 2022 Aug 1;5(8):e2227109. doi: 10.1001/jamanetworkopen.2022.27109. Erratum in: JAMA Netw Open. 2022 Sep 1;5(9):e2237810. PMID: 35972739; PMCID: PMC9382443.
259. McLaren JR, Jing J, **Westover MB**, Nascimento FA. Journal Club: Criteria for Defining Interictal Epileptiform Discharges in EEG. Neurology. 2022 Jul 19;99(10):430–2. doi: 10.1212/WNL.0000000000200991. Epub ahead of print. PMID: 35853743; PMCID: PMC9519249.
260. Song JL, **Westover MB**, Zhang R. A mechanistic model of calcium homeostasis leading to occurrence and propagation of secondary brain injury. J Neurophysiol. 2022 Nov 1;128(5):1168-1180. doi: 10.1152/jn.00045.2022. Epub 2022 Oct 5. PMCID: PMC9621713.
261. Jones DK, Eckhardt CA, Sun H, Tesh RA, Malik P, Quadri S, Firme MS, van Sleuwen M, Jain A, Fan Z, Jing J, Ge W, Nascimento FA, Sheikh IS, Jacobson C, Frigault M, Kimchi EY, Cash SS, Lee JW,

Dietrich J, **Westover MB**. "EEG-based grading of immune effector cell-associated neurotoxicity syndrome." *Scientific Reports*. 2022 Nov 21;12(1):20011. doi: 10.1038/s41598-022-24010-1. PMCID: PMC9681864.

262. Amidi Y, Eckhardt CA, Quadri SA, Malik P, Firme MS, Jones DK, Jain A, Danish HH, Rubin DB, Jacobson CA, Cash SS, Lee JW, Dietrich J, **Westover MB**. Forecasting immune effector cell-associated neurotoxicity syndrome after chimeric antigen receptor t-cell therapy. J Immunother Cancer. 2022 Nov;10(11):e005459. doi: 10.1136/jitc-2022-005459. PMCID: PMC9716920.
263. Zheng WL, Kim JA, Elmer J, Zafar SF, Ghanta M, Moura Junior V, Patel A, Rosenthal E, **Westover MB**. "Automated EEG-based prediction of delayed cerebral ischemia after subarachnoid hemorrhage." Clin Neurophysiol. 2022 Sep 11;143:97-106. doi: 10.1016/j.clinph.2022.08.023. Epub ahead of print. PMCID: PMC9847346.
264. Liang F, Baldyga K, Quan Q, Khatri A, Choi S, Wiener-Kronish J, Akeju O, **Westover MB**, Cody K, Shen Y, Marcantonio ER, Xie Z. "Preoperative Plasma Tau-PT217 and Tau-PT181 Are Associated With Postoperative Delirium." Ann Surg. 2022 Jul 6. doi: 10.1097/SLA.0000000000005487. Epub ahead of print. PMCID: PMC9875943.
265. Fernandes M, Donahue MA, Hoch D, Cash S, Zafar S, Jacobs C, Hosford M, Voinescu PE, Fureman B, Buchhalter J, McGraw CM, **Westover MB***, Moura LMVR*. "A replicable, open-source, data integration method to support national practice-based research & quality improvement systems." Epilepsy Res. 2022 Oct;186:107013. doi: 10.1016/j.eplepsyres.2022.107013. Epub 2022 Aug 18. ****co-senior authors.** *Epilepsy Research.* PMCID: PMC9810436.
266. Nascimento FA, Jing J, Strowd R, Sheikh IS, Weber D, Gavvala JR, Maheshwari A, Tanner A, Ng M, Vinayan KP, Sinha SR, Yacubian EM, Rao VR, Perry MS, Fountain NB, Karakis I, Wirrell E, Yuan F, Friedman D, Tankisi H, Rampp S, Fasano R, Wilmshurst JM, O'Donovan C, Schomer D, Kaplan PW, Sperling MR, Benbadis S, **Westover MB**, Beniczky S. "Competency-based EEG education: a list of "must-know" EEG findings for adult and child neurology residents." *Epileptic Disorders.* 2022 Oct 1;24(5):1-4. doi: 10.1684/epd.2022.1476. PMCID: PMC9812628.
267. Noori A, Magdamo C, Liu X, Tyagi T, Li Z, Kondepudi A, Alabsi H, Rudmann E, Wilcox D, Brenner L, Robbins GK, Moura L, Zafar S, Benson NM, Hsu J, R Dickson J, Serrano-Pozo A, Hyman BT, Blacker D, **Westover MB**, Mukerji SS, Das S. Development and Evaluation of a Natural Language Processing Annotation Tool to Facilitate Phenotyping of Cognitive Status in Electronic Health Records: Diagnostic Study. J Med Internet Res. 2022 Aug 30;24(8):e40384. doi: 10.2196/40384. PMID: 36040790; PMCID: PMC9472045.
268. Nascimento FA, Jing J, Beniczky S, Olandoski M, Benbadis SR, Cole AJ, **Westover MB**. EEG reading with or without clinical information - a real-world practice study. Neurophysiol Clin. 2022 Oct;52(5):394-397. doi: 10.1016/j.neucli.2022.08.002. Epub 2022 Sep 17. PMCID: PMC9815944.
269. Martinez P, Sheikh I, **Westover MB**, Zafar SF. Implications of stimulus-induced, rhythmic, periodic, or ictal discharges (SIRPIDs) in hospitalized patients. Front Neurol. 2023 Jan 23;13:1062330. doi: 10.3389/fneur.2022.1062330. PMID: 36756343; PMCID: PMC9899805.
270. Goldenholz DM, **Westover MB**. Flexible realistic simulation of seizure occurrence recapitulating statistical properties of seizure diaries. Epilepsia. 2023 Feb;64(2):396-405. doi: 10.1111/epi.17471. Epub 2022 Dec 4. PMID: 36401798; PMCID: PMC9905290.
271. Qu JZ, Mueller A, McKay TB, **Westover MB**, Shelton KT, Shaefi S, D'Alessandro DA, Berra L, Brown EN, Houle TT, Akeju O; MINDDS Study Team. Nighttime dexmedetomidine for delirium prevention in non-mechanically ventilated patients after cardiac surgery (MINDDS): A single-centre, parallel-arm, randomised, placebo-controlled superiority trial. EClinicalMedicine. 2022 Dec 24;56:101796. doi: 10.1016/j.eclinm.2022.101796. PMID: 36590787; PMCID: PMC9800196.
272. Fernandes MB, Valizadeh N, Alabsi HS, Quadri SA, Tesh RA, Bucklin AA, Sun H, Jain A, Brenner LN, Ye E, Ge W, Collens SI, Lin S, Das S, Robbins GK, Zafar SF, Mukerji SS, **Westover MB.** Classification of neurologic outcomes from medical notes using natural language processing. Expert Syst Appl. 2023 Mar 15;214:119171. doi: 10.1016/j.eswa.2022.119171. Epub 2022 Nov 6. PMID: 36865787; PMCID: PMC9974159.

273. Bucklin AA, Ganglberger W, Quadri SA, Tesh RA, Adra N, Da Silva Cardoso M, Leone MJ, Krishnamurthy PV, Hemmige A, Rajan S, Panneerselvam E, Paixao L, Higgins J, Ayub MA, Shao YP, Ye EM, Coughlin B, Sun H, Cash SS, Thompson BT, Akeju O, Kuller D, Thomas RJ, **Westover MB.** High prevalence of sleep-disordered breathing in the intensive care unit - a cross-sectional study. Sleep Breath. 2022 Aug 16;. doi: 10.1007/s11325-022-02698-9. [Epub ahead of print] PubMed PMID: 35971023; PMCID: PMC9931933.
274. Chen Y, Li S, Ge W, Jing J, Chen HY, Doherty D, Herman A, Kaleem S, Ding K, Osman G, Swisher CB, Smith C, Maciel CB, Alkhachroum A, Lee JW, Dhakar MB, Gilmore EJ, Sivaraju A, Hirsch LJ, Omay SB, Blumenfeld H, Sheth KN, Struck AF, Edlow BL, **Westover MB**, Kim JA. Quantitative epileptiform burden and electroencephalography background features predict post-traumatic epilepsy. J Neurol Neurosurg Psychiatry. 2023 Mar;94(3):245-249. doi: 10.1136/jnnp-2022-329542. Epub 2022 Oct 14. PMID: 36241423; PMCID: PMC9931627.
275. Ye EM, Sun H, Krishnamurthy PV, Adra N, Ganglberger W, Thomas RJ, Lam AD, **Westover MB.** Dementia detection from brain activity during sleep. Sleep. 2023 Mar 9;46(3). doi: 10.1093/sleep/zsac286. PMID: 36448766; PMCID: PMC9995788.
276. Sun H, Ye E, Paixao L, Ganglberger W, Chu CJ, Zhang C, Rosand J, Mignot E, Cash SS, Gozal D, Thomas RJ, **Westover MB.** The sleep and wake electroencephalogram over the lifespan. Neurobiol Aging. 2023 Apr;124:60-70. doi: 10.1016/j.neurobiolaging.2023.01.006. Epub 2023 Jan 19. PMID: 36739622; PMCID: PMC9957961.
277. Ganglberger W, Krishnamurthy PV, Quadri SA, Tesh RA, Bucklin AA, Adra N, Da Silva Cardoso M, Leone MJ, Hemmige A, Rajan S, Panneerselvam E, Paixao L, Higgins J, Ayub MA, Shao YP, Coughlin B, Sun H, Ye EM, Cash SS, Thompson BT, Akeju O, Kuller D, Thomas RJ, **Westover MB**. Sleep staging in the ICU with heart rate variability and breathing signals. An exploratory cross-sectional study using deep neural networks. Front Netw Physiol. 2023 Feb 27;3:1120390. doi: 10.3389/fnetp.2023.1120390. PMID: 36926545; PMCID: PMC10013021.
278. Chwalisz BK, Le VK, Cheng JR, Jain A, **Westover MB**, Cheng HT. COVID-19-induced headache in Boston and the vicinity. J Clin Virol Plus. 2023 Apr 7:100148. doi: 10.1016/j.jcvp.2023.100148. Epub ahead of print. PMCID: PMC10079317.
279. Jing J, Ge W, Struck AF, Fernandes MB, Hong S, An S, Fatima S, Herlopian A, Karakis I, Halford JJ, Ng MC, Johnson EL, Appavu BL, Sarkis RA, Osman G, Kaplan PW, Dhakar MB, Jayagopal LA, Sheikh Z, Taraschenko O, Schmitt S, Haider HA, Kim JA, Swisher CB, Gaspard N, Cervenka MC, Rodriguez Ruiz AA, Lee JW, Tabaeizadeh M, Gilmore EJ, Nordstrom K, Yoo JY, Holmes MG, Herman ST, Williams JA, Pathmanathan J, Nascimento FA, Fan Z, Nasiri S, Shafi MM, Cash SS, Hoch DB, Cole AJ, Rosenthal ES, Zafar SF, Sun J, **Westover MB**. Interrater Reliability of Expert Electroencephalographers Identifying Seizures and Rhythmic and Periodic Patterns in EEGs. Neurology. 2023 Apr 25;100(17):e1737-e1749. doi: 10.1212/WNL.0000000000201670. Epub 2022 Dec 2. PMID: 36460472; PMCID: PMC10136018.
280. Coelho LMG, Blacker D, Hsu J, Newhouse JP, **Westover MB,** Zafar SF, Moura LMVR. Association of Early Seizure Prophylaxis With Posttraumatic Seizures and Mortality: A Meta-analysis With Evidence Quality Assessment. Neurol Clin Pract. 2023 Jun;13(3):e200145. doi: 10.1212/CPJ.0000000000200145. Epub 2023 Mar 30. PMID: 37066107; PMCID: PMC10101717.
281. Boncompte G, Sun H, Elgueta MF, Benavides J, Carrasco M, Morales MI, Calderón N, Contreras V, Westover MB, Cortínez LI, Akeju O, Pedemonte JC. Intraoperative electroencephalographic marker of preoperative frailty: A prospective cohort study. J Clin Anesth. 2023 Jun;86:111069. doi: 10.1016/j.jclinane.2023.111069. Epub 2023 Feb 2. PMID: 36738630; PMCID: PMC10074446.
282. McInnis RP, Ayub MA, Jing J, Halford JJ, Mateen FJ, **Westover MB**. Epilepsy diagnosis using a clinical decision tool and artificially intelligent electroencephalography. Epilepsy Behav. 2023 Apr;141:109135. doi: 10.1016/j.yebeh.2023.109135. Epub 2023 Mar 3. PMID: 36871319; PMCID: PMC10082472.
283. Goldenholz DM, Sun H, Ganglberger W, **Westover MB**. Sample Size Analysis for Machine Learning Clinical Validation Studies. Biomedicines. 2023 Feb 23;11(3):685. doi: 10.3390/biomedicines11030685. PMID: 36979665; PMCID: PMC10045793.

284. Jing J, Ge W, Hong S, Fernandes MB, Lin Z, Yang C, An S, Struck AF, Herlopian A, Karakis I, Halford JJ, Ng MC, Johnson EL, Appavu BL, Sarkis RA, Osman G, Kaplan PW, Dhakar MB, Arcot Jayagopal L, Sheikh Z, Taraschenko O, Schmitt S, Haider HA, Kim JA, Swisher CB, Gaspard N, Cervenka MC, Rodriguez Ruiz AA, Lee JW, Tabaeizadeh M, Gilmore EJ, Nordstrom K, Yoo JY, Holmes MG, Herman ST, Williams JA, Pathmanathan J, Nascimento FA, Fan Z, Nasiri S, Shafi MM, Cash SS, Hoch DB, Cole AJ, Rosenthal ES, Zafar SF, Sun J, **Westover MB**. Development of Expert-Level Classification of Seizures and Rhythmic and Periodic Patterns During EEG Interpretation. Neurology. 2023 Apr 25;100(17):e1750-e1762. doi: 10.1212/WNL.0000000000207127. Epub 2023 Mar 6. PMID: 36878708; PMCID: PMC10136013.
285. Fernandes M, Cardall A, Jing J, Ge W, Moura LMVR, Jacobs C, McGraw C, Zafar SF, **Westover MB**. Identification of patients with epilepsy using automated electronic health records phenotyping. Epilepsia. 2023 Jun;64(6):1472-1481. doi: 10.1111/epi.17589. Epub 2023 Apr 04. PMID: 36934317; PMCID: PMC10239346.
286. Fernandes M, **Westover MB**, Zafar SF. Identifying inpatient hospitalizations with continuous electroencephalogram monitoring from administrative data. Res Sq [Preprint]. 2023 May 8:rs.3.rs-2882806. doi: 10.21203/rs.3.rs-2882806/v1. PMID: 37214908; PMCID: PMC10197757.
287. Chen PM, Stekhoven SS, Haider A, Jing J, Ge W, Rosenthal ES, **Westover MB,** Zafar SF. Association of Epileptiform Activity With Outcomes in Toxic-Metabolic Encephalopathy. Crit Care Explor. 2023 May 5;5(5):e0913. doi: 10.1097/CCE.0000000000000913. PMID: 37168691; PMCID: PMC10166342.
288. Lin Z, Glass L, **Westover MB**, Xiao C, Sun J. SCRIB: Set-Classifier with Class-Specific Risk Bounds for Blackbox Models. Proc AAAI Conf Artif Intell. 2022 Jun 30;36(7):7497-7505. doi: 10.1609/aaai.v36i7.20714. PMID: 37144139; PMCID: PMC10155818.
289. Nasiri S, Ganglberger W, Sun H, Thomas RJ, **Westover MB**. Exploiting labels from multiple experts in automated sleep scoring. Sleep. 2023 May 10;46(5):zsad034. doi: 10.1093/sleep/zsad034. PMID: 36795078; PMCID: PMC10171620.
290. Chinappen DM, Xiao G, Jing J, Spencer ER, Eden UT, Kramer MA, **Westover MB***, Chu CJ*. Spike height improves prediction of future seizure risk. Clin Neurophysiol. 2023 Jun;150:49-55. doi: 10.1016/j.clinph.2023.02.180. Epub 2023 Mar 21. PMID: 37002980; PMCID: PMC10192090. *** co-senior authors.**
291. Adra N*, Dümmer L, Paixao L*, Tesh RA*, Sun H, Ganglberger W, Westmeijer M, Cardoso MDS, Kumar A, Ye E, Henry J, Cash SS, Kitchener E, Leveroni C, Au R, Rosand J, Salinas J, Lam A, Thomas RJ**, **Westover MB****. "Decoding information about cognitive health from the brainwaves of sleep." 2023 Jul 15;13(1):11448. doi: 10.1038/s41598-023-37128-7. PMID: 37454163; PMCID: PMC10349883. * Co-first authors, ** Co-senior authors.
292. Rathmell CS, Akeju O, Inouye SK, **Westover MB**. Estimating the number of cases of dementia that might be prevented by preventing delirium. Br J Anaesth. 2023 Jun;130(6):e477-e478. doi: 10.1016/j.bja.2023.03.001. Epub 2023 Apr 6. PMID: 37031027; PMCID: PMC10329187.
293. Wilcox DR, Rudmann EA, Ye E, Noori A, Magdamo C, Jain A, Alabsi H, Foy B, Triant VA, Robbins GK, **Westover MB**, Das S, Mukerji SS. Cognitive concerns are a risk factor for mortality in people with HIV and coronavirus disease 2019. AIDS. 2023 Aug 1;37(10):1565-1571. doi: 10.1097/QAD.0000000000003595. Epub 2023 May 9. PMID: 37195278; PMCID: PMC10355333.
294. Sinha N, Dauwels J, Kaiser M, Cash SS, **Westover MB**, Wang Y, Taylor PN. Reply: Computer models to inform epilepsy surgery strategies: prediction of postoperative outcome. Brain. 2017 May 1;140(5):e31. doi: 10.1093/brain/awx068. PMID: 28334902; PMCID: PMC10448005.
295. Ramantani G, **Westover MB**, Gliske S, Sarnthein J, Sarma S, Wang Y, Baud MO, Stacey WC, Conrad EC. Passive and active markers of cortical excitability in epilepsy. Epilepsia. 2023 Mar 10:10.1111/epi.17578. doi: 10.1111/epi.17578. Epub ahead of print. PMID: 36897228; PMCID: PMC10512778.
296. Amorim E, Zheng WL, Jing J, Ghassemi MM, Lee JW, Wu O, Herman ST, Pang T, Sivaraju A, Gaspard N, Hirsch L, Ruijter BJ, Tjepkema-Cloostermans MC, Hofmeijer J, van Putten MJAM, **Westover MB**. Neurophysiology State Dynamics Underlying Acute Neurologic Recovery After Cardiac Arrest.

Neurology. 2023 Aug 29;101(9):e940-e952. doi: 10.1212/WNL.0000000000207537. Epub 2023 Jul 6. PMID: 37414565; PMCID: PMC10501085.

297. Berger M, Ryu D, Reese M, McGuigan S, Evered LA, Price CC, Scott DA, **Westover MB**, Eckenhoff R, Bonanni L, Sweeney A, Babiloni C. A Real-Time Neurophysiologic Stress Test for the Aging Brain: Novel Perioperative and ICU Applications of EEG in Older Surgical Patients. Neurotherapeutics. 2023 Jul;20(4):975-1000. doi: 10.1007/s13311-023-01401-4. Epub 2023 Jul 12. PMID: 37436580; PMCID: PMC10457272.

298. Ryan SL, Liu X, McKenna V, Ghanta M, Muniz C, Renwick R, **Westover MB**, Kimchi EY. Associations between early in-hospital medications and the development of delirium in patients with stroke. J Stroke Cerebrovasc Dis. 2023 Sep;32(9):107249. doi: 10.1016/j.jstrokecerebrovasdis.2023.107249. Epub 2023 Aug 1. PMID: 37536017; PMCID: PMC10529367.

299. Sound and light levels in intensive care units in a large urban hospital in the United States. Leone MJ, Dashti HS, Coughlin B, Tesh RA, Quadri SA, Bucklin AA, Adra N, Krishnamurthy PV, Ye EM, Hemmige A, Rajan S, Panneerselvam E, Higgins J, Ayub MA, Ganglberger W, Paixao L, Houle TT, Thompson BT, Johnson-Akeju O, Saxena R, Cash SS, Thomas RJ, **Westover MB**. Chronobiology International. 2023 May 5:1-10. doi: 10.1080/07420528.2023.2207647. Online ahead of print. PMID: 37144470. PMCID: PMC10524721.

300. Namirembe GE, Baker S, Albanese M, Mueller A, Qu JZ, Mekonnen J, Wiredu K, **Westover MB**, Houle TT, Akeju O; Minimizing Intensive Care Unit Neurological Dysfunction with Dexmedetomidine-Induced Sleep Study Team. Association Between Postoperative Delirium and Long-Term Subjective Cognitive Decline in Older Patients Undergoing Cardiac Surgery: A Secondary Analysis of the Minimizing Intensive Care Unit Neurological Dysfunction with Dexmedetomidine-Induced Sleep Trial. J Cardiothorac Vasc Anesth. 2023 Sep;37(9):1700-1706. doi: 10.1053/j.jvca.2023.04.035. Epub 2023 Apr 30. PMID: 37217424; PMCID: PMC10524446.

301. Parikh H, Hoffman K, Sun H, Zafar SF, Ge W, Jing J, Liu L, Sun J, Struck A, Volfovsky A, Rudin C, **Westover MB.** Effects of epileptiform activity on discharge outcome in critically ill patients in the USA: a retrospective cross-sectional study. Lancet Digit Health. 2023 Aug;5(8):e495-e502. doi: 10.1016/S2589-7500(23)00088-2. Epub 2023 Jun 7. PMID: 37295971; PMCID: PMC10528143.

302. Eckhardt CA, Sun H, Malik P, Quadri S, Santana Firme M, Jones DK, van Sleuwen M, Jain A, Fan Z, Jing J, Ge W, Danish HH, Jacobson CA, Rubin DB, Kimchi EY, Cash SS, Frigault MJ, Lee JW, Dietrich J, **Westover MB**. Automated detection of immune effector cell-associated neurotoxicity syndrome via quantitative EEG. Ann Clin Transl Neurol. 2023 Oct;10(10):1776-1789. doi: 10.1002/acn3.51866. Epub 2023 Aug 6. PMID: 37545104; PMCID: PMC10578889.

303. Singh SD, Oreskovic T, Carr S, Papier K, Conroy M, Senff JR, Chemali Z, Gutierrez-Martinez L, Parodi L, Mayerhofer E, Marini S, Nunley C, Newhouse A, Ouyang A, Brouwers HB, **Westover B**, Rivier C, Falcone G, Howard V, Howard G, Pikula A, Ibrahim S, Sheth KN, Yechoor N, Lazar RM, Anderson CD, Tanzi RE, Fricchione G, Littlejohns T, Rosand J. The predictive validity of a Brain Care Score for dementia and stroke: data from the UK Biobank cohort. Front Neurol. 2023 Dec 1;14:1291020. doi: 10.3389/fneur.2023.1291020. PMID: 38107629; PMCID: PMC10725202.

304. Ge W, Rice HJ, Sheikh IS, **Westover MB**, Weathers AL, Jones LK, Moura L. Improving Neurology Clinical Care With Natural Language Processing Tools. Neurology. 2023 Nov 27;101(22):1010-1018. doi: 10.1212/WNL.0000000000207853. PMID: 37816638; PMCID: PMC10727205.

305. Yang C, Qian C, Singh N, Xiao C, **Westover MB**, Solomonik E, Sun J. ATD: Augmenting CP Tensor Decomposition by Self Supervision. Adv Neural Inf Process Syst. 2022;35:32039-32052. PMID: 37994346; PMCID: PMC10664864.

306. Palepu K, Sadeghi K, Kleinschmidt DF, Donoghue J, Chapman S, Arslan AR, **Westover MB**, Cash SS, Pathmanathan J. An examination of sleep spindle metrics in the Sleep Heart Health Study: superiority of automated spindle detection over total sigma power in assessing age-related spindle decline. BMC Neurol. 2023 Oct 06;23(1):359. doi: 10.1186/s12883-023-03376-3. PMID: 37803266; PMCID: PMC10557170.

307. Amorim E, Zheng WL, Ghassemi MM, Aghaeeaval M, Kandhare P, Karukonda V, Lee JW, Herman ST, Sivaraju A, Gaspard N, Hofmeijer J, van Putten MJAM, Sameni R, Reyna MA, Clifford GD, **Westover MB**. The International Cardiac Arrest Research Consortium Electroencephalography Database. Crit Care Med. 2023 Dec 01;51(12):1802-1811. doi: 10.1097/CCM.0000000000006074. Epub 2023 Oct 19. PMID: 37855659; PMCID: PMC10841086.
308. Amerineni R, Sun H, Fernandes MB, **Westover MB**, Moura L, Patorno E, Hsu J, Zafar SF. Real-World Continuous EEG Utilization and Outcomes in Hospitalized Patients With Acute Cerebrovascular Diseases. J Clin Neurophysiol. 2025 Jan 01;42(1):20-27. doi: 10.1097/WNP.0000000000001043. Epub 2023 Oct 30. PMID: 37938032; PMCID: PMC11058112.
309. Yang C, Xiao C, Westover MB, Sun J. Self-Supervised Electroencephalogram Representation Learning for Automatic Sleep Staging: Model Development and Evaluation Study. JMIR AI. 2023 Jan-Dec;2(1):e46769. doi: 10.2196/46769. Epub 2023 Jul 26. PMID: 38090533; PMCID: PMC10715804.
310. Sun H, Adra N, Ayub MA, Ganglberger W, Ye E, Fernandes MB, Paixao L, Fan Z, Gupta A, Ghanta M, Junior VM, Rosand J, **Westover MB***, Thomas RJ* *co-senior authors. "Assessing Risk of Health Outcomes from Brain Activity in Sleep: A Retrospective Cohort Study." *Neurol Clin Pract*. 2024 Feb;14(1):e200225. doi: 10.1212/CPJ.0000000000200225. Epub 2023 Dec 8. PMID: 38173542; PMCID: PMC10759032.
311. Ke SY, Wu H, Sun H, Zhou A, Liu J, Zheng X, Liu K, **Westover MB***, **Xu H, Kong XJ***. Classification of autism spectrum disorder using electroencephalography in Chinese children: a cross-sectional retrospective study. Front Neurosci. 2024 Jan 25;18:1330556. doi: 10.3389/fnins.2024.1330556. PMID: 38332856; PMCID: PMC10850305. ***co-senior authors**.
312. Fernandes M, Sun H, Chemali Z, Mukerji SS, M V R Moura L, Zafar SF, Sonni A, Biffi A, Rosand J, **Westover MB**. Brain health scores to predict neurological outcomes from electronic health records. Int J Med Inform. 2023 Dec;180:105270. doi: 10.1016/j.ijmedinf.2023.105270. Epub 2023 Oct 24. PMID: 37890202; PMCID: PMC10842359.
313. Wei R, Ganglberger W, Sun H, Hadar PN, Gollub RL, Pieper S, Billot B, Au R, Eugenio Iglesias J, Cash SS, Kim S, Shin C, **Westover MB*,** Thomas RJ*. Linking brain structure, cognition, and sleep: insights from clinical data. Sleep. 2024 Feb 8;47(2):zsad294. doi: 10.1093/sleep/zsad294. PMID: 37950486; PMCID: PMC10851868. ***co-senior authors.**
314. Dhakal K, Rosenthal ES, Kulpanowski AM, Dodelson JA, Wang Z, Cudemus-Deseda G, Villien M, Edlow BL, Presciutti AM, Januzzi JL, Ning M, Taylor Kimberly W, Amorim E, Brandon **Westover MB**, Copen WA, Schaefer PW, Giacino JT, Greer DM, Wu O. Increased task-relevant fMRI responsiveness in comatose cardiac arrest patients is associated with improved neurologic outcomes. J Cereb Blood Flow Metab. 2024 Jan;44(1):50-65. doi: 10.1177/0271678X231197392. Epub 2023 Sep 20. PMID: 37728641; PMCID: PMC10905635.
315. Song J, **Westover MB**, Zhang R. A neural mass model for disturbance of alpha rhythm in the minimal hepatic encephalopathy. Mol Cell Neurosci. 2024 Mar;128:103918. doi: 10.1016/j.mcn.2024.103918. Epub 2024 Feb 01. PMID: 38296121.
316. McCullar KS, Abbaspour S, Wang W, Aguirre AD, **Westover MB**, Klerman EB. Timing of diuretic administration effects on urine volume in hospitalized patients. Front Physiol. 2023;14:1208324. doi: 10.3389/fphys.2023.1208324. Epub 2024 Jan 23. PMID: 38321985; PMCID: PMC10844419.
317. Goldenholz DM, Karoly PJ, Viana PF, Nurse E, Loddenkemper T, Schulze-Bonhage A, Vieluf S, Bruno E, Nasseri M, Richardson MP, Brinkmann BH, **Westover MB**. Minimum clinical utility standards for wearable seizure detectors: A simulation study. Epilepsia. 2024 Apr;65(4):1017-1028. doi: 10.1111/epi.17917. Epub 2024 Feb 17. PMID: 38366862; PMCID: PMC11018505.
318. Sun H, Li P, Gao L, Yang J, Yu L, Buchman AS, Bennett DA, **Westover MB**, Hu K. Altered Motor Activity Patterns within 10-Minute Timescale Predict Incident Clinical Alzheimer's Disease. J Alzheimers Dis. 2024;98(1):209-220. doi: 10.3233/JAD-230928. PMID: 38393904; PMCID: PMC10977378.
319. Barnett AJ*, Guo Z*, Jing J*, Ge W, Kaplan PW, Kong WY, Karakis I, Herlopian A, Jayagopal LA, Taraschenko O, Selioutski O, Osman G, Goldenholz D, Rudin C†, **Westover MB**†. Improving Clinician Performance in Classifying EEG Patterns on the Ictal-Interictal Injury Continuum Using Interpretable

Machine Learning. NEJM AI. 2024 Jun;1(6):10.1056/aioa2300331. doi: 10.1056/aioa2300331. Epub 2024 May 23. PMID: 38872809; PMCID: PMC11175595. Co-first authors, †co-senior authors.

320. Goldenholz DM, Eccleston C, Moss R, Westover MB. Prospective validation of a seizure diary forecasting falls short. Epilepsia. 2024 Jun;65(6):1730-1736. doi: 10.1111/epi.17984. Epub 2024 Apr 12. PMID: 38606580; PMCID: PMC11166505.
321. Goldenholz D, Brinkmann BH, **Westover MB.** How accurate do self-reported seizures need to be for effective medication management in epilepsy? Epilepsia. 2024 Jul;65(7):e104-e112. doi: 10.1111/epi.18019. Epub 2024 May 22. PMID: 38776216; PMCID: PMC11251847.
322. Parikh H, Sun H, Amerineni R, Rosenthal ES, Volfovsky A, Rudin C, **Westover MB**, Zafar SF. How many patients do you need? Investigating trial designs for anti-seizure treatment in acute brain injury patients. Ann Clin Transl Neurol. 2024 Jul;11(7):1681-1690. doi: 10.1002/acn3.52059. Epub 2024 Jun 12. PMID: 38867375; PMCID: PMC11251465.
323. Ouyang A, Zhang C, Adra N, Tesh RA, Sun H, Lei D, Jing J, Fan P, Paixao L, Ganglberger W, Briggs L, Salinas J, Bevers MB, Wrann CD, Chemali Z, Fricchione G, Thomas RJ, Rosand J, Tanzi RE, **Westover MB**. Effects of Aerobic Exercise on Brain Age and Health in Middle-Aged and Older Adults: A Single-Arm Pilot Clinical Trial. Life (Basel). 2024 Jul 8;14(7):855. doi: 10.3390/life14070855. PMID: 39063609; PMCID: PMC11278044.
324. Singh SD, Rivier CA, Papier K, Chemali Z, Gutierrez-Martinez L, Parodi L, Mayerhofer E, Senff J, Clocchiatti-Tuozzo S, Nunley C, Newhouse A, Ouyang A, **Westover MB**, Tanzi RE, Lazar RM, Pikula A, Ibrahim S, Brouwers HB, Howard VJ, Howard G, Yechoor N, Littlejohns T, Sheth KN, Rosand J, Fricchione G, Anderson CD, Falcone GJ. The predictive validity of a Brain Care Score for late-life depression and a composite outcome of dementia, stroke, and late-life depression: data from the UK Biobank cohort. Front Psychiatry. 2024 Jul 23;15:1373797. doi: 10.3389/fpsyt.2024.1373797. PMID: 39109366; PMCID: PMC11301016.
325. Singh SD, Rivier CA, Papier K, Chemali Z, Gutierrez-Martinez L, Parodi L, Mayerhofer E, Senff J, Clocchiatti-Tuozzo S, Nunley C, Newhouse A, Ouyang A, **Westover MB**, Tanzi RE, Lazar RM, Pikula A, Ibrahim S, Brouwers HB, Howard VJ, Howard G, Yechoor N, Littlejohns T, Sheth KN, Rosand J, Fricchione G, Anderson CD, Falcone GJ. The predictive validity of a Brain Care Score for late-life depression and a composite outcome of dementia, stroke, and late-life depression: data from the UK Biobank cohort. Front Psychiatry. 2024 Jul 23;15:1373797. doi: 10.3389/fpsyt.2024.1373797. PMID: 39109366; PMCID: PMC11301016.
326. Nascimento FA, Jing J, Traner C, Kong WY, Olandoski M, Kapur S, Duhaime E, Strowd R, Moeller J, **Westover MB**. A randomized controlled educational pilot trial of interictal epileptiform discharge identification for neurology residents. Epileptic Disord. 2024 Aug;26(4):444-459. doi: 10.1002/epd2.20229. Epub 2024 Apr 26. PMID: 38669007; PMCID: PMC11329359.
327. Marcinski Nascimento KJ, **Westover MB**, Nascimento FA. Teaching the 6 EEG Spectrogram Patterns Using an Infographic. Neurol Educ. 2024 Sep 25;3(4):e200158. doi: 10.1212/NE9.0000000000200158. PMID: 39360183; PMCID: PMC11436318.
328. Luccarelli J, Smith JR, Fricchione G, **Westover MB**. Generalized Periodic Discharges Associated With Catatonia and Delirium: A Case Series. J Neuropsychiatry Clin Neurosci. 2024 Fall;36(4):340-343. doi: 10.1176/appi.neuropsych.20230174. Epub 2024 May 9. PMID: 38720623; PMCID: PMC11479820.
329. Fernandes M, Cardall A, Moura LM, McGraw C, Zafar SF, **Westover MB.** Extracting seizure control metrics from clinic notes of patients with epilepsy: A natural language processing approach. Epilepsy Res. 2024 Nov;207:107451. doi: 10.1016/j.eplepsyres.2024.107451. Epub 2024 Sep 10. PMID: 39276641; PMCID: PMC11499027.
330. Sanches PR, Tabaeizadeh M, Moura LMVR, Rosenthal ES, Caboclo LO, Hsu J, Patorno E, **Westover MB**, Zafar SF. Anti-seizure medication treatment and outcomes in acute ischemic stroke patients undergoing continuous EEG monitoring. Neurol Sci. 2022 Sep;43(9):5441-5449. doi: 10.1007/s10072-022-06183-9. Epub 2022 Jun 17. PMID: 35713732; PMCID: PMC11550088.
331. Thangavel P, Thomas J, Sinha N, Peh WY, Yuvaraj R, Cash SS, Chaudhari R, Karia S, Jing J, Rathakrishnan R, Saini V, Shah N, Srivastava R, Tan YL, **Westover MB**, Dauwels J. Improving automated

diagnosis of epilepsy from EEGs beyond IEDs. J Neural Eng. 2022 Nov 24;19(6):10.1088/1741-2552/ac9c93. doi: 10.1088/1741-2552/ac9c93. PMID: 36270485; PMCID: PMC11549972.

332. McGraw CM, Rao S, Manjunath S, Jing J, **Westover MB.** Automated quantification of periodic discharges in human electroencephalogram. Biomed Phys Eng Express. 2024 Sep 20;10(6):10.1088/2057-1976/ad6c53. doi: 10.1088/2057-1976/ad6c53. PMID: 39111323; PMCID: PMC11580140.
333. Ng MC, Zafar S, Foreman B, Kim J, Struck AF, **Westover MB**. Commentary on stimulus-induced arousal with transient electroencephalographic improvement distinguishes nonictal from ictal generalized periodic discharges. Epilepsia. 2024 Dec;65(12):3484-3487. doi: 10.1111/epi.18159. Epub 2024 Oct 18. PMID: 39422357; PMCID: PMC11649443.
334. Chang CY, Zhang B, Moss R, Picard R, **Westover MB**, Goldenholz D. Necessary for seizure forecasting outcome metrics: Seizure frequency and benchmark model. Epilepsy Res. 2024 Dec;208:107474. doi: 10.1016/j.eplepsyres.2024.107474. Epub 2024 Nov 8. PMID: 39522392; PMCID: PMC11614381.
335. Sun H, Shen S, Thomas RJ, **Westover MB,** Zhang C. Sleep as a window to understand and regulate Alzheimer's disease: emerging roles of thalamic reticular nucleus. Neural Regen Res. 2025 Jun 1;20(6):1711-1712. doi: 10.4103/NRR.NRR-D-24-00351. Epub 2024 Jul 10. PMID: 39104106; PMCID: PMC11688547.
336. He H, Li C, Ganglberger W, Gallagher K, Hristov R, Ouroutzoglou M, Sun H, Sun J, **Westover MB,** Katabi D. What radio waves tell us about sleep! Sleep. 2025 Jan 13;48(1):zsae187. doi: 10.1093/sleep/zsae187. PMID: 39155830; PMCID: PMC11725512.
337. Hadar PN, Westmeijer M, Sun H, Meulenbrugge EJ, Jing J, Paixao L, Tesh RA, Da Silva Cardoso M, Arnal P, Au R, Shin C, Kim S, Thomas RJ, Cash SS, **Westover MB.** Epilepsy is associated with the accelerated aging of brain activity in sleep. Front Physiol. 2024 Nov 28;15:1458592. doi: 10.3389/fphys.2024.1458592. PMID: 39668843; PMCID: PMC11634596.
338. Fernandes M, Gallagher K, Turley N, Gupta A, **Westover MB,** Singhal AB, Zafar SF. Automated extraction of post-stroke functional outcomes from unstructured electronic health records. Eur Stroke J. 2025 Jan 22:23969873251314340. doi: 10.1177/23969873251314340. Epub ahead of print. PMID: 39838914; PMCID: PMC11752148.
339. Rivier CA, Singh S, Senff J, Tack RW, Marini S, Clocchiatti-Tuozzo S, Huo S, Renedo D, Papier K, Conroy M, Littlejohns TJ, Chemali Z, Kourkoulis C, Payabvash S, Newhouse A, **Westover MB**, Lazar RM, Pikula A, Ibrahim S, Howard VJ, Howard G, Brouwers HB, Van Duijn CM, Fricchione G, Tanzi RE, Yechoor N, Sheth KN, Anderson CD, Rosand J, Falcone GJ. Brain Care Score and Neuroimaging Markers of Brain Health in Asymptomatic Middle-Age Persons. Neurology. 2024 Aug 27;103(4):e209687. doi: 10.1212/WNL.0000000000209687. Epub 2024 Jul 25. Erratum in: Neurology. 2024 Dec 24;103(12):e210123. doi: 10.1212/WNL.0000000000210123. PMID: 39052961; PMCID: PMC11760050.
340. Shivdat S, Zhan T, De Palma A, Zheng WL, Krishnamurthy P, Paneerselvam E, Snider S, Bevers M, O'Reilly UM, Lee JW, **Westover MB**, Amorim E. Early Burst Suppression Similarity Association with Structural Brain Injury Severity on MRI After Cardiac Arrest. Neurocrit Care. 2024 Jul 24:10.1007/s12028-024-02047-6. doi: 10.1007/s12028-024-02047-6. Epub ahead of print. PMID: 39043984; PMCID: PMC11757804.
341. Nassi TE, Oppersma E, Labarca G, Donker DW, **Westover MB***, Thomas RJ*. Morphological Prediction of CPAP Associated Acute Respiratory Instability. Ann Am Thorac Soc. 2024 Sep 17;22(1):138–49. doi: 10.1513/AnnalsATS.202311-979OC. Epub ahead of print. PMID: 39288402; PMCID: PMC11708763.***co-senior authors.**
342. Sun S, Lomachinsky V, Smith LH, Newhouse JP, **Westover MB**, Blacker DL, Schwamm LH, Haneuse S, Moura LMVR. Benzodiazepine Initiation and the Risk of Falls or Fall-Related Injuries in Older Adults Following Acute Ischemic Stroke. Neurol Clin Pract. 2025 Jun;15(3):e200452. doi: 10.1212/CPJ.0000000000200452. Epub 2025 Mar 18. PMID: 40144887; PMCID: PMC11936338.

343. Fernandes M, **Westover MB,** Singhal AB, Zafar SF. Automated Extraction of Stroke Severity From Unstructured Electronic Health Records Using Natural Language Processing. J Am Heart Assoc. 2024 Nov 5;13(21):e036386. doi: 10.1161/JAHA.124.036386. Epub 2024 Oct 25. PMID: 39450737; PMCID: PMC11935650.
344. Nascimento FA, Katyal R, Kass NR, Yuan D, Sirven JI, **Westover MB**, Beniczky S. Electroencephalography in Clinical Practice: Neurology Professionals' Views on Optimal Standards of Care. J Clin Neurophysiol. 2025 Nov 01;42(7):639-642. doi: 10.1097/WNP.0000000000001142. Epub 2025 Jan 17. PMID: 39820182.
345. Akras Z, Jing J, **Westover MB**, Zafar SF. Using artificial intelligence to optimize anti-seizure treatment and EEG-guided decisions in severe brain injury. Neurotherapeutics. 2025 Jan;22(1):e00524. doi: 10.1016/j.neurot.2025.e00524. Epub 2025 Jan 23. PMID: 39855915; PMCID: PMC11840355.
346. Freedman IG, Boncompte G, Qu JZ, Khawaja ZQ, Turco I, Mueller A, Wiredu K, McKay TB, **Westover MB**, Pedemonte JC, Akeju O. Anesthesia-induced electroencephalogram oscillations and perioperative outcomes in older adults undergoing cardiac surgery. J Clin Anesth. 2025 Mar;102:111770. doi: 10.1016/j.jclinane.2025.111770. Epub 2025 Feb 7. PMID: 39921932; PMCID: PMC11953626.
347. Goldenholz DM, Goldenholz SR, Habib S, **Westover MB.** Inductive reasoning with large language models: A simulated randomized controlled trial for epilepsy. Epilepsy Res. 2025 Mar;211:107532. doi: 10.1016/j.eplepsyres.2025.107532. Epub 2025 Feb 24. PMID: 40020525; PMCID: PMC11908886.
348. Ge W, Godeiro Coelho LM, Donahue MA, Rice HJ, Blacker D, Hsu J, Newhouse JP, Hernández-Díaz S, Haneuse S, **Westover MB**, Moura LMVR. Automated identification of fall-related injuries in unstructured clinical notes. Am J Epidemiol. 2025 Apr 8;194(4):1097-1105. doi: 10.1093/aje/kwae240. PMID: 39060160; PMCID: PMC11978607. *** co-senior authors.**
349. Azarian M, Ramezani A, Sharafkhaneh A, Maghsoudi A, Kryger M, Thomas RJ, **Westover MB,** Razjouyan J. The Association between All-Cause Mortality and Obstructive Sleep Apnea in Adults: A U-Shaped Curve. Ann Am Thorac Soc. 2025 Apr;22(4):581-590. doi: 10.1513/AnnalsATS.202407-755OC. PMID: 39746198; PMCID: PMC12005042.
350. Shaw KM, Shao YP, Ghanta M, Junior VM, Kimchi EY, Houle TT, Akeju O, **Westover MB.** Daily Automated Prediction of Delirium Risk in Hospitalized Patients: Model Development and Validation. JMIR Med Inform. 2025 Apr 18;13:e60442. doi: 10.2196/60442. PMID: 39721068; PMCID: PMC12048784.
351. Pathmanathan J, **Westover MB**, Sivakumaran S, Donoghue J, Puryear CB. The role of sleep in Alzheimer's disease: a mini review. Front Neurosci. 2025;19:1428733. doi: 10.3389/fnins.2025.1428733. Epub 2025 Feb 05. PMID: 39975973; PMCID: PMC11835855.
352. McLaren JR, Yuan D, Beniczky S, **Westover MB**, Nascimento FA. The future of EEG education in the era of artificial intelligence. Epilepsia. 2025 Jun;66(6):1838-1842. doi: 10.1111/epi.18326. Epub 2025 Mar 04. PMID: 40035709; PMCID: PMC12169395.
353. Sumsion D, Davis E, Fernandes M, Wei R, Milde R, Veltink JM, Kong WY, Xiong Y, Rao S, Westover T, Petersen L, Turley N, Singh A, Buss S, Mukerji S, Zafar S, Das S, Junior VM, Ghanta M, Gupta A, Kim J, Stone K, Mignot E, Hwang D, Trotti LM, Clifford GD, Katwa U, Thomas R, **Westover MB*,** Sun H*. Identification of Patients With Congestive Heart Failure From the Electronic Health Records of Two Hospitals: Retrospective Study. JMIR Med Inform. 2025 Apr 10;13:e64113. doi: 10.2196/64113. PMID: 40208662; PMCID: PMC12022513. *** co-senior authors**.
354. Sun H, Dang R, Haack M, Hauser K, Scott-Sutherland J, **Westover MB**, Parthasarathy S, Redline S, Thomas RJ, Mullington JM. Facility-measured nocturnal hypoxemia and sleep among adults with long COVID versus age- and sex-matched healthy adults: a preliminary observational study. Sleep Adv. 2025 Mar 22;6(2):zpaf017. doi: 10.1093/sleepadvances/zpaf017. PMID: 40365527; PMCID: PMC12070477.
355. Zhou S, Song G, Sun H, Zhang D, Leng Y, **Westover MB**, Hong S. Continuous sleep depth index annotation with deep learning yields novel digital biomarkers for sleep health. NPJ Digit Med. 2025 Apr 11;8(1):203. doi: 10.1038/s41746-025-01607-0. PMID: 40216900; PMCID: PMC11992070.
356. Ganglberger W, Nasiri S, Sun H, Kim S, Shin C, **Westover MB*,** Thomas RJ*. Refining sleep staging accuracy: transfer learning coupled with scorability models. Sleep. 2024 Nov 8;47(11):zsae202. doi:

10.1093/sleep/zsae202. PMID: 39215679; PMCID: PMC12104150. ***co-first authors; **co-senior authors.**

357. Baizer L, Bures R, Nadkarni G, Reyes-Guzman C, Ladwa S, Cade B, **Westover MB**, Durmer J, de Zambotti M, Desai M, Parekh A, Si B, Fernandez-Mendoza J, Minor K, Mazzotti DR, Lee S, Katabi D, Kiss O, Spira AP, Morris J, Seixas A, Kioumourtzoglou MA, Bridges JFP, Brown M, Hale L, Purcell S. Big data approaches for novel mechanistic insights on sleep and circadian rhythms: a workshop summary. Sleep. 2025 Jun 13;48(6):zsaf035. doi: 10.1093/sleep/zsaf035. PMID: 39945146; PMCID: PMC12163129.
358. Reese M, Wright MC, Roberts KC, Browndyke JN, Bennett M, Acker L, Devinney MJ, Reekes TH, Waligorska T, Shaw LM, Blennow K, Zetterberg H, Cohen HJ, Mathew JP, Whitson HE, **Westover MB**, Woldorff MG, Berger M, MADCO-PC, INTUIT, PRIME, and MARBLE study groups. Associations between anaesthetic dose-adjusted intraoperative EEG alpha power, processing speed, and postoperative delirium: analysis of data from three prospective studies. Br J Anaesth. 2025 Jul;135(1):109-120. doi: 10.1016/j.bja.2024.12.041. Epub 2025 Apr 11. PMID: 40221315; PMCID: PMC12226762.
359. Wei R, Buss SS, Milde R, Fernandes M, Sumsion D, Davis E, Kong WY, Xiong Y, Veltink J, Rao S, Westover TM, Petersen L, Turley N, Singh A, Das S, Junior VM, Ghanta M, Gupta A, Kim J, Lam AD, Stone KL, Mignot E, Hwang D, Trotti LM, Clifford GD, Katwa U, Thomas RJ, Mukerji S, Zafar SF, **Westover MB**, Sun H. Automated phenotyping of mild cognitive impairment and Alzheimer's disease and related dementias using electronic health records. Int J Med Inform. 2025 Aug;200:105917. doi: 10.1016/j.ijmedinf.2025.105917. Epub 2025 Apr 11. PMID: 40222334.
360. Haider MA, Khalil MH, Fernandes MB, **Westover MB**, Zafar SF. Association of Time to Continuous EEG Initiation With Outcomes in Critically Ill Patients. J Clin Neurophysiol. 2026 Jan 01;43(1):17-22. doi: 10.1097/WNP.0000000000001161. Epub 2025 Apr 16. PMID: 40237584.
361. **Westover MB**, Westover AM. General AI May Revolutionize Neurology-Or It Might Be Bad. JAMA Neurol. 2025 Oct 01;82(10):977-978. doi: 10.1001/jamaneurol.2025.0905. PMID: 40323622.
362. Krumm L, Kranz DD, Halimeh M, Nelde A, Amorim E, Zafar S, Jing J, Thomas RJ, **Westover MB***, Meisel C*. Toward a Universal Map of EEG: A Semantic, Low-Dimensional Manifold for EEG Classification, Clustering, and Prognostication. Ann Neurol. 2025 Aug;98(2):357-368. doi: 10.1002/ana.27260. Epub 2025 Jun 20. PMID: 40539771; PMCID: PMC12278028. *** co-senior authors.**
363. Westover AM, Westover TM, **Westover MB.** Interrater Reliability Estimation via Maximum Likelihood for Gwet's Chance Agreement Model. Open J Stat. 2024 Oct;14(5):481-491. doi: 10.4236/ojs.2024.145021. Epub 2024 Oct 28. PMID: 40342797; PMCID: PMC12061032.
364. Choksi D, Gutiérrez-Martínez L, Rist PM, Buring JE, Senff JR, Marini S, Kourkoulis C, Chemali Z, Newhouse A, **Westover MB**, Tanzi RE, Fricchione G, Singh S, Rosand J, Anderson CD, Yechoor N. Use of the Brain Care Score to Estimate the Risk of Incident Cerebrovascular Events in Middle-Aged Women. Neurology. 2025 Jun 10;104(11):e213674. doi: 10.1212/WNL.0000000000213674. Epub 2025 May 16. PMID: 40378376; PMCID: PMC12089658.
365. Luccarelli J, Smith JR, Turley N, Rogers JP, Sun H, Kohrman SI, Fricchione G, **Westover MB**. Electrographic Features of Catatonia With or Without Comorbid Delirium. J Neuropsychiatry Clin Neurosci. 2026;38(1):61-67. doi: 10.1176/appi.neuropsych.20240215. Epub 2025 Jun 13. PMID: 40509792; PMCID: PMC12357770.
366. Banks JC, Hariri S, Kveraga K, Ouyang A, Gallagher K, Quadri SA, Tesh RA, Reed PU, Thomas RJ, **Westover MB**, Sun H, Subramaniam B. Sleep-Based Brain Age Is Reduced in Advanced Inner Engineering Meditators. Mindfulness (N Y). 2025;16(6):1675-1692. doi: 10.1007/s12671-025-02583-y. Epub 2025 May 16. PMID: 40535579; PMCID: PMC12170783.
367. Li J, Goldenholz DM, Alkofer M, Sun C, Nascimento FA, Halford JJ, Dean BC, Galanti M, Struck AF, Greenblatt AS, Lam AD, Herlopian A, Nwankwo C, Weber D, Maus D, Haider HA, Karakis I, Yoo JY, Ng MC, Selioutski O, Taraschenko O, Osman G, Katyal R, Schmitt SE, Benbadis S, Cash SS, Tatum WO, Sheikh Z, Kong WY, Bayas G, Turley N, Hong S, **Westover MB*,** Jing J*. Expert-Level Detection of Epilepsy Markers in EEG on Short and Long Timescales. NEJM AI. 2025 Jul;2(7):10.1056/aioa2401221. doi: 10.1056/aioa2401221. Epub 2025 Jun 26. PMID: 40689158; PMCID: PMC12276842. ***co-senior authors.**

368. Leng Y, He Y, Amini S, Magdamo C, Paschalidis I, Mukerji SS, Moura LMVR, **Westover MB,** Vranceanu AM, Ritchie CS, Blacker D, Dickson JR, Das S. A GPT-4o-powered framework for identifying cognitive impairment stages in electronic health records. NPJ Digit Med. 2025 Jul 3;8(1):401. doi: 10.1038/s41746-025-01834-5. PMID: 40610683; PMCID: PMC12229571.
369. McDevitt WM, Jones TJ, Quinn L, Easter CL, Jing J, **Westover MB,** Scholefield BR, Seri S, Drury NE. Surgery on the aortic arch and feasibility of electroencephalography (SAFE) monitoring in neonates: protocol for a prospective observational cohort study. BMJ Open. 2025 Jul 10;15(7):e106423. doi: 10.1136/bmjopen-2025-106423. PMID: 40639843; PMCID: PMC12258354.
370. Alkofer M, Yang C, Ganglberger W, Beal J, Hegde M, Kang JY, Yoo JY, Gelfand MA, Thio LL, Kutluay E, Campbell Z, Schmitt S, Gleichgerrcht E, Waterhouse E, Lopez MR, Eisenschenk S, Galanti M, Singh RK, Wills KE, Meulenbrugge EJ, Dlugos D, Dean B, Halford JJ, Goldenholz D, Jing J, Thomas R, **Westover MB.** Automated detection of interictal epileptiform discharges with few electroencephalographic channels. Epilepsia. 2025 May 3;66(7):e114–20. doi: 10.1111/epi.18431. Epub ahead of print. PMID: 40317534; PMCID: PMC12291011.
371. Li J, Aguirre AD, Moura V, Jin J, Liu C, Zhong L, Sun C, Clifford G, **Westover MB**, Hong S. An Electrocardiogram Foundation Model Built on over 10 Million Recordings. NEJM AI. 2025 Jul;2(7):10.1056/aioa2401033. doi: 10.1056/aioa2401033. Epub 2025 Jun 26. PMID: 40771651; PMCID: PMC12327759.
372. Yuan D, Katyal R, Sheikh I, Karakis I, Benbadis S, Amin U, Vinayan KP, Barot N, Weber D, Greenblatt A, Beniczky S, **Westover MB,** Nascimento FA. Utility of the IFCN criteria for identifying interictal epileptiform discharges by experts: A decision hygiene approach to improve inter-rater reliability. Clin Neurophysiol. 2025 May;173:138-146. doi: 10.1016/j.clinph.2025.02.275. Epub 2025 Mar 8. PMID: 40117757; PMCID: PMC12335889.
373. Tesh RA, Zahoor A, Banks J, Gallagher K, Eckhardt CA, Sun H, Karakis I, Katyal R, Williams J, Nayak C, Herlopian A, Ng MC, Greenblatt AS, Meyers E, Westmeijer M, Harrison DS, Ganglberger W, Galina Gheihman, Fan T, Struck AF, Sheikh IS, Nascimento FA, **Westover MB**. Inter-Rater Reliability of EEG-Based Encephalopathy Grading. J Clin Neurophysiol. 2025 Jul 2:10.1097/WNP.0000000000001185. doi: 10.1097/WNP.0000000000001185. Epub ahead of print. PMID: 40601962; PMCID: PMC12335890.
374. Caffarelli M, Simmons R, Tolokh I, Karukonda V, Guterman EL, Smith W, Fox CK, **Westover MB,** Amorim E. A Quantitative Electroencephalographic Index for Stroke Detection in Adults. J Clin Neurophysiol. 2025 Mar 5;. doi: 10.1097/WNP.0000000000001151. [Epub ahead of print] PubMed PMID: 40048377; PubMed Central PMCID: PMC12360042.
375. Sun C, Jing J, Turley N, Alcott C, Kang WY, Cole AJ, Goldenholz DM, Lam A, Amorim E, Chu C, Cash S, Junior VM, Gupta A, Ghanta M, Nearing B, Nascimento FA, Struck A, Kim J, Sartipi S, Tauton AM, Fernandes M, Sun H, Bayas G, Gallagher K, Wagenaar JB, Sinha N, Lee-Messer C, Silvers CT, Gunapati B, Rosand J, Peters J, Loddenkemper T, Lee JW, Zafar S, **Westover MB.** Harvard Electroencephalography Database: A comprehensive clinical electroencephalographic resource from four Boston hospitals. Epilepsia. 2025 Sep;66(9):3411-3425. doi: 10.1111/epi.18487. Epub 2025 Jun 4. PubMed PMID: 40464151; PubMed Central PMCID: PMC12455399.
376. Nasiri, S., Ganglberger, W., Nassi, T., Meulenbrugge, E. J., Moura Junior, V., Ghanta, M., Gupta, A., Stone, K. L., Kjaer, M. R., Sum-Ping, O., Mignot, E., Hwang, D., Trotti, L. M., Clifford, G. D., Katwa, U., Sun, H.††, Thomas, R. J.††, **Westover, M. B.**†† "CAISR: Achieving Human-Level Performance in Automated Sleep Analysis Across All Clinical Sleep Metrics." *Sleep. 2025 Aug 14;48(8). doi: 10.1093/sleep/zsaf134. PubMed PMID: 40554678; PubMed Central PMCID: PMC12341759.*
††Co-senior authors.
377. Nassi T, Amidi Y, Oppersma E, Donker DW, Redeker NS, **Westover MB***, Thomas RJ*. Unravelling Sleep Apnea Dynamics: Quantifying Loop Gain Using Dynamical Modelling of Ventilatory Control. Sleep. 2025 Jul 24;. doi: 10.1093/sleep/zsaf213. [Epub ahead of print] PubMed PMID: 40704699; PubMed Central PMCID: PMC12398341. *** co-senior authors.**

378. Meulenbrugge EJ, Sun H, Ganglberger W, Nasiri S, Thomas RJ, **Westover MB.** Ordinal Sleep Depth: A Data-Driven Continuous Measurement of Sleep Depth. J Sleep Res. 2025 Apr 25:e70074. doi: 10.1111/jsr.70074. Epub ahead of print. PMID: 40276961; PMCID: PMC12353260.
379. Brooks JD, de Medeiros RC, Sun S, Sankaranarayanan M, **Westover MB**, Schwamm LH, Newhouse JP, Haneuse S, Moura LMVR. Choice of antiseizure medications and associated outcomes in Medicare beneficiaries after acute ischemic stroke. Epilepsia. 2025 Dec;66(12):4857-4868. doi: 10.1111/epi.18594. Epub 2025 Aug 06. PMID: 40770930; PMCID: PMC13649624.
380. Singh A, Sartipi S, Sun H, Milde R, Quinn C, Harrold GK, Gillani R, Turbett S, Das S, Zafar S, Fernandes M, **Westover MB*,** Mukerji SS*. "A Machine Learning Approach for Identifying People with Neuroinfectious Diseases in Electronic Health Records: Algorithm Development and Validation." JMIR Med Inform. 2025 Aug 29;13:e63157. doi: 10.2196/63157. PMID: 40882236; PMCID: PMC12396800. * **co-first authors.**
381. Huang GB, **Westover MB**, Tan EK, Wang H, Cui D, Ma WY, Wang T, He Q, Wei H, Wang N, Tian Q, Lam KY, Yao X, Wong TY. Artificial intelligence without restriction surpassing human intelligence with probability one: Theoretical insight into secrets of the brain with AI twins of the brain. Neurocomputing (Amst). 2025 Feb 28;619:129053. doi: 10.1016/j.neucom.2024.129053. Epub 2024 Dec 16. PMCID: PMC12656881.
382. Kalkach-Aparicio M, Fatima S, Selte A, Sheikh IS, Cormier J, Gallagher K, Avagyan G, Cespedes J, Krishnamurthy PV, Elazim AA, Khan N, Hussein OM, Maganti R, Larocque J, Holla S, Desai M, **Westover MB**, Hirsch LJ, Struck AF. Seizure Assessment and Forecasting With Efficient Rapid-EEG: A Retrospective Multicenter Comparative Effectiveness Study. Neurology. 2024 Jul 23;103(2):e209621. doi: 10.1212/WNL.0000000000209621. Epub 2024 Jun 14. Erratum in: Neurology. 2024 Nov 12;103(9):e209840. doi: 10.1212/WNL.0000000000209840. Erratum in: Neurology. 2024 Dec 24;103(12):e210123. doi: 10.1212/WNL.0000000000210123. PMID: 38875512; PMCID: PMC12530332.
383. Siddiquee AT, Hwang YH, Kim S, Lee SK, Lee MH, Kim HJ, Kim YJ, Kim BJ, Hadar PN, **Westover MB**, Thomas RJ, Kim NH, Shin C. Obstructive Sleep Apnea and Cerebral Microbleeds in Middle-Aged and Older Adults. JAMA Netw Open. 2025 Oct 1;8(10):e2539874. doi: 10.1001/jamanetworkopen.2025.39874. PMID: 41148136; PMCID: PMC12569715.
384. Tripathi A, Nasiri S, Ganglberger W, Nassi T, Meulenbrugge EJ, Sun H, Stone KL, Mignot E, Hwang D, Trotti LM, Reyna MA, Clifford GD, Katwa U, Thomas RJ, **Westover MB**. Automated analysis of the American Academy of Sleep Medicine Inter-Scorer Reliability gold-standard polysomnogram dataset. J Clin Sleep Med. 2025 Nov 1;21(11):1821-1829. doi: 10.5664/jcsm.11848. PMID: 40790924; PMCID: PMC12582208.
385. Donahue MA, Brooks JD, Hsu J, Price M, Blacker D, Schwamm LH, Newhouse JP, **Westover MB,** Haneuse S, Moura LMVR. Differences in patterns of outpatient epilepsy-specific medication initiation after acute ischemic stroke in the Medicare population. Epilepsia. 2025 Aug;66(8):2728-2742. doi: 10.1111/epi.18396. Epub 2025 Apr 4. PMID: 40184019; PMCID: PMC12353908.
386. Bayas G, Bayas G, Bayas A, Bayas Cardenas EO, Franco N, Sigcha CM, Shapiro K, Moskowitz S, Espinosa PS, Dey RK, Camia CS, Coates S, Stow C, Turley N, Jing J, **Westover MB.** Decreasing the Epilepsy Treatment Gap in Tena, Ecuador: A Report From 2021 to 2023. Cureus. 2025 Oct 14;17(10):e94581. doi: 10.7759/cureus.94581. PMID: 41246783; PMCID: PMC12614661.
387. Pathmanathan J, Little D, Sadeghi K, Di Marco T, Kleinschmidt D, Nerrie J, Tjiptarto N, Arslan A, Hubbard J, Olivieri A, Puryear CB, **Westover MB,** Donoghue J. The insomnia EEG score (IES): a new tool for the classification of people with poor sleep. Sleep. 2025 Nov 11:zsaf353. doi: 10.1093/sleep/zsaf353. Epub ahead of print. PMID: 41217192.
388. **Westover MB**, Kaplan PW, Husain AM. Controversies: Periodic discharges in critically ill patients -" urgent treatment is essential". Clin Neurophysiol Pract. 2025 Jul 5;10:286-291. doi: 10.1016/j.cnp.2025.06.010. PMID: 40703850; PMCID: PMC12284540.
389. Vlasac IM, Bormes GW, Do E, Benkhoukha SS, Diallo N, Fryou NL, Gioia S, Akeju O, Joseph C, Kuan A, Lapan J, Oluwadara D, Team TP, Rahman SA, Saxena R, Scheer FAJL, **Westover MB**, Winkelman JW, Woodson F, Lane JM. A Novel Home-Based Study of Circadian Rhythms: Design,

Rationale, and Methods for the Circadia Study. Sleep. 2023 Aug 9:zsad197. doi: 10.1093/sleep/zsad197. Epub ahead of print. PMID: 37555446.

390. Wang Z, Kulpanowski AM, Copen WA, Rosenthal ES, Dodelson JA, McCrory DE, Edlow BL, Taylor Kimberly W, Amorim E, **M. Brandon Westover**, Ning M, Zabihi M, Schaefer PW, Malhotra R, Giacino JT, Greer DM, Wu O. Automated detection of severe cerebral edema using explainable deep transfer learning after hypoxic ischemic brain injury. Resuscitation. 2025 Sep;214:110652. doi: 10.1016/j.resuscitation.2025.110652. Epub 2025 May 23. PMID: 40414341; PMCID: PMC12715856.

391. Sun H, Ganglberger W, Nasiri S, Nassi TE, Meulenbrugge EJ, Lam AD, Zafar S, Gupta A, Ghanta M, Moura Junior VF, Shin C, Au R, Cash SS, Thomas RJ, **Westover MB**. Health-Oriented Sleep States: Making Sleep States Reflect Health Conditions. Sleep. 2025 Aug 7:zsaf229. doi: 10.1093/sleep/zsaf229. Epub ahead of print. PMID: 40795277.*

392. Tripathi A, Ganglberger W, Sun H, Alcott C, Turley N, Waters S, Gupta A, Ghanta M, Mouda Junior V, Nasiri S, Nearing B, Stone KL, Mignot E, Hwang D, Reyna MA, Koscova Z, Robichaux C, Zhang Z, Li Q, Ganjoo G, Trotti LM, Clifford GD, Tsien Silvers C, Gunapati B, Thomas RJ†, **Westover MB*,** Maski K*, Katwa U*. "The Boston Children's Hospital Sleep Corpus: A collection of 15,695 Annotated Pediatric Polysomnograms." Sleep. 2025 Sep 16:zsaf273. doi: 10.1093/sleep/zsaf273. Epub ahead of print. PMID: 40971987. ***Co-senior authors**.*

393. Ning MH, Rodionov A, Ross JM, Ozdemir RA, Burch M, Lian SJ, Alsop D, Cavallari M, Dickerson B, Fong T, Jones RN, Libermann T, Marcantonio E, Santarnecchi E, Schmitt EM, Touroutoglou A, Travison T, Acker L, Smith MR, Sun H, **Westover MB**, Pascual-Leone Á, Berger M, Inouye SK, Shafi M. Prediction of postoperative delirium in older adults from preoperative cognition and occipital alpha power from resting-state electroencephalogram. Age Ageing. 2025 Oct 30;54(11):afaf330. doi: 10.1093/ageing/afaf330. PMID: 41222553; PMCID: PMC13171208.

394. Cheng D, Fernandes MB, **Westover MB**, Moura L, Zafar SF. Real-World Antiseizure Medication Prophylaxis and Outcomes in Hospitalized Adults with Acute Brain Injuries. Neurocrit Care. 2026 Apr;44(2):656-665. doi: 10.1007/s12028-025-02345-7. Epub 2025 Oct 1. PMID: 41034682.

395. Chang CY, Moss R, **Westover MB**, Goldenholz DM. Rigorous evaluation of five models for e-diary-only seizure forecasting: Retrospective and prospective datasets do not outperform the Napkin method. Epilepsia. 2026 Feb;67(2):753-761. doi: 10.1111/epi.18677. Epub 2025 Oct 14. PMID: 41085335; PMCID: PMC12959853.

396. Habib M, Cazé de Medeiros R, Ahsan SM, McDonald Wojciechowski A, Donahue MA, Blacker D, Newhouse JP, Schwamm LH, **Westover MB**, Moura LMVR. Claims-Based Machine Learning Classifier of Modified Rankin Scale in Acute Ischemic Stroke. J Am Heart Assoc. 2025 Oct 21;14(20):e041635. doi: 10.1161/JAHA.125.041635. Epub 2025 Oct 14. PMID: 41085187; PMCID: PMC12684556.

397. Nascimento FA, Yuan D, Sheikh IS, Hirsch LJ, Beniczky S, **Westover MB.** The EEG Talk experience: Lessons in e-teaching EEG. Epileptic Disord. 2025 Sep 4. doi: 10.1002/epd2.70096. Epub ahead of print. PMID: 40904123.**

398. Kong WY, Nascimento FA, Struck A, Duhaime E, Kapur S, Amorim E, Kapinos G, Rodriguez A, Thomas B, Desai M, Lee JW, Westover MB, Jing J. Evaluating crowdsourcing for ICU EEG annotation: A comparison with expert performance. Epilepsia. 2025 Aug 6. doi: 10.1111/epi.18547. Epub ahead of print. PMID: 40768345.**

399. Suen C, Ma J, Lin X, Thomas RJ, **Westover MB**, Stone K, Yaffe K, Ensrud KE, Redline S, Tanzi RE, Sun H, Zhang C. Linking sleep microstructure, blood markers of inflammation and metabolism, and cognition: mediation analysis in the osteoporotic fractures in men study. Geroscience. 2026 Feb;48(1):1277-1289. doi: 10.1007/s11357-025-01730-6. Epub 2025 Jun 14. PMID: 40515807; PMCID: PMC12972494.

400. Lam AD, Johnson EL, Sarkis RA, Blank LJ, Gaston TE, Shafi MM, Zepeda R, Pellerin KR, Jette N, Greve DN, Chibnik LB, Amariglio RE, Marshall GA, **Westover MB**. Late-onset unexplained epilepsy as a risk factor for cognitive impairment and dementia: Protocol for a multi-center prospective longitudinal observational study (ELUCID). Epilepsia Open. 2026 Feb;11(1):363-375. doi: 10.1002/epi4.70184. Epub 2025 Nov 29. PMID: 41317246; PMCID: PMC12903818.

401. Tăuțan AM, Jing J, Basovic L, Hadar PN, Sartipi S, Fernandes MP, Kim J, Struck AF, **Westover MB,** Zafar SF. Automated estimation of frequency and spatial extent of periodic and rhythmic epileptiform activity from continuous electroencephalography data. J Neural Eng. 2025 Dec 16;22(6). doi: 10.1088/1741-2552/ae2716. PMID: 41330044.
402. Clive C, Singh A, Overmeer B, Boris S, Peterson L, Searle J, Hooke G, Turley N, Fernandes M, Gupta A, Ghanta M, Junior VM, Mukeriji S, Zafar S, Amorim E, **Westover MB**, Sun H. Large-scale automated phenotyping of cardiac arrest and withdrawal of life-sustaining therapy using electronic health record data. Resuscitation. 2026 Jan;218:110919. doi: 10.1016/j.resuscitation.2025.110919. Epub 2025 Dec 8. PMID: 41371332.
403. Chang CY, Moss R, **Westover MB**, Goldenholz DM. Deployable seizure forecasting requires clinically meaningful performance: Response to Stirling et al. Epilepsia. 2026 Apr;67(4):1587-1588. doi: 10.1002/epi.70083. Epub 2025 Dec 31. PMID: 41474369; PMCID: PMC12854151.
404. Wiredu K, Sun H, Boncompte G, **Westover MB**, Pedemonte JC, Akeju O. The Predictive Power of Intraoperative EEG and Clinical Characteristics for Postoperative Delirium Following Cardiac Surgery. J Clin Neurophysiol. 2026 Jan 1;43(1):32-38. doi: 10.1097/WNP.0000000000001146. Epub 2025 Jan 28. PMID: 41481100.
405. Goldenholz DM, Cheng JC, Chang CY, Moss R, **Westover MB.** Does Missing Medication Acutely Change Seizure Risk? A Prospective Study. Ann Neurol. 2026 Apr;99(4):1076-1082. doi: 10.1002/ana.78134. Epub 2026 Jan 5. PMID: 41489031; PMCID: PMC12782288.
406. Thapa R, Kjaer MR, He B, Covert I, Moore Iv H, Hanif U, Ganjoo G, **Westover MB**, Jennum P, Brink-Kjaer A, Mignot E, Zou J. A multimodal sleep foundation model for disease prediction. Nat Med. 2026 Feb;32(2):752-762. doi: 10.1038/s41591-025-04133-4. Epub 2026 Jan 6. PMID: 41495409; PMCID: PMC12920147.
407. Nascimento FA, McLaren JR, Zhao W, Katyal R, Sheikh IS, Kong WY, Aljaafari D, Barot N, Benbadis S, Friedman D, Gavvala JR, Halford J, Hogan RE, Kaplan PW, Karakis I, Maheshwari A, Matthews R, O'Donovan C, Rampp S, Schuele S, Sirven J, Tatum WO, Williams J, Yacubian EM, Yuan D, Beniczky S, Sibony O, **Westover MB**. Noise in the diagnosis of epilepsy by experts. Epileptic Disord. 2026 Apr;28(2):457-469. doi: 10.1002/epd2.70181. Epub 2026 Jan 20. PMID: 41556879.
408. Nascimento FA, **Westover MB**. The indispensable human factor in EEG-based artificial intelligence. Epilepsia. 2026 Mar;67(3):1517-1518. doi: 10.1002/epi.70085. Epub 2026 Jan 24. PMID: 41579049.
409. Zhao W, Ting YL, Marcinski Nascimento KJ, Poll SR, Pineda Alvarez DE, **Westover MB**, Nascimento FA. Conventional clinical characteristics do not predict the result of genetic testing in adults with epilepsy. Seizure. 2025 Dec;133:157-160. doi: 10.1016/j.seizure.2025.10.015. Epub 2025 Oct 22. PMID: 41637156.
410. Liu K, Sun B, Wang BK, Chen J, **Westover MB,** Tian FY, Sun H, Kong XJ. An Electroencephalographic Study of Sleep Spindle and Infraslow Oscillation in Children With Autism Spectrum Disorder. J Sleep Res. 2026 Feb 10:e70309. doi: 10.1111/jsr.70309. Epub ahead of print. PMID: 41668405.
411. Zafar SF, Park H, Letourneau AR, Rock AE, **Westover MB,** Mukerji SS. Strategies for Safer Cefepime Use to Prevent Neurotoxicity Using the Electronic Health Record. Crit Care Explor. 2026 Feb 11;8(2):e1380. doi: 10.1097/CCE.0000000000001380. PMID: 41670408; PMCID: PMC12900184.
412. Savietto SF, Guillot A, Pathmanathan J, Harris M, Nordstrom K, **Westover MB**, Lemoine D, Donoghue J. Usability and stability of longitudinal at-home sleep evaluation using the Waveband electroencephalogram headband in an insomnia population. Sleep Adv. 2026 Jan 21;7(1):zpag001. doi: 10.1093/sleepadvances/zpag001. PMID: 41710032; PMCID: PMC12910617.
413. Koscova Z, Li Q, Robichaux C, Junior VM, Ghanta M, Gupta A, Rosand J, Aguirre AD, Reinertsen E, Song S, Hong S, Albert DE, Xue J, Parekh A, Sameni R, Reyna MA, **Westover MB,** Clifford GD. The Harvard-Emory ECG Database. Sci Data. 2026 Feb 25;13(1):516. doi: 10.1038/s41597-026-06861-9. PMID: 41741499; PMCID: PMC13046864.
414. Pedemonte JC, Sun H, Freedman IG, Turco I, Wiredu K, Penna A, Egana JI, Gutierrez R, Ibacache M, Cortinez LI, **Westover MB**, Akeju O, Boncompte G. Inferring preoperative cognitive function from

intraoperative electroencephalography in elderly patients using machine learning. IEEE Trans Biomed Eng. 2026 Mar 6;PP. doi: 10.1109/TBME.2026.3671187. Epub ahead of print. PMID: 41790820.

415. Tripathi A, Gupta A, Ganglberger W, Waters S, Sun H, Nasiri S, Mitra A, Stone KL, Mignot E, Hwang D, Reyna MA, Trotti LM, Clifford GD, Maski K, Katwa U, Thomas RJ, **Westover MB**. Pediatric SleepNet: A Deep Learning Network for Reliable Pediatric Sleep Staging Across Developmental Stages. Sleep. 2026 Mar 10:zsag064. doi: 10.1093/sleep/zsag064. Epub ahead of print. PMID: 41804802.
416. Sun H, Milton S, Fang Y, Taha HB, Shiju S, Thomas RJ, Ganglberger W, Pase MP, Hughes T, Purcell S, Redline S, Stone KL, Yaffe K, **Westover MB**, Leng Y. Machine Learning-Based Sleep Electroencephalographic Brain Age Index and Dementia Risk: An Individual Participant Data Meta-Analysis. JAMA Netw Open. 2026 Mar 2;9(3):e261521. doi: 10.1001/jamanetworkopen.2026.1521. PMID: 41854616; PMCID: PMC13003368.
417. Savietto SF, Guillot A, Harris M, Pathmanathan J, Chan AM, **Westover MB**, Hill D, Bertaina-Anglade V, Viardot G, Arnal P, Donoghue J. Multicenter evaluation of the Waveband System for automated sleep assessment in patients with insomnia symptoms. Sleep. 2026 Mar 28:zsag069. doi: 10.1093/sleep/zsag069. Epub ahead of print. PMID: 41903179.
418. Runcie M, Nolan N, Yoo O, Silbergleit R, Nascimento FA, Kennedy M, **Westover MB**, Goldstein JN. EEG in patients with altered mental status in the emergency department. Int J Emerg Med. 2026 Apr 2;19(1):137. doi: 10.1186/s12245-026-01200-6. PMID: 41928084; PMCID: PMC13173918.
419. Ganglberger W, Sun H, Turley N, Tripathi A, Hadar P, Gupta A, Gallagher K, Tesh R, Kim S, Nasiri S, Leng Y, Harrison S, Stone KL, Hughes T, Redline S, Au R, Manoach DS, Landolt HP, Huber R, Mignot E, Shin C, Cash SS, Thomas RJ, **Westover MB.** Brain Health from Sleep EEG: A Multicohort, Deep Learning Biomarker for Cognition, Disease, and Mortality. NEJM AI. 2026 Mar;3:10.1056/aioa2500487. doi: 10.1056/aioa2500487. Epub 2026 Feb 26. PMID: 41953210; PMCID: PMC13055923.
420. Marino FR, Ganglberger W, Sun H, Liu Y, Ding H, Thomas RJ, Au R, **Westover MB**. Better sleep now, better cognition later? Predicting cognitive function using a machine learning-based sleep EEG brain health score. Sleep. 2026 Aug 14;49(8). doi: 10.1093/sleep/zsag100. PMID: 41964500.
421. Tăuțan AM, Sarami M, Sartipi S, Turley N, Gupta A, Ghanta M, Fernandes MP, Mitra A, Kim J, Struck AF, **Westover MB,** Zafar SF. Electroencephalogram Monitoring in Critical Care: Multicenter Analysis of Timing, Duration, and Readmissions. Crit Care Explor. 2026 Apr 22;8(4):e1402. doi: 10.1097/CCE.0000000000001402. PMID: 42012873; PMCID: PMC13102442.
422. Sun H, Dang R, Li P, Xiao W, Scott-Sutherland J, Sassower KC, **Westover MB,** Felsenstein D, Thomas RJ, Haack M, Mullington JM. Facility-Measured Sleep Electroencephalographic Microstructures in Long COVID. Sleep. 2026 Apr 22:zsag090. doi: 10.1093/sleep/zsag090. Epub ahead of print. PMID: 42017829.
423. Hilger SG, Rosenthal ES, Kulpanowski AM, Dodelson JA, Cudemus-Deseda G, Villien M, Edlow BL, Januzzi JL, Ning M, Kimberly WT, Amorim E, **Westover MB**, Copen WA, Schaefer PW, Giacino JT, Greer DM, Wu O. Connectome disruptions after hypoxic-ischaemic injury associate with consciousness disorder severity. Brain Commun. 2026 Apr 2;8(2):fcag117. doi: 10.1093/braincomms/fcag117. PMID: 42051852; PMCID: PMC13111490.
424. Husain AM, Beniczky S, Bensalem-Owen M, Bland J, Brinkman BH, Hanajima R, Holub M, Jehi L, Nandedkar S, Shahrizaila N, Tankisi H, Tannemaat MR, **Westover MB**, Paulus W. IFCN position statement: use of artificial intelligence in clinical neurophysiology. Clin Neurophysiol. 2026 Apr 16:2111893. doi: 10.1016/j.clinph.2026.2111893. Epub ahead of print. PMID: 42067415.
425. Blank LJ, Johnson EL, Pellerin KR, Sarkis RA, Gaston TE, Shafi MM, Zepeda R, Camitan M, Anderson T, Hankerson A, Jan A, Jung K, Leake M, **Westover MB**, Lam AD. Late-Onset Seizures: Etiology and Demographics in US Tertiary Care Epilepsy Centers. Neurology. 2026 Jun 9;106(11):e214948. doi: 10.1212/WNL.0000000000214948. Epub 2026 May 7. PMID: 42096671.
426. Maschke C, Hadar PN, Zhang Y, Li J, Ganjoo G, Hoopes A, Guazzo A, Gupta A, Ghanta M, Nearing B, Silvers CT, Gunapati B, Thomas R, Kim JA, Mukerji SS, Dalca A, Zafar S, Lam AD, Mignot E, **Westover MB.** The Brain Imaging and Neurophysiology Dataset of large-scale multimodal neural data. Sci Data. 2026 May 21. doi: 10.1038/s41597-026-07421-x. Epub ahead of print. PMID: 42168237.

427. Mendu M, Tesh RA, Pellerin K, Steward GE, Cerda IH, Williams M, Colman M, Shah S, Lam AD, Cash SS, **Westover MB**, Kimchi EY. Video-based detection of Delirium in hospitalized adults. PLOS Digit Health. 2026 May 29;5(5):e0001462. doi: 10.1371/journal.pdig.0001462. PMID: 42213702; PMCID: PMC13221075.
428. Goldenholz DM, Goldenholz SR, Mullan H, Moss R, Drislane FW, **Westover MB**. Self-Reported Seizure Durations. JAMA Neurol. 2026 Sep 01;83(9):933-935. doi: 10.1001/jamaneurol.2026.1644. PMID: 42258203; PMCID: PMC13247841.
429. Safavynia SA, Barra ME, Der Nigoghossian C, Shinnick D, Waldrop G, Choi JM, Ganglberger W, Shen Q, Doyle K, Walline MC, **Westover MB**, Brown EN, Fins JJ, Victor J, Edlow BL, Claassen J, Schiff ND, Thaweethai T. Determinants of Delayed Recovery of Consciousness After Analgosedation Discontinuation in the ICU: Insights From Patients With COVID-19 Hypoxemic Respiratory Failure. Crit Care Med. 2026 Aug 01;54(8):2076-2089. doi: 10.1097/CCM.0000000000007205. Epub 2026 Jun 15. PMID: 42294965.
430. Nassi T, Oppersma E, Donker DW, **Westover MB**, Thomas RJ. Quantification of ventilatory control in sleep apnea: from physiological insight to computable loop gain. Sleep. 2026 Sep 09;49(9). doi: 10.1093/sleep/zsag155. PMID: 42315124; PMCID: PMC13553299.
431. Fernandes M, Turley N, Sun H, Mukerji SS, Moura LMVR, **Westover MB**, Zafar SF. Automated Prediction of Glasgow Coma Scale Scores From Unstructured Electronic Health Records Using Natural Language Processing: Development and Validation Study. J Med Internet Res. 2026 Jun 29;28:e81245. doi: 10.2196/81245. PMID: 42372230; PMCID: PMC13313573.
432. Chen S, Chen X, Huang W, Lei F, Shan C, Jin Z, Shi Y, Wang Y, Zhao R, Xu X, Lv D, Li Y, Pan S, Chiang A, **Westover MB**, Hong S, Zhang C, Yin S, Liu CF, Yi H, Tang X, Leng Y. BCGNet: an AI model trained on 600 K hours of sleep data for a novel under-pillow contactless monitoring device. NPJ Digit Med. 2026 Jul 03. doi: 10.1038/s41746-026-02885-y. Epub ahead of print. PMID: 42399358.
433. Gupta A, Tripathi A, Ganglberger W, Waters S, Sun H, Nasiri S, Stone KL, Mignot E, Hwang D, Reyna MA, Trotti LM, Clifford GD, Maski K, Katwa U, Thomas RJ, **Westover MB**. PANDA pediatric arousal neural detection architecture. NPJ Digit Med. 2026 Jul 07. doi: 10.1038/s41746-026-02967-x. Epub ahead of print. PMID: 42414584.
434. Hwang N, **Westover MB**, Mazzotti DR, Thomas RJ, Trotti LM, Hwang D, Gratton M, Shi J, Yan R, Mignot E, Clifford GD, Katwa UA, Becker KA, Klimper MT, Jara JK, Sun H. Large language model derived regular expressions for sleep phenotyping from electronic health record: a feasibility study. Sleep Adv. 2026;7(3):zpag059. doi: 10.1093/sleepadvances/zpag059. Epub 2026 Jun 11. PMID: 42460030; PMCID: PMC13372092.
435. Wen S, Ferri R, Li Q, Deng W, **Westover MB**, Thomas RJ, Nasiri S. The Respiration-Limb Coactivation Index (RLCI): A Continuous Signal-Based Biomarker for Stratifying Cardiovascular Disease Risk in a Cross-Sectional Sleep Cohort. Sleep. 2026 Jul 28:zsag204. doi: 10.1093/sleep/zsag204. Epub ahead of print. PMID: 42518228.
436. Ganglberger W, Sun H, Quinn TM, **Westover MB**, Thomas RJ. Dynamic instability of breathing during sleep predicts cognition, disease, and mortality. Sleep. 2026 Jul 31:zsag212. doi: 10.1093/sleep/zsag212. Epub ahead of print. PMID: 42536396.
437. Li Q, Wen S, Sun H, Ganglberger W, Tripathi A, Turley N, Waters S, Gupta A, Gupta A, Ghanta M, Nearing B, Wu H, Stone KL, Robichaux C, Zhang Z, Li Q, Ganjoo G, Silvers CT, Gunapati B, Maski K, Nasiri S, Hwang D, Trotti LM, Katwa U, Clifford GD, Mignot E, Thomas RJ, **Westover MB**. The Human Sleep Project: A Multi-Center Clinical Polysomnography Dataset Across the Human Lifespan. Sleep. 2026 Aug 05:zsag215. doi: 10.1093/sleep/zsag215. Epub ahead of print. PMID: 42555111.
438. Sun C, Karakis I, Herlopian A, Ng MC, Osman G, Sheikh Z, Taraschenko O, Yoo JY, Appavu BL, Arcot Jayagopal L, Kaplan PW, Lee JW, Selioutski O, Haider HA, Halford JJ, Hoch DB, Lam A, Nascimento FA, Pathmanathan J, Schmitt S, De Stefano P, Fisch U, Gijs J, Castro-Lima H, Kong WY, Cervenka MC, Dhakar MB, Fatima S, Gaspard N, Gilmore EJ, Herman ST, Holmes MG, Johnson EL, Muniz CF, Rosenthal ES, Rodriguez Ruiz AA, Sarkis RA, Shafi MM, Swisher CB, Tabaeizadeh M, Benbadis SR, Chan F, Chu CJ, Dolatshahi M, Greenblatt AS, Katyal R, Nwankwo C, Amorim E, Tatum

WO, Weber D, Loddenkemper T, Peters JM, Katwa U, Maski K, Thomas RJ, Hong S, Yuan D, Cash SS, Cole AJ, Goldenholz DM, Stow C, Kim JA, Zafar SF, Struck AF, **Westover MB**, Jing J. Toward unified and comprehensive automated electroencephalogram interpretation: a multicentre development and validation of an electroencephalogram foundation model. Lancet Digit Health. 2026 Sep;8(9):101039. doi: 10.1016/j.landig.2026.101039. Epub 2026 Aug 25. PMID: 42642264.

439. Dey RK, Yaramis A, Turley N, Mullan H, Chougule S, Gallagher K, Bayas G, Barot N, Drislane FW, Shafi MM, Pang T, Goldenholz DM, Struck AF, Kim J, Zafar S, Lee JW, Amorim E, **Westover MB**. Prognostic Significance of EEG Patterns in Post-Cardiac Arrest Coma: A Multicenter Retrospective Cohort Study of 1,000 Patients. Neurol Open Access. 2026 Sep;2(3). doi: 10.1212/wn9.0000000000000099. Epub 2026 Aug 25. PMID: 42677185; PMCID: PMC13528602.
440. Goldenholz DM, Goldenholz SR, Bhansali R, Kaptchuk TJ, **Westover MB**. Counterintuitive: reducing regression-to-the-mean worsens efficiency in simulated clinical trials. Seizure. 2026 Nov;142:168-174. doi: 10.1016/j.seizure.2026.09.002. Epub 2026 Sep 03. PMID: 42715842; PMCID: PMC13580671.
441. Marino FR, Ganglberger W, Sun H, Liu Y, Ding H, Thomas RJ, Au R, **Westover MB**. Sleep electroencephalography-derived brain health and long-term cognitive decline in community-dwelling adults. Sleep Adv. 2026;7(3):zpag089. doi: 10.1093/sleepadvances/zpag089. Epub 2026 Aug 06. PMID: 42769852; PMCID: PMC13592772.
442. Furlow S, Houston P, Bao K, Zhou B, Velasquez G, Peterson K, Hayman A, Bains A, Lieberman O, Hendrickson CM, Zier L, Singhal NS, Hemphill JC, Zafar S, **Westover MB**, Fernandes M, Amorim E. Automated Derivation of Cerebral Performance Category at Hospital Discharge After Cardiac Arrest Using Natural Language Processing and Machine Learning. Neurocrit Care. 2026 Sep 28. doi: 10.1007/s12028-026-02650-9. Epub ahead of print. PMID: 42806261.

Peer-reviewed publications (accepted, in press)

1. Sartipi S, Godfrey DS, Tauțan AM, Fernandes MP, Ghanta M, Gupta A, Nearing B, Kim J, Struck AF, Loddenkemper T, Peters J, Lee JW, **Westover MB**, Zafar SF. Automated Extraction of Seizures and Ictal-Interictal Continuum Patterns from EEG Reports to Enable Large-Scale Neurophysiology and Neurocritical Care Research. Available at SSRN. doi: 10.2139/ssrn.5489575. *SSRN*.**

Peer-reviewed publications (other - 10 total)

1. **Westover MB**. “Review of Peter Dayan & L. F. Abbott's Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems.” Philosophical Psychology. 2002. 15: 563-567.
2. Bianchi MT and **Westover MB**. (2009) “Comment on: Analysis of SPARCL trial data for reduction of future stroke risk in older stroke patients with recent stroke or TIA.” Neurology 73(10):817-818.
3. **Westover MB, Gaspard N. “**Not a simple plumbing problem: Updating our understanding of delayed cerebral ischemia in aneurysmal subarachnoid hemorrhage.” J Clin Neurophysiol. 2016 Jun;33(3):171-3. PMCID: PMC4894327. ***Invited review as Guest Editor for a special issue of JCN on the neurophysiology of subarachnoid hemorrhage.***
4. Dreier J, Fabricius M, Cenk A, Sakowitz O, Shuttleworth C, Dohmen C, Graf R, Vajkoczy P, Helbok R, Suzuki M; Schiefecker A,; Major S, Winkler M, Kang EJ, Milakara D, Oliveira-Ferreira A, Reiffurth C, Revankar G, Sugimoto K, Dengler N, Hecht N, Foreman B, Feyen B, Kondziella D, Friberg C. Piilgaard H, Rosenthal ES, **Westover MB**, Maslarova A, Santos E, Hertle D, Sanchez-Porras R, Jewell S, Balança B, Platz J, Hinzman J, Luckl J, Schoknecht K, Schöll M, Drenckhahn C, Feuerstein D, Eriksen N, Horst V, Bretz J, Jahnke P, Scheel M, Bohner G, Rostrup E, Pakkenberg B, Heinemann U, Claassen J, Carlson A, Kowoll C, Lublinsky S, Chassidim Y, Shelef I, Friedman A, Brinker G, Reiner M, Kirov S, Andrew R, Farkas E, Gueresir, E, Vatter H, Chung L, Brennan K Lieutaud T, Marinesco S, Maas A, Sahuquillo J,

Dahlem M, Richter F, Herreras O, Boutelle M, Okonkwo D, Bullock R, Witte OW, Martus P, Van den Maagdenberg A, Ferrari M, Dijkhuizen R, Shutter L, Andaluz N, Schulte A, MacVicar B, Watanabe T, Woitzik J, Lauritzen M, Strong A, Hartings J. "Recording, analysis, and interpretation of spreading depolarizations in neurointensive care: review and recommendations of the COSBID research group." Journal of Cerebral Blood Flow and Metabolism. 2016 Jun 17. pii: 0271678X16654496. [Epub ahead of print] PMID: 27317657.

5. **Westover MB,** Halford JJ, Bianchi MT. What it should mean for an algorithm to pass a statistical Turing test for detection of epileptiform discharges. Clin Neurophysiol. 2017 Jul;128(7):1406-1407. doi: 10.1016/j.clinph.2017.02.026. Epub 2017 Mar 16. PMID: 28495216; PMCID: PMC9359665.
6. Bianchi MT, Thomas RJ, **Westover MB**. An open request to epidemiologists: please stop querying self-reported sleep duration. Sleep Med. 2017 Jul;35:92-93. doi: 10.1016/j.sleep.2017.02.001. Epub 2017 Feb 17. PMCID: PMC9371612.
7. Singla S, Garcia GE, Rovenolt GE, Soto AL, Gilmore EJ, Hirsch LJ, Blumenfeld H, Sheth KN, Omay SB, Struck AF, **Westover MB**, Kim JA. "Detecting Seizures and Epileptiform Abnormalities in Acute Brain Injury." Curr Neurol Neurosci Rep. 2020 Jul 27;20(9):42. doi: 10.1007/s11910-020-01060-4. Review. PMC9353690.
8. Nasiri S, Ganglberger W, Sun H, Thomas RJ, **Westover MB**. Exploiting Labels from Multiple Experts in Automated Sleep Scoring. Sleep. 2023 Feb 16:zsad034. doi: 10.1093/sleep/zsad034. Epub ahead of print. PMID: 36795078.
9. Amorim E, **Westover MB**. Reply to "Continuous EEG in patients with extracorporeal membrane oxygenation support: Clinical need in multidisciplinary collaboration and standardized monitoring". Clin Neurophysiol. 2022 Oct;142:275-276. doi: 10.1016/j.clinph.2022.07.488. Epub 2022 Jul 26. PMID: 35933303.
10. Nascimento FA, Beniczky S, **Westover MB**, Yuan D. The quest for an ideal EEG competency assessment. Clin Neurophysiol. 2026 Oct;190:2111972. doi: 10.1016/j.clinph.2026.2111972. Epub 2026 Jun 10. PMID: 42287920.

Non-Peer-reviewed Articles

1. **Westover MB**, O'Sullivan JA. "Towards an information theoretic framework for object recognition." IEEE 2004 International Symposium on Information Theory, Chicago, Illinois, June 27-July 2, 2004.
2. **Westover MB**, O'Sullivan JA. "Achievable rates for pattern recognition: Binary and Gaussian cases." IEEE 2005 International Symposium on Information Theory, Adelaide, Australia, Sept 4-9, 2005.
3. O'Sullivan JA, Singla N, and **Westover MB.** "Successive Refinement for Pattern Recognition," IEEE Information Theory Workshop, Punta del Este, Uruguay, Feb 2006.
4. Liddle S, Zhang R, Grover L, Khitrov M, Brown J, Cobb JP, Goldman J, Chou J, **Westover MB**, Yagoda D, Reisner A, "Safety Evaluation of a Medical Device Data System". International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC'12), San Diego, California, USA on August 28 -September 1, 2012.
5. Bianchi MT, Thomas RJ, **Westover MB**. "Response." Sleep Med. 2017 Aug 3. pii: S1389-9457(17)30316-7. doi: 10.1016/j.sleep.2017.07.017. [Epub ahead of print] PMID: 2884338
6. Amorim E, Zheng WL, Ghassemi MM, Aghaeeaval M, Kandhare P, Karukonda V, Lee JW, Herman ST, Sivaraju A, Gaspard N, Hofmeijer J, van Putten MJAM, Sameni R, Reyna MA, Clifford GD, Westover MB. "The International Cardiac Arrest Research (I-CARE) Consortium Electroencephalography Database." medRxiv. 2023 Aug 28:2023.08.28.23294672. doi: 10.1101/2023.08.28.23294672. Preprint. PMID: 37693458.

7. Sun H, Ganglberger W, **Westover MB**, Thomas RJ. "Sleep as a Window into Brain Health: Clinical Implications from the Sleeping Brain Wave-Structure Associations." SLEEP. 2025 Aug;48(8):zsaf127. doi: 10.1093/sleep/zsaf127. PMID: 40355550; PMCID: PMC12351272. (Editorial).**

Book Chapters

1. **Westover MB**, Thomas TT, Greer DM, "Delirium" in Pocket Neurology, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 86-89.
2. **Westover MB**, Choi E, Awad KM, Greer DM, Editors. Pocket Neurology. Lippincott Williams & Wilkins, 2010.
3. **Westover MB**, Peters JM, Bromfield EB, "Seizures and Other Spells" in Pocket Neurology, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 57-79.
4. **Westover MB,** Xia, Z, Cho TA, "Meningitis, Encephalitis, & Brain Abscesses" in Pocket Neurology, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 136-149.
5. Thomas TT, **Westover MB**, "Rapidly Progressive Dementia" in Pocket Neurology, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 89-96.
6. Kummer TT, **Westover MB**, "Central Nervous System Vasculitis" in Pocket Neurology, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 175-177.
7. **Westover MB,** Awad K, "Ischemic Stroke, Muscular Dystrophy and Friedreich's Ataxia" in MGH Cardiology Board Review. Springer-Verlag, London, 2013.
8. **Westover MB,** Edlow BL, Greer DM, "Coma After Cardiac Arrest: Management and Neurological Prognostication" in MGH Cardiology Board Review. Springer-Verlag, London, 2013.
9. **Westover MB,** Shafi MM, "Seizures" in Pocket ICU, Lippincott Williams & Wilkins, 2013.

VI. Grant Funding

Current:

09/2021 – 05/2026
Funder: NIH-NINDS (R01 AG073410)
Title: Comparative Safety of Seizure Prophylaxis within the Medicare Program
Role: Co-Investigator (PI: Lidia Maria Moura)

05/2021 – 04/2026
Funder: NIH-NINDS (R01 NS120947)
Title: Establishing a Brain Health Index from the Sleep Electroencephalogram
Role: PI

09/2022 – 08/2027
Funder: NIH-NHLBI (R01 HL161253)
Title: Data-Driven Sleep Biomarkers of Brain Health, Heart Health, and Mortality
Role: PI

08/2022 – 04/2027
Funder: NIH-NIA (R01 AG073598)
Title: Low Neurophysiologic Resistance to Anesthetics as a Marker of Preclinical/Prodromal Alzheimer's Disease and Neurovascular Pathology, Delirium Risk and Inattention
Role: Co-Investigator (PI: Miles Berger, via Duke subcontract)

01/2023 – 12/2027
Funder: NIH-NINDS (R01 NS126282)
Title: Prospective Validation of Neurophysiologic Outcome Prediction in Acute Brain Injury
Role: PI

09/2023 – 05/2028
Funder: NIH-NINDS (R01 NS130119)
Title: Late-Onset Unexplained Epilepsy as a Risk Factor for Dementia
Role: Co-PI (with Alice D. Lam)

12/2023 – 11/2028
Funder: NIH-NINDS (R01 NS131347)
Title: Comparative Effectiveness of EEG-Guided Anti-Seizure Treatment in Acute Brain Injury
Role: Co-Investigator (PI: Sahar F. Zafar)

11/2023 – 10/2028
Funder: NIH-NIA (R01 AG078261)
Title: Delirium Dynamics: Understanding Causes and Effects
Role: Co-Investigator (PI: Eyal Kimchi, via Northwestern subcontract)

09/2023 – 09/2025
Funder: NIH-NIA (R21 AG085495)
Title: Development of Novel Polysomnography-Based Digital Biomarkers to Predict Alzheimer's Disease and Parkinson's Disease
Role: Co-Investigator (PI: Yue Leng, via UCSF subcontract)

02/2025 – 02/2026
Funder: Takeda (PO8001791544)
Title: Analysis of Risk of Narcolepsy through Clinical Notes
Role: PI

Submitted:

12/2025 – 11/2030
Funder: NIH (Pending)
Title: Leveraging Clinical Data and Automation to Predict Epilepsy After TBI
Role: Co-Investigator (PI: Jennifer Kim)

12/2025 – 11/2030
Funder: NIH (Pending)
Title: De-Biasing Effect Assessment of Seizure Effects on Neurologic Outcomes (DEBIAS)
Role: Co-Investigator (PI: Adrian Dalca)

12/2025 – 11/2030
Funder: NIH-NINDS (Pending)
Title: Automating Critical Care Electroencephalography Using AI Foundation Models
Role: PI

12/2027 – 11/2030
Funder: NIH (Pending)
Title: *Tana Mu Na?* Feasibility of Mobile EEG and Automated Intelligence to Diagnose Epilepsy in Children in Guinea-Bissau
Role: Co-Investigator (PI: Farrah Mateen)

07/2025 – 06/2030
Funder: NIH-NINDS (1UG3NS143749-01)
Title: Continuous Monitoring of Pain Using Facial Expressions in the ICU Environment (COMPANION)
Role: Co-Investigator (PI: Hao Deng)

Completed:

2012 – 2014
Funder: American Brain Foundation Clinical Research Training Fellowship

	Title: Early Detection of Delayed Cerebral Ischemia After Subarachnoid Hemorrhage Using Context-Sensitive Computational EEG Role: PI
2014 – 2019	Funder: NIH-NINDS (K23 NS090900) Title: Quantitative Neurophysiological Monitoring and Control of Sedation-Analgesia in the ICU Environment Role: PI
2017 – 2019	Funder: SAGE Therapeutics Title: Automated Detection of Seizure-Spectrum Patterns and Quantitative Neurological Outcome Prediction Role: PI
2018 – 2019	Funder: Partners Healthcare Innovation Title: Artificial Intelligence-Based Seizure Detection and Classification Role: PI
2018 – 2019	Funder: MGH-MIT Grand Challenge Title: Big Data and Deep Learning for the Interictal-Ictal-Injury Continuum Role: PI
2018 – 2021	Funder: Glenn Foundation for Medical Research Title: Electroencephalogram-Based Brain Age and Its Relationship with Cognitive Function and Sleep Quality Role: PI
2018 – 2023	Funder: NIH-NINDS (R01 NS102190) Title: Investigation of Sleep in the Intensive Care Unit Role: PI
2019 – 2024	Funder: NIH-NINDS (R01 NS107291) Title: Deep Learning for the Ictal-Interictal-Injury Continuum Role: PI
2018 – 2023	Funder: NIH-NINDS (R01 NS102574) Title: Multimodal Network Connectivity Architecture (MOCA) of the Brain and Its Role in Recovery of Consciousness in Comatose Cardiac Arrest Patients Role: Co-Investigator (PI: Ona Wu)
2019 – 2024	Funder: NIH-NIA (R01 AG060987) Title: Treating Hyperexcitability in Alzheimer's Disease with Levetiracetam to Improve Brain Function and Cognition Role: Co-Investigator (PI: Mouhsin Shafi)
2019 – 2024	Funder: NIH-NIA (RF1 AG064312) Title: Integrated Motor Activity Biomarker for Risk of Alzheimer's Dementia Role: Co-Investigator (PI: Kun Hu)

VII. Professional Activities

Editorial Service

2010	Editor, Pocket Neurology, Lippincott Williams & Wilkins
2016	Editor, Pocket Neurology, 2nd Edition; Wolters Kluwer
2019-2024	Editorial Board Member, Journal of Clinical Neurophysiology

Ad Hoc Reviewer

2010- present	Neurology
2010- present	JAMA Neurology (formerly Archives of Neurology)
2006- present	Philosophical Psychiatry
2010- present	Journal of Clinical Epidemiology
2010- present	BMJ Case Reports
2010- present	IEEE Transactions on Information Theory
2010- present	IEEE Transactions on Signal Processing
2010- present	Journal of Medical Engineering
2010- present	Journal of Clinical Neurophysiology
2010- present	Journal of Neuroscience
2010- present	European Journal of Neurology
2010- present	Critical Care Medicine
2010- present	Annals of Clinical and Translational Neurology
2010- present	Journal of Neural Engineering
2010- present	Journal of Cerebral Blood Flow & Metabolism
2010- present	Computers in Biology and Medicine
2010- present	Proceedings of the National Academy of Sciences (PNAS)
2010- present	Brain
2010- present	Epilepsy Research
2010- present	American Journal of Respiratory and Critical Care Medicine
2010- present	SLEEP

Service as Grant Reviewer

2018-2020	Ad Hoc Member, Acute Neural Injury & Epilepsy (ANIE), NIH
2021- 2025	Permanent Member, Acute Neural Injury & Epilepsy (ANIE), NIH

University Administrative Service

2011-2017	Critical Care Informatics Committee Massachusetts General Hospital
2011-2022	Neurocritical Care Clinical Review Committee Massachusetts General Hospital
2011-2022	Co-director, MGH Neuromonitoring Committee Massachusetts General Hospital
2010-2016	Member, MGH EEG Reporting System Redesign Committee, MGH
2013-2017	MGH Infusion Pump Committee Massachusetts General Hospital
2014-2022	MGH Cooling Committee Massachusetts General Hospital
2015-2018	Member, PhD Thesis Committee (Candidate: Mohammad Ghassemi), MIT
2015-2019	Member, PhD Thesis Committee (Candidate: Jing Zhi An), MIT
2017-2019	Chair, Data Safety Monitoring Board (DSMB) for R01 (PI: Ken Solt), MGH
2018-2020	Member, DSMB for R01 (PI: Alvaro Pascual Leone), BIDMC

Service to Professional Organizations

2011-2014	TRENdS (Treatment of Recurrent Nonconvulsive SeizureS) Clinical Trial Investigators Committee, Duke University / Duke Clinical Research Institute (DCRI)

2015-2017	Member at Large Critical Care EEG Monitoring Research Consortium (CCEMRC)
2016	Co-Chair, 7th International Conference on Extreme Learning Machines (ELM2016). Singapore, December 13-15, 2016.
2017-2019	Vice-Chair, Critical Care EEG Monitoring Research Consortium (CCEMRC)
2017	Co-Chair, The 8th International Conference on Extreme Learning Machines (ELM2017). Yantai, China, October 4-7, 2017.
2019-2021	Chair, Critical Care EEG Monitoring Research Consortium (CCEMRC)
2025-2029	Chair, International League Against Epilepsy (ILAE) – Commission on Big Data and AI
2025-	Committee Member, NINDS Common Data Elements (CDE) in Epilepsy Research Oversight Committee.

Community Service

2012 Christ Lutheran Nursery School, Belmont, MA	Educational Speaker
2014 Boy Scout Troop 66, Belmont, MA. Belmont, MA	Educational Speaker
2014 Butler Elementary School, Belmont, MA,	Educational Speaker
2016 Butler Elementary School, Belmont, MA,	Educational Speaker

VIII. Invited Presentations (Since 2020)

Grand Rounds:

1. "Monitoring and Closed-Loop Control of Burst Suppression." Neurology Grand Rounds, Massachusetts General Hospital, Boston, MA (2013)
2. "Continuous Brain Monitoring in Critical Care." Anesthesia, Critical Care and Pain Medicine Grand Rounds, Massachusetts General Hospital, Boston, MA (2013)
3. "Why Brains Need Computers." Neurology Grand Rounds, Massachusetts General Hospital, Boston, MA (2016)
4. "Towards Personalized Critical Care Brain Monitoring." Neurology Grand Rounds, Stanford University, Palo Alto, CA (2017)
5. "Machine Learning in Neurology." Neurology Grand Rounds, University of Wisconsin, Madison, WI (2019)
6. "Early Ischemia Detection Using EEG in Subarachnoid Hemorrhage." Neurology Grand Rounds, University Hospitals Cleveland Medical Center, Cleveland, OH (2019)
7. "Automating Clinical Neurophysiology." Neurology Grand Rounds, Stanford University, Palo Alto, CA (2019)
8. "AI in Clinical Neurophysiology." Neurology Grand Rounds, Massachusetts General Hospital, Boston, MA (2020)
9. "Covid-19 Outcome Risk Prediction." Neurology Grand Rounds, Massachusetts General Hospital, Boston, MA (2020)
10. "Automated EEG Analysis in Clinical Care." Neurology Grand Rounds, University of California San Diego, San Diego, CA (2020)
11. "Denoising the Diagnosis of Epilepsy." Neurology Grand Rounds, Einstein Medical Center Philadelphia, Philadelphia, PA (2022)
12. "Quantifying the Ictal-Interictal-Injury Continuum." Neurocritical Care Grand Rounds, University of Washington, Seattle, WA (2022)
13. "Superhuman Management of Seizures in the ICU." Neurology Grand Rounds, Yale School of Medicine, New Haven, CT (2023)
14. "Optimizing Detection and Treatment of Nonconvulsive Status Epilepticus." Geschwind Pan-Harvard Grand Rounds, Harvard Medical School, Boston, MA (2023)
15. "Big Data and AI to Achieve Superhuman Detection and Treatment of Seizures in ICU Patients." Healthcare Delivery Science Grand Rounds, Beth Israel Deaconess Medical Center / Harvard Medical School, Boston, MA (2023)

16. “Toward Optimal Treatment of Harmful Brain Activity in Critical Illness.” Neurology Grand Rounds, University Hospital Cleveland Medical Center, Cleveland, OH (2024)
17. “Reducing Noise in the Diagnosis of Epilepsy.” Neurology Grand Rounds, University of Texas Houston, Houston, TX (2024)
18. “Reducing Noise in the Diagnosis of Epilepsy.” Neurology Grand Rounds, University of Pennsylvania, Philadelphia, PA (2024)
19. “Reducing Diagnostic Noise in Neurology: Lessons from Epilepsy.” Neurology Grand Rounds, Beth Israel Deaconess Medical Center / Harvard Medical School, Boston, MA (2025)
20. “Reducing Noise in Neurology: Lessons from Epilepsy.” Neurology Grand Rounds, University of Michigan, Ann Arbor, MI (2025)
21. “Reducing Diagnostic Noise in Neurology: Lessons from Epilepsy.” Neurology Grand Rounds, Stanford University, Palo Alto, CA (2025)
22. “Designing Randomized Trials for Nonconvulsive Status Epilepticus Using Observational Data.” Neurology Grand Rounds, Montefiore Einstein, Bronx, NY (2026)

Local Invited Presentations:

1. “What is monkey cortex layer four for?” Neuroanatomy & Neurobiology Department, Washington University School of Medicine, St. Louis, MO (2002)
2. “EEG Patterns in the Neurological ICU.” MGH Neurosciences ICU, Boston, MA (2010)
3. “Should Statins Be Avoided After Intracerebral Hemorrhage? Implications of SPARCL.” MGH Stroke Research Center, Boston, MA (2010)
4. “Fundamentals of ICU EEG Monitoring.” Nursing Skills Day, MGH Neurosciences ICU, Boston, MA (2012)
5. “Status Epilepticus and the Ictal–Interictal Continuum: Introduction to ICU EEG.” MGH Critical Care Center, Boston, MA (2012)
6. “Neurological Prognostication in Coma After Cardiac Arrest.” MGH Critical Care Center, Boston, MA (2013)
7. “Monitoring and Closed-Loop Control of Burst Suppression.” Department of Brain & Cognitive Sciences, MIT, Cambridge, MA (2014)
8. “EEG-Related Skin Lesions: Statistical Process Monitoring & Risk Reduction.” MGHfC Quality & Safety Operations Lead Team Meeting, Boston, MA (2014)
9. “Fundamentals of Continuous EEG Monitoring in Neurocritical Care; Quantitative Clinical EEG Fundamentals.” MGH Neurosciences ICU & BWH/MGH Clinical Neurophysiology Fellows Didactic Conference, Boston, MA (2014)
10. “Automated Surveillance for Ventilator-Associated Pneumonia.” MGH Infection Control Committee, Boston, MA (2015)
11. “The Role of Continuous EEG in Coma After Cardiac Arrest: Prognostic Value.” MGH EEG Technologist Training Meeting, Boston, MA (2015)
12. “Quantitative Electroencephalography.” Epilepsy Fellows Training Conference, MGH Neurology, Boston, MA (2017)
13. “CDAC: Bringing Big Clinical Data to Life.” MGH Neurosurgery Faculty Meeting, Boston, MA (2018)
14. “Landmark Project #4: Clinical Time Series Data.” Partners HealthCare EDMT Meeting, Boston, MA (2018)
15. “Validating the Brain Age Index: Modifying Brain Aging Through Exercise.” McCance Center for Brain Health Faculty Meeting, Boston, MA (2019)
16. “Automating Clinical Neurophysiology.” Special Seminar, Department of Brain & Cognitive Sciences, MIT, Cambridge, MA (2020)
17. “Automatic Detection of Nonconvulsive Seizures.” BIDMC Research Day, Boston, MA (2023)
18. “Are We Still Worried AI Might Kill Us All? Yeah.” (with Alek Westover) BIDMC NeuroAI Special Interest Group, Boston, MA (2026)

National and Regional Meetings:

1. "Automatic Detection of Beta Amyloid in Digital Images Using Visual Texture." Alzheimer's Disease Research Center, Washington University, St. Louis, MO (2005)
2. "EEG Monitoring and Closed-Loop Control of Burst Suppression." American Clinical Neurophysiology Society Annual Meeting, Miami, FL (2013)
3. "Early Detection of Delayed Cerebral Ischemia Using EEG." Greater Boston Epilepsy Society, Boston, MA (2015)
4. "Why Brains Need Computers: How Computer Science and Engineering Can Improve Neurology." Computational Health Distinguished Lecture Series, Georgia Institute of Technology, Atlanta, GA (2016)
5. "Polysomnographic Signatures of Brain Age." Massachusetts Sleep Society, Westborough, MA (2017)
6. "Artificially Intelligent Brain Monitoring." World Medical Innovation Forum, Boston, MA (2019)
7. "Expert-Level Automatic Detection of Epileptiform Discharges." American Neurological Association, St. Louis, MO (2019)
8. "Quantitative EEG for Ischemia Detection." American Clinical Neurophysiology Society Annual Meeting, Las Vegas, NV (2019)
9. "EEG Monitoring for Early Detection of Cerebral Ischemia." American Clinical Neurophysiology Society Annual Meeting, New Orleans, LA (2020)
10. "Automating Clinical Neurophysiology." Duke Comprehensive Epilepsy Center Quarterly Research Symposium, Durham, NC (2020)
11. "Big Data and Machine Learning to Advance Status Epilepticus Research." American Neurological Association, Virtual Meeting (2020)
12. "Automated Detection of Seizures and Ictal–Interictal Continuum Patterns Using Deep Learning." American Clinical Neurophysiology Society Annual Meeting, Virtual (2021)
13. "Sleep as a Window into Brain Health." Harvard Sleep Grand Rounds, Virtual (2021)
14. "Avoiding Pitfalls in EEG Interpretation." American Academy of Neurology Annual Meeting, Seattle, WA (2022)
15. "Artificial Intelligence in Sleep Medicine." American Neurological Association Annual Meeting, Chicago, IL (2022)
16. "Addressing Problems in Critical Care Neurophysiology Using Machine Learning." Machine Learning in Medicine Seminar Series, Carnegie Mellon University, Pittsburgh, PA (2022)
17. "Electroencephalogram-Based Brain Age and Its Relation with Cognitive Function and Sleep Quality." American Federation for Aging Research (AFAR) Grantee Conference, Santa Barbara, CA (2022)
18. "Spectrograms in ICU EEG." American Clinical Neurophysiology Society Fall Course: ICU EEG Monitoring, San Diego, CA (2022)
19. "Denoising and Debiasing the Diagnosis of Epilepsy." NESET Fall Educational Meeting, Quincy, MA (2022)
20. "Epileptiform and Non-Epileptiform Findings in ICU Electroencephalography." American Academy of Neurology Annual Meeting, Boston, MA (2023)
21. "Neurophysiological Grading of Encephalopathy." American Academy of Neurology Annual Meeting, Boston, MA (2023)
22. "Identifying and Overcoming Barriers for Clinical Use of Artificial Intelligence in Sleep Medicine." American Academy of Sleep Medicine Annual Meeting, Indianapolis, IN (2023)
23. "Optimizing Treatment of Nonconvulsive Status Epilepticus." American Epilepsy Society Annual Meeting, Orlando, FL (2023)
24. "Multimodal Monitoring and Brain Monitoring for Ischemia." American Clinical Neurophysiology Society Annual Meeting, Austin, TX (2023)
25. "Hot Topics in AI and ICU EEG." American Clinical Neurophysiology Society Annual Meeting, Austin, TX (2023)

26. “Collaborating with Industry.” American Clinical Neurophysiology Society Annual Meeting, Orlando, FL (2024)
27. “Quantitative EEG.” American Clinical Neurophysiology Society Annual Meeting, Orlando, FL (2024)
28. “The EEG Map of Everything (MOE).” Critical Care EEG Monitoring Research Consortium Meeting, Orlando, FL (2024)
29. “AI in ICU EEG.” American Epilepsy Society Annual Meeting, Los Angeles, CA (2024)
30. “Machine Learning Approaches to Studying Neurophysiology in LoUE.” Defining the Landscape of Late-Onset Unexplained Epilepsy Conference, South Casco, ME (2025)
31. “Data-Driven Biomarkers for Brain Health from Sleep.” SLEEP 2025 Annual Meeting, Seattle, WA (2025)
32. “Big Data in Epilepsy Research.” NINDS Curing the Epilepsies Conference, Bethesda, MD (2026)

International Meetings:

1. “Opportunities for Intelligent Computing in Clinical Neurophysiology.” Nanyang Technical University Symposium on Novel Technologies for Clinical Practice in Neurology, Singapore (2012)
2. “Real-Time Measurement and Closed-Loop Control of Burst Suppression.” IEEE Engineering in Medicine and Biology Society Conference, Osaka, Japan (2013)
3. “Optimizing the Treatment of Refractory Status Epilepticus.” Networks in Epilepsy Conference, Beijing, China (2015)
4. “Engineering in Neurocritical Care: Modeling & Control of Refractory Status Epilepticus.” International Symposium on EEG Signal Analysis, Xi’an, China (2015)
5. “Quantitative Electroencephalography for Detection of Seizures and Ischemia.” Beijing ICU Neuroelectrophysiological Monitoring Forum, Beijing, China (2015)
6. “Why Brains Need Computers: How Machine Learning and Big Data Can Help Neurology.” Extreme Learning Machines Annual Conference, Singapore (2016)
7. “Machine Learning in Neurocritical Care.” Northwestern University International Forum on Big Data in Healthcare, Xi’an, China (2017)
8. “Quantitative EEG: Detecting Ischemia and Seizures.” International Congress of Clinical Neurophysiology, Washington, DC (2018)
9. “Removing the Confusion from Toxic-Metabolic Encephalopathy.” University of Twente Mini-Symposium, Enschede, Netherlands (2018)
10. “Mapping the Ictal–Interictal–Injury Continuum.” Neurocritical Care Society Annual Meeting, Vancouver, Canada (2019)
11. “Superhuman Detection of Seizures, Rhythmic and Periodic Patterns in ICU EEG.” International Congress on Clinical Neurophysiology (ICCN), Geneva, Switzerland (2022)
12. “Quantifying the Ictal–Interictal Continuum.” International Conference on Technology and Analysis for Seizures (ICTALS), Bern, Switzerland (2022)
13. “The Brain Data Science Platform.” (with Junior Moura) International Epilepsy Data Ecosystem Workshop, Neuroscience School of Advanced Studies, Crans-Montana, Switzerland (2023)
14. “Quantitative EEG Primer.” Teaching Course: Continuous EEG Monitoring, International Congress of Clinical Neurophysiology (ICCN), Jakarta, Indonesia (2024)
15. “Urgent Treatment for Periodic Discharges in Critically Ill Patients Is Essential.” Controversies Session, International Congress of Clinical Neurophysiology (ICCN), Jakarta, Indonesia (2024)
16. “Using AI in Continuous EEG Monitoring.” Symposium: Using Artificial Intelligence in Clinical EEG, International Congress of Clinical Neurophysiology (ICCN), Jakarta, Indonesia (2024)
17. “Denoising the Diagnosis of Epilepsy.” AI in Epilepsy Conference, Breckenridge, CO (2025)
18. “MORGOTH: Automating the Interpretation of Clinical Neurophysiology Tests.” International Conference on Technology and Analysis for Seizures (ICTALS), Montreal, Canada (2025)
19. “Artificial Intelligence in Neurology: Promise and Peril.” Donald Baxter Lecture, Montreal Neurological Institute, Montreal, Canada (2025)

IX. Teaching

Teaching during Stanford appointment:

None.

Teaching prior to Stanford:

Courses

2001–2002	Instructor for Computational Neuroscience, Washington University School of Medicine, Department of Anatomy and Neurobiology.
2003	Instructor for Detection and Estimation Theory, Washington University, Department of Electrical and Systems Engineering.
2004	Instructor for Information Theory, Washington University, Department of Electrical and Systems Engineering.
2008–2009	Instructor for Neuroanatomy Laboratory, Harvard Medical School.
2012–2025	Lecturer for Medical Decision Analysis and Diagnostic Test Interpretation (HST-192.0), Harvard–MIT Program in Health Sciences and Technology.
2013–2017	Course Director and Lecturer for Medical Decision Analysis and Diagnostic Test Interpretation (HST-192.0), Harvard–MIT Program in Health Sciences and Technology.

Tutorials and Lectures

10/2009	Granger Causality Analysis Techniques for EEG Data. Clinical Neurophysiology Research Group, MGH, Boston, MA.
5/2010	Synchronization Likelihood for Functional Connectivity Analysis of EEG Data. Clinical Neurophysiology Research Group, MGH, Boston, MA.
7/2008	Probability in Medical Reasoning. Neurology Residents, Brigham and Women's Hospital, Boston, MA.
9/2009	Invited lecturer, Bayes' Rule for Physicians. Partners Neurology, MGH/BWH, Boston, MA.
6/2010	EEG Alphabet Soup: GCSE, NCSE, PLEDs, BIPLEDs, GPEDs, & DEADs. Partners Neurology, MGH/BWH, Boston, MA.
8/2011	Status Epilepticus: Diagnosis and Management. Partners Neurology, MGH/BWH, Boston, MA.
9/2012	Introduction to ICU EEG: Seizures, Spikes, PEDs, Ischemia, and Coma. Partners Neurology, MGH/BWH, Boston, MA.
7/2018	Quantitative Electroencephalography. Neurophysiology and Epilepsy Fellows, MGH Epilepsy Division, Boston, MA.
7/2019	Recognizing Seizures and Ischemia in Critical Care EEG Monitoring. Neurophysiology and Epilepsy Fellows, MGH Epilepsy Division, Boston, MA.
6/2020	Emergency Electroencephalography. Emergency Medicine Residents, Massachusetts General Hospital, Boston, MA.
7/2020	Introduction to Spectrograms in ICU EEG. Neurophysiology and Epilepsy Fellows, MGH Epilepsy Division, Boston, MA.

Clinical Supervisory and Training

2012–	Supervisor, MGH Adult Epilepsy Clinic (½-day session per week).
2012–	Attending Physician, MGH Inpatient Neurology Service (2-week block per year), supervising neurology residents and Harvard medical students.
2012–	Attending Physician, MGH General Neurology Consult Service (2-week block per year), supervising neurology residents.
2012–	Director, MGH Critical Care EEG Monitoring Service (≈10 weeks per year), supervising Epilepsy and Clinical Neurophysiology Fellows.
2012–	Attending, MGH Epilepsy Monitoring Unit (≈6 weeks per year), supervising Epilepsy and Clinical Neurophysiology Fellows.
2012–	Faculty, Outpatient EEG Reading Service (≈2 days per month), supervising Epilepsy and Clinical Neurophysiology Fellows.

Research Supervisory and Training Responsibilities

2012–	Supervisor of postdoctoral research fellows (average 5–10 fellows per year), Massachusetts General Hospital.
2012–	Supervisor of graduate students (average 2–4 per year), Massachusetts General Hospital and MIT.

X. Trainees

Postdoctoral Fellows

Sahar Zafar, MD (K23)	2015–present
Jennifer Kim, MD, PhD (K23)	2014–2020
Edilberto Amorim, MD (K23)	2014–2019
Marta Priscila Bento Fernandes, PhD	2019–present
Rajesh Amerineni, PhD	2020–2022
Wendong Ge, PhD	2019–2022
Weilong Zheng, PhD	2019–2022
Yalda Amidi, PhD	2022–2024
Samaneh Nasiri, PhD	2021–2023
Wolfgang Ganglberger, PhD	2019–present
Shadi Sartipi, PhD	2024–present
Alexandra Tautan, PhD	2024–present
Ayush Tripathi, PhD	2024–present
Samuel Waters, PhD	2024–present
ChenXi Sun, PhD	2024–present
Charlotte Maschke, PhD	2024–present
Bruce Nearing, PhD	2024–present
Ayan Mitra, PhD	2024–present
Sungtae An, MS / PhD	2016–2018
Haoqi Sun, MD	2016–present
Jing Jin, PhD	2013–present

Graduate Students

Jingzhi An, PhD	2014–2020
Mohammad Ghassemi, PhD	2014–2019
Jefferson T. Oliva, PhD	2016–2017
Qiang Li, PhD	2018–2020
Arnav Gupta	2024–present
Thijs Nassi, MSc	2021–2025

Medical Students / Residents / Fellows

Elliott Akama-Garren	2014–2015
Jennifer Kim, MD, PhD	2014–2020
Eyal Kimchi, MD, PhD (K23)	2018–present
Daniel Goldenholz, MD, PhD (K23)	2022–present
Peter Hadar, MD, PhD	2022–present
Wan Yee Kong, MD, MSc	2022–2025
James Lucarelli, MD, PhD	2022–present
Callison Alcott, MD, PhD	2023–present
Sarah Wahlster, MD	2011–2012

Craig Williamson, MD	2011–2012
Lidia Moura, MD	2013–2014
Marcus Ng, MD	2013–2014
Patrick Chen, MD	2013–2014
Deirdre O'Rourke, MD	2013–2014
Sandipan Pati, MD	2013–2014
Ammar Kheder, MD	2013–2014
Aaron Berkowitz, MD, PhD	2013–2014
Luis Paixao, MD	2017–2019
Aline Herlopian, MD	2017–present
Nitish Harid, MD	2020–2021

Undergraduate Students

Nathaniel Eiseman	2010–2011
Wade Whitt	2017–2019
Jacob Hogan	2018–2020
Alek Westover	2018–2023
Danny Jones	2021–2022
Jaden Barfuss	2021–2022
Jayme Banks	2021–2023
Leia Jekel	2023–2024
Rebecca Milde	2023–2024
Moritz Alkofer	2023–2024
Elijah Davis	2023–2024
Dan Sumsion	2023–present
Lydia Petersen	2023–present
Kiwi (Yiwen) Xiong	2023–present
Luke Jackson	2024–present
Carlos Nuñez-Sosa	2024–present
Amber Peng	2024–present
Mitchell McCauley	2024–present
Patrick Wynn	2024–present
Elijah Keldsen	2024–present

High School Students

Sophia Bechek	2016–2017
Samvrit Rao	2023–2024